Carbon Capture Container Design for Ambient Pressure Carbon Dioxide Removal from Climate–Friendly Gensets

Rohit Ray

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Main Supervisor: Dr. Antonius Kies (Scania R&D)
Co-Supervisor: Dr. Henrik Birgersson (Traton Group)
Academic Examiner: Prof. Matthäus Bäbler (KTH Royal Institute of Technology)

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Abstract

The rapid deployment of Battery-electric Heavy-duty vehicles is increasing the need for high-power charging infrastructure around Europe, while reinforcement of local electricity grids can require several years. A stopgap solution to this can be Diesel Generator sets, which can provide temporary, decentralised charging capacity, but they also create a concentrated source of fossil Carbon Dioxide. This study investigates whether these emissions can be substantially reduced through a compact Carbon Capture Container (CCC) integrated with one or two Scania DC13-505A Generator sets operating at Continuous Power Operation (CPO) and Maximum Power Operation (MPO) modes. The design target was to achieve 90% CO_2 Capture Efficiency with low exhaust line backpressure, efficient use of Engine Waste Heat, and delivery of Transport-ready liquid CO_2.

A moderate–temperature and ambient–pressure Regenerative Absorption process using CESAR-1, an aqueous blend of 3 M 2-Amino-2-Methyl-1-Propanol (AMP) and 1.5 M Piperazine (PZ), was modelled in Aspen Plus V14 using the ENRTL-RK property method and Rate-based packed-column calculations. The modelling framework was benchmarked against published data from the NCCC, CESAR Esbjerg, SINTEF for Monoethanolamine (MEA) solvent and University of Kaiserslautern Pilot Plant for CESAR–1 solvent. Key indicators namely CO_2 Capture Rate, Solvent CO_2 loading and Specific Reboiler Duty were reproduced within a relative error margin of 5–8%, although higher deviations in the internal absorber-temperature profile were observed at lower liquid-to-gas (L/G) ratios below 4.0 mol Solvent/mol Flue Gas for MEA solvent based model validations. Following model validation, a functional flowsheet model for CESAR-1 based CO_2 Capture unit was constructed for Scania Generators. Sensitivity analyses were then used to optimise Absorber height, total Stripper height, Lean-Solvent CO_2 loading, Recuperator area and the number of Compression Stages. Waste-heat recovery and upgrading, CO_2 Compression, Liquefaction and Equipment costs were evaluated separately using analytical calculations, Microsoft Excel and MATLAB.

The selected common column design comprises a 0.77 m internal-diameter Absorber with 1.95 m of packing and two 2.0 m Stripper sections with internal diameters of 0.38 and 0.55 m. A Lean loading of 0.10 mol CO_2 per mol AMP+PZ and a three-stage Compression unit were selected. Under normal climatic conditions, the system captured approximately 178–260 kg CO_2 hr^{-1} for Single Genset and 357–519 kg CO_2 hr^{-1} for Two Gensets Operation, with specific Reboiler duties of 2.97–3.20 MJ_{th} $kg_{CO_2}^{-1}$. Simulated Exhaust Gas pressure drops were only 2.6–11.4 mbar(a), well below the 50 mbar(a) design limit. Integrated Exhaust,Coolant and Charge-air heat, upgraded by a Type II Absorption Heat Pump, supplied 3.51–3.55 MJ_{th} $kg_{CO_2}^{-1}$ under normal conditions. Under hot-summer conditions, the two-genset Maximum Power Operation mode case required a small supplementary heat input to retain 90% capture. The conditioned product reached 99.4–99.7 wt% CO_2 and was liquefied to $-25\,^{\circ}C$ and 20 bar(a) for Road Transport by Tank Trailer Trucks.

The estimated fixed capital investments were € 1.65 million and € 2.36 million for the Single- and Two-Genset systems, respectively (estimated for year 2024). At 2000 operating hours per year, the corresponding specific capture costs were approximately € 1200 and € 860 per tonne of CO_2 (2024 estimate), excluding transport and storage costs. The results therefore indicate strong technical feasibility but weak present-day economics at this small-scale under-design operating condition. Higher annual CO_2 utilisation, serial modular production, engine-side heat-recovery modifications and future operation with biofuels could materially improve the business and climate case.

Keywords: Carbon Capture Container; Diesel Generator Set; CESAR–1; AMP/PZ; Aspen Plus; Waste Heat Integration; Absorption Heat Transformer; CO_2 Compression and Liquefaction; Techno-economic Assessment.

Sammanfattning

Den snabba introduktionen av batterielektriska tunga fordon ökar behovet av infrastruktur för högeffektsladdning i Europa, samtidigt som förstärkning av lokala elnät kan ta flera år. En möjlig övergångslösning är dieseldrivna generatoraggregat, vilka kan tillhandahålla tillfällig och decentraliserad laddningskapacitet. Dessa aggregat utgör emellertid även en koncentrerad källa till fossil koldioxid. I denna studie undersöks om dessa utsläpp kan minskas avsevärt genom en kompakt containeriserad koldioxidavskiljningsenhet, benämnd Carbon Capture Container (CCC), integrerad med ett eller två Scania DC13-505A-generatoraggregat som drivs i driftlägena Continuous Power Operation (CPO) och Maximum Power Operation (MPO). Designmålet var att uppnå en avskiljningsgrad på 90% för CO_2, samtidigt som ett lågt mottryck i avgasledningen bibehölls, motorns spillvärme utnyttjades effektivt och transportfärdig flytande CO_2 producerades.

En regenerativ absorptionsprocess vid förhöjd temperatur med CESAR-1, en vattenbaserad blandning av 3 M 2-amino-2-metyl-1-propanol (AMP) och 1,5 M piperazin (PZ), modellerades i Aspen Plus V14 med egenskapsmetoden ENRTL-RK och hastighetsbaserade beräkningar för packade kolonner. Modelleringsramverket validerades mot publicerade data från NCCC, CESAR-pilotanläggningen i Esbjerg, SINTEF:s pilotanläggning för lösningsmedlet monoetanolamin (MEA) samt pilotanläggningen vid universitetet i Kaiserslautern för CESAR-1. Nyckelindikatorer, nämligen CO_2-avskiljningsgrad, CO_2-belastning i lösningsmedlet och specifik reboilerkokarbehov (SRD), återgavs inom en relativ felmarginal på 5–8%, även om större avvikelser i den interna absorbertemperaturprofilen observerades vid lägre vätska-till-gas-förhållanden (L/G) under 4,0 mol lösningsmedel/mol rökgas vid modellvalideringar baserade på MEA-lösningsmedel. Efter modellvalideringen utvecklades en funktionell flödesschemamodell för en CESAR-1-baserad CO_2-avskiljningsenhet anpassad till Scanias generatoraggregat. Känslighetsanalyser genomfördes därefter för att optimera absorbertornets höjd, den totala stripperhöjden, CO_2-belastningen i den magra lösningen, rekuperatorns värmeöverföringsarea samt antalet kompressionssteg. Återvinning och temperaturhöjning av spillvärme, CO_2-kompression, förvätskning och utrustningskostnader utvärderades separat med analytiska beräkningar, Microsoft Excel och MATLAB.

Den valda gemensamma kolonnutformningen består av ett absorbertorn med en innerdiameter på 0,77 m och en packningshöjd på 1,95 m samt två strippersektioner med en packningshöjd på 2,0 m vardera och innerdiametrar på 0,38 respektive 0,55 m. En magerlösningsbelastning på 0,10 mol CO_2 per mol AMP+PZ samt en trestegskompressionsenhet valdes. Under normala klimatförhållanden avskilde systemet cirka 178–260 kg CO_2 hr^{-1} vid drift med ett generatoraggregat och 357–519 kg CO_2 hr^{-1} vid drift med två generatoraggregat. Det specifika återkokarvärmebehovet uppgick till 2,97–3,20 MJ_{th} $kg_{CO_2}^{-1}$. De simulerade tryckfallen i avgassystemet var endast 2,6–11,4 mbar(a), vilket ligger klart under designgränsen på 50 mbar. Integrerad värme från avgaser, motorkylvätska och laddluft, vars temperaturnivå höjdes med hjälp av en absorptionsvärmepump av typ II, tillhandahöll 3,51–3,55 MJ_{th} $kg_{CO_2}^{-1}$ under normala förhållanden. Under varma sommarförhållanden krävde MPO-fallet med två generatoraggregat ett mindre tillskott av värme för att bibehålla en avskiljningsgrad på 90%. Den konditionerade produktströmmen uppnådde en CO_2-halt på 99,4–99,7 viktprocent och förvätskades till $-25\,°C$ och 20 bar(a) för vägtransport med tanktrailer.

Den uppskattade fasta kapitalinvesteringen uppgick till € 1,65 miljoner för systemet med ett generatoraggregat och € 2,36 miljoner för systemet med två generatoraggregat, baserat på kostnadsnivån för år 2024. Vid 2 000 drifttimmar per år beräknades de motsvarande specifika avskiljningskostnaderna till cirka € 1 200 respektive € 860 per ton CO_2 i 2024 års kostnadsnivå, exklusive kostnader för transport och lagring. Resultaten visar således att konceptet har god teknisk genomförbarhet men svag ekonomisk konkurrenskraft under dagens småskaliga utformning och den antagna årliga drifttiden. En högre årlig utnyttjandegrad, serieproduktion av modulära enheter, modifieringar som förbättrar värmeåtervinningen på motorsidan samt framtida drift med biobränslen skulle kunna förbättra både affärsnyttan och systemets klimatprestanda avsevärt.

Nyckelord: containeriserad koldioxidavskiljning; dieselgeneratoraggregat; CESAR-1; AMP/PZ; Aspen Plus; spillvärmeintegrering; absorptionsvärmetransformator; CO_2-kompression och förvätskning; teknoekonomisk analys.

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Declaration

Artificial intelligence was used in this thesis in a responsible and limited manner, with the sole purpose of supporting the clarity, quality, and efficiency of the writing process without compromising the originality or scientific integrity of the work. Specifically, ChatGPT 5.5 was used to assist with grammar checking, language refinement, interactive image generation and the identification of relevant peer-reviewed literature in an efficient manner. In this sense, it served as a practical search aid, complementary to conventional literature-search tools such as Google and academic databases.

No AI tool was used to develop the methodology, perform calculations, generate results, interpret findings, or draw conclusions. All technical content, analysis, discussions, and final judgments presented in this thesis were made by the author. The author therefore takes full responsibility for the accuracy, originality, and integrity of the work presented in this report.

Where AI-assisted support was used, the author carefully reviewed and verified the output before inclusion in the thesis. The use of AI was kept within ethical and academic boundaries, and it was applied only as a support tool for drafting and information retrieval.

List of Abbreviations

Abbreviation	Definition
ABPG	Amine Best Practices Group
AHP	Absorption Heat Pump
AHT	Absorption Heat Transformer
AMP	2-Amino-2-methyl-1-propanol
ASME	American Society of Mechanical Engineers
BE	Brake Efficiency
BEV	Battery-Electric Vehicle
BMC	Bare Module Cost
CAC	Charge-Air Cooler
CAPEX	Capital Expenditure
CCC	Carbon Capture Container
CCS	Carbon Capture and Storage
CCU	Carbon Capture Unit
CCUS	Carbon Capture, Utilisation and Storage
CE	Capture Efficiency
CEPCI	Chemical Engineering Plant Cost Index
CESAR-1	Non-proprietary aqueous solvent blend containing 3 M AMP and 1.5 M PZ
CHP	Compression Heat Pump
CO_2	Carbon dioxide
COP	Coefficient of Performance
CPO	Continuous Power Operation
CR	Capture Rate
CRF	Capital Recovery Factor
CW	Cooling Water
DLE_{Diesel}	Diesel Litre Equivalent
DYNAMIS	European Sixth Framework Programme DYNAMIS project
e-NRTL	Electrolyte Non-Random Two-Liquid model
ENCAP	Enhanced Capture of CO_2 project
ENRTL-RK	Electrolyte NRTL liquid-phase model combined with the Redlich–Kwong vapour-phase equation of state
EOR	Enhanced Oil Recovery
EU	European Union
FCI	Fixed Capital Investment
FEC	Final Equipment Cost
FG	Flue Gas
FOAK	First of its Kind
GHG	Greenhouse Gas
GUI	Graphic User Interface
H/D	Height-to-Diameter Ratio
HDV	Heavy-Duty Vehicle
HPC	Hot Potassium Carbonate
ICCT	International Council on Clean Transportation
IEA	International Energy Agency
IEAGHG	IEA Greenhouse Gas R&D Programme
IPCC	Intergovernmental Panel on Climate Change
ISO	International Organization for Standardization
KPI	Key Performance Indicator
KTH	KTH Royal Institute of Technology

Abbreviation	Definition
L/G	Liquid-to-Gas Ratio
LH	Long Haul
LHV	Lower Heating Value
LLM	Lean Liquid Mix
LM	Liquid Mix
MDEA	Methyldiethanolamine
MEA	Monoethanolamine
MPO	Maximum Power Operation
NCCC	National Carbon Capture Center
NETL	National Energy Technology Laboratory
NIST	National Institute of Standards and Technology
NMVOC	Non-Methane Volatile Organic Compound
NO_x	Nitrogen oxides
NPV	Net Present Value
NRTL	Non-Random Two-Liquid model
OPEX	Operational Expenditure
P&ID	Process and Instrumentation Diagram
PEC	Purchased Equipment Cost
PFD	Process Flow Diagram
PM	Particulate Matter
PPBV	Parts Per Billion by Volume
PPMV	Parts Per Million by Volume
PPW	Pressurised Process Water
PW	Process Water
PZ	Piperazine
R&D	Research and Development
R290	Propane refrigerant designation
REFPROP	Reference Fluid Thermodynamic and Transport Properties database
RK	Redlich–Kwong equation of state
RLM	Rich Liquid Mix
SINTEF	Norwegian research organisation and operator of the Tiller CO_2 pilot facility
SO_x	Sulfur oxides
SOP	Standard Operating Procedure
SRD	Specific Reboiler Duty
TCI	Total Capital Investment
TDC	Total Direct Cost
TEA	Techno-Economic Assessment
TPC	Total Production Cost
TR	Total Revenue
UKL	University of Kaiserslautern pilot plant
UNFCCC	United Nations Framework Convention on Climate Change
VLE	Vapour–Liquid Equilibrium
VOC	Volatile Organic Compound
WC	Working Capital
WCI	Working Capital Investment

List of Symbols

Symbol	Definition	Unit
A	Equipment size or capacity parameter	Equipment-specific
A^0	Capacity of the required equipment	Equipment-specific
A^1	Capacity of the reference equipment	Equipment-specific
C_p^0	Purchased cost of the required equipment	€ or $
C_p^1	Purchased cost of the reference equipment	€ or $
C_{2024}^{Euro}	Cost converted to 2024 euros	$€_{2024}$
$C_{\text{CO}_2,\text{capture}}$	Specific CO_2 capture cost	€ $\text{tonne}_{\text{CO}_2}^{-1}$
D	Vessel or column internal diameter	m
E	Activation energy	kJ mol^{-1}
F_{LV}	Packed-column flow parameter	–
F_p	Packing factor	m^{-1}
$F_{P,\text{vessel}}$	Vessel pressure factor	–
H	Packed height or vessel height	m
H/D	Height-to-diameter ratio	–
i	Annual discount rate	– or %
K_1–K_8	Empirical equipment-cost correlation constants	Correlation-specific
$K_{4,\text{gas}}$	Modified gas-load factor used in packed-column sizing	–
$(K_{\text{eq}})_j$	Equilibrium constant for reaction j	–
k_0	Arrhenius pre-exponential factor	Reaction-specific
L	Liquid flow rate	kg hr^{-1} or kmol hr^{-1}
L/G	Liquid-to-gas flow ratio	kg/kg or mol/mol
$L_{\min}$	Minimum liquid-mix flow rate	kg hr^{-1}
L_w^*	Liquid mass flux per unit column area	kg m^{-2} s^{-1}
m	Exponent in the nonlinear power cost correlation	–
$\dot{m}_{\text{CO}_2}$	CO_2 mass capture or desorption rate	kg s^{-1} or kg hr^{-1}
N	Number of components included in a kinetic expression	–
n	Temperature exponent, cost-scaling exponent, or project lifetime, depending on the equation	– or years
P	Absolute or gauge pressure, as specified	bar(a), bar(g), or Pa
p_{CO_2}	Partial pressure of CO_2	bar(a) or Pa
Q	Heat duty or thermal-energy flow	kW_{th}
Q_a	Heat released in the AHT absorber	kW_{th}
Q_c	Heat rejected in the AHT condenser	kW_{th}
Q_e	Heat supplied to the AHT evaporator	kW_{th}
Q_g	Heat supplied to the AHT generator	kW_{th}
Q_{CAC}	Recoverable heat from the charge-air cooler	kW_{th}
Q_{coolant}	Recoverable heat from the engine coolant	kW_{th}
Q_{exhaust}	Recoverable heat from the exhaust gas	kW_{th}
Q_{latent}	Latent-heat contribution to solvent regeneration	kW_{th}
Q_{reaction}	Reaction-enthalpy contribution to solvent regeneration	kW_{th}
Q_{reboiler}	Reboiler thermal duty	kW_{th}
$Q_{\text{regeneration}}$	Total solvent-regeneration duty	kW_{th}
Q_{sensible}	Sensible-heat contribution to solvent regeneration	kW_{th}
R	Universal gas constant	J mol^{-1} K^{-1}
r	Reaction rate in the kinetic model or discount rate in the NPV equation, depending on context	Model-specific or –
T	Absolute temperature	K

Symbol	Definition	Unit
T_0	Heat-sink temperature in the AHT COP expression	K
T_1	Low-grade waste-heat source temperature in the AHT COP expression	K
T_2	Upgraded useful-heat temperature in the AHT COP expression	K
t_{vessel}	Vessel wall thickness	m
V	Vapour or gas flow rate	kg hr^{-1} or kmol hr^{-1}
V_w^*	Gas mass flux per unit column area	kg m^{-2} s^{-1}
x_i	Liquid-phase mole fraction of component i	–
α_i	Reaction-order coefficient of component i	–
α_{lean}	Lean-solvent CO_2 loading	mol CO_2/mol amine
α_{rich}	Rich-solvent CO_2 loading	mol CO_2/mol amine
γ_i	Activity coefficient of component i	–
ΔG_j°	Standard Gibbs free-energy change of reaction j	J mol^{-1}
ΔP	Pressure drop	Pa or mbar(a)
ΔP_c	Pressure drop across the packed column section	Pa m^{-1} or mbar(a) m^{-1}
ΔT_{min}	Minimum heat-exchanger approach temperature	K or °C
ε	Packing void fraction	–
η_{is}	Compressor isentropic efficiency	– or %
λ	Engine excess-air ratio	–
μ_L	Liquid dynamic viscosity	Pa s
ρ_L	Liquid density	kg m^{-3}
ρ_V	Vapour or gas density	kg m^{-3}
CE	CO_2 capture efficiency	%
COP	Coefficient of performance	–
$\text{COP}_{\text{AHT,max}}$	Maximum theoretical AHT coefficient of performance	–
COP_{real}	Estimated real AHT coefficient of performance	–
SRD	Specific reboiler duty	MJ_{th} $\text{kg}_{\text{CO}_2}^{-1}$

Chemical-Species notation

Species	Definition
CO_2	Carbon dioxide
H_2O	Water
H_2S	Hydrogen sulfide
H_3O^+	Hydronium ion
OH^-	Hydroxide ion
HCO_3^-	Bicarbonate ion
CO_3^{2-}	Carbonate ion
N_2	Nitrogen
O_2	Oxygen
MEA	Monoethanolamine
$MEAH^+$	Protonated monoethanolamine
$MEACOO^-$	Monoethanolamine carbamate
AMP	2-Amino-2-methyl-1-propanol

Species	**Definition**
$AMPH^+$	Protonated AMP
$AMPCOO^-$	AMP carbamate
PZ	Piperazine
PZH^+	Monoprotonated piperazine
PZH_2^{2+}	Diprotonated piperazine
$PZCOO^-$	Piperazine monocarbamate
$PZ(COO^-)_2$	Piperazine dicarbamate
HPZCOO	Protonated piperazine carbamate

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1 Introduction

The need to mitigate greenhouse gas (GHG) emissions has been recognized at the highest levels of international forum and policy making. The ever-increasing demand for energy coupled with the need for a fast-paced transition to cleaner energy alternatives is driving countries and industries to adopt sustainable and low-carbon technologies. Achieving this balance is essential to address climate change while ensuring energy security and supporting sustainable economic growth. The United Nations Framework Convention on Climate Change (UNFCCC), which entered into force in 1994, marked a significant step in the global effort to address climate change by aiming to stabilize atmospheric greenhouse gas concentrations at levels that would prevent dangerous anthropogenic interference with the climate system. This goal was reinforced at the Copenhagen Climate Change Conference in 2009, which emphasized the importance of limiting the rise in global average temperature to below 2 °C relative to preindustrial levels (Dessens & Weyant, 2016). The Paris Agreement, adopted in 2015, reiterated the global commitment to climate action by setting the goal of limiting the increase in global average temperature to "well below" 2^oC while pursuing efforts to restrict it to 1.5 °C (Dessens & Weyant, 2016). Meeting these goals requires a rapid and large-scale transformation of the global energy system, which continues to be highly dependent on fossil fuels. Some Strategies to reduce GHG emissions could be rapid switch to cleaner resources, stricter surveillance of the implementation of zero-emission or negative-emission goals at sectoral and national levels, enhancing the energy efficiency of the process, creating stronger emphasis on Carbon Capture, Storage, and Utilization (CCUS) technologies (Kotagodahetti et al., 2021).

Although the renewable energy sector is growing rapidly, fossil based energy resources continues to suffice a vast portion of the global energy demand, with fossil Carbon Dioxide (CO_2) emissions reaching 33 Gigatonnes (Gt) in 2019 (Plazas et al., 2022; Kotagodahetti et al., 2021). The transport and energy sector is a major source of GHG emissions including fossil CO_2, driven by both continuous economic growth and heavy dependence on fossil fuels. As per statistics the transport sector alone is responsible for 15% of total and 23% of global energy related anthropogenic GHG emissions which is roughly 8.7 Gt CO_2-eq (Ferrer & Thomé, 2023).

Carbon dioxide capture and utilization offers dual benefit of mitigating GHG emissions and the possibility of generating value-added products from the captured CO_2. Research efforts focused on converting the captured CO_2 into value added fuels and chemicals, such as alcohols, carboxylic acids, organic carbonates, and hydrocarbons, are made extensively thereby paving the pathway towards a sustainable CO_2 economy (C. Xu et al., 2025). Despite significant growth and advancements in CO_2 utilization, a significant gap still exists in large-scale applications of utilization techniques for various sectors (Fu et al., 2022). The transport sector, specifically the heavy-duty road transport is one such promising area where there is much scope for fossil CO_2 capture. Although the recent increase in demand for battery electric vehicles (BEVs) indicates growing willingness among individuals and organizations to adopt cleaner transport technologies, electricity grid infrastructure in many regions worldwide, including parts of the European Union (EU), may lack the capacity and flexibility required to accommodate the additional peak loads associated with widespread vehicle charging (Gonzalez Venegas et al., 2021).

This work takes into consideration the issue of persistent fossil CO_2 emissions from the transportation sector due to the energy demand and supply gap that still exists specifically in the heavy-duty road transport sector and investigates a potential stopgap solution for cleaner fossil fuel-based charging of BEVs through the Carbon Capture route. The main motivation of this project is focused around performing a Techno-Economic assessment of moderate–temperature and ambient–pressure CO_2 capture from the Scania generator set (Genset) units using a Carbon Capture Container (CCC), which can be described as a stationary, point source and stand-alone CO_2 capture process. The future scope of the study also includes retrofitting the CCC Unit for future Scania Biofuel Gensets to to capture biogenic CO_2 and achieve a net-negative CO_2 footprint. By evaluating both process efficiency and economic viability, this study provides critical information on the potential role of industrial stage CO_2 Capture in achieving carbon neutrality and advancing a sustainable fossil-free economy.

1.1 Aim and Objectives

Regulation (EU) 2019/1242 established long-term CO_2-emission performance standards for new heavy-duty vehicles in the European Union, while its subsequent amendment through Regulation (EU) 2024/1610 strengthened the targets applicable from 2030 onwards until year 2040 (European Parliament and Council, 2024). These targets are intended to expedite the introduction of zero-emission heavy-duty vehicles, including Battery-Electric trucks, and therefore the subsequent expansion of high-power charging infrastructure must be completed to handle the peak charging loads of the Battery-Electric trucks. The high power demand associated with Heavy Duty Truck-charging facilities in many cases requires a new connection to the Medium-Voltage distribution grid or a substantial reinforcement of the existing grid connection. Grid upgrades can introduce considerable additional capital and operational expenses and project-development delays, making early coordination among fleet operators, charging-infrastructure providers, and distribution-system operators essential (Bernard et al., 2022). Recent European research similarly identifies the capacity of local grid infrastructure as one of the principal constraints on the construction of high-power Charging Stations and suggests that the completion of suitable grid infrastructure may require several years, depending on the location and the availability of nearby substations (Hacker et al., 2025). The other potential option could be a stationary Stand-Alone High-Power Charging Backup systems installed at the Charging Station. This second option typically involves the use of a Genset unit; in this case, a Scania diesel operated Genset is considered for charging Battery-Electric Trucks and other Heavy-Duty Vehicles. However, the use of fossil fuel based Gensets also leads to the release of fossil CO_2 to the atmosphere. To address this issue, Scania has begun investigating a retrofitted Carbon Capture Container (CCC) design capable of capturing approximately 90% of the tailpipe CO_2 emissions from the Genset system, thereby complying with regulations and emission standards fixed by the EU Regulations.

The main objective of this study is to develop and optimize a High Temperature Regenerative Amine absorption Carbon Capture process that can be integrated with a diesel Genset based Battery Electric Vehicle Charging Station to achieve the set target of 90% of tailpipe CO_2 capture. The Charging Station setup will primarily consist of one or two Scania diesel-based 13-Liter engine DC13-505A Genset(s). The Gensets would operate on one generator for one engine, DC13-505A, and with one generator in between for two engines, DC13-505A, depending on the average and peak charging loads for different charging stations at different locations throughout the EU. Additionally, the Genset unit(s) operates at either Continuous Power Operation (CPO) mode or Maximum Power Operation (MPO) mode. During the CPO mode, also known as the Prime Power mode, the output mechanical power from the Genset is 312 kW-mech (1500 rpm $\times$ 1986 Nm), which is roughly 70% of the rated power output. The CPO mode is the proposed usual operation point of the Gensets to meet the daily Charging Station power demands. During the MPO mode, also known as the Standby Power mode, the output mechanical power from the Genset is 437 kW-mech (1500 rpm $\times$ 2782 Nm), which is 100% of the rated power output. The MPO mode is also the proposed Carbon Capture Container (CCC) design point. The goal is to make this Carbon Capture Unit (CCU) design technically feasible, and economically viable. The combined Scania Genset–CCC system is proposed as a temporary, transitional, decentralized power-generation solution for charging Long Haul (LH) Heavy-Duty Vehicles (HDV) in locations where grid capacity or charging infrastructure is insufficient. When combined with a streamlined transportation facility for the captured CO_2 and injection into permanent geological storage with future utilization plans, this configuration has the potential to achieve net-zero fossil CO_2 emissions in the future.

1.1.1 Research Questions

To achieve the above-mentioned research objectives, the study is guided by the following five research questions, which are addressed throughout this thesis:

Research Question 1: Which solvent is most suitable for capturing CO_2 from Genset Exhaust Gas considering Capture Performance, Capture Efficiency (CE), Solvent Regeneration Energy Requirements and Operating Conditions?

Research Question 2: To what extent can a unified Absorber–Stripper Column design handle and efficiently perform across Flue Gas flowrate conditions from a Single Genset and Two Gensets engine setup (Hydrodynamic Study)?

Research Question 3: How does the change in Absorber height, Stripper height and Lean Solvent CO_2 loading affect the Specific Reboiler Heat Duty? What combination of Absorber height, Stripper height and Lean Solvent CO_2 loading values minimizes the Specific Reboiler Duty (SRD) while satisfying the specified 90% CO_2-capture target?

Research Question 4: Is there enough useful Waste Heat available from the Genset Exhaust and Genset Coolant that can be used for Solvent regeneration in the High Temperature Carbon Capture circuit? What is the most suitable heat-upgrading technology to upgrade the available Waste Heat to the desired temperature?

Research Question 5: What is the estimated Specific Cost of CO_2 captured for the Process-Heat-Integrated and Standalone System configurations, expressed per unit mass (tonne) of CO_2 captured?

1.2 Approach

During both Maximum Power and Continuous Power operation modes for the Scania Gensets, a significant amount of CO_2 is produced from diesel combustion. Typically, the CO_2 content in the outlet hot exhaust gas is approximately 14.60 wt% (wet basis) from Maximum Power Operation (MPO) mode and around 12.99 wt% (wet basis) from Continuous Power Operation (CPO) mode. The CO_2 emitted from the Scania genset operation is considered fossil CO_2 since it's a product of fossil diesel combustion with charge air from the environment. The extent to which the charging station can approach net-zero operation depends on the achieved Capture Efficiency, Capture Rate (CR), the energy requirements of the capture process for different environmental conditions, and emissions associated with the broader fuel use, CO_2 transport, and storage supply chains. The composition of the Genset Exhaust Gas under the provided operating conditions is presented in Tables 4 and 5.

Table 4: Genset Exhaust Gas mixture composition (wet basis) for Maximum Power Operation (MPO) mode

Component	**Conc. (mol ppm)**	**Conc. (mol%)**	**Weight (kg) (100 kmol gas basis)**	**Conc. (wt%)**	**Molecular weight (kg/kmol)**
CO_2	95500	9.55	420.20	14.60	44
O_2	77000	7.70	264.40	8.56	32
H_2O	106000	10.60	190.80	6.63	18
N_2	721500	72.15	2020.20	70.20	28
Total	**1000000**	**100.00**	**2877.60**	**100.00**	

Table 5: Genset Exhaust Gas mixture composition (wet basis) for Continuous Power Operation (CPO) mode

Component	**Conc. (mol ppm)**	**Conc. (mol%)**	**Weight (kg) (100 kmol gas basis)**	**Conc. (wt%)**	**Molecular weight (kg/kmol)**
CO_2	85100	8.51	374.44	12.99	44
O_2	91200	9.12	291.84	10.12	32
H_2O	90000	9.00	162.00	5.62	18
N_2	733700	73.37	2054.36	71.27	28
Total	**1000000**	**100.00**	**2882.64**	**100.00**	

Various established technologies are available for CO_2 capture and some have also been implemented in large, medium and scale industrial applications, primarily in fossil-fuel based combustion exhaust gas aftertreatment. Technologies such as absorption, adsorption, membrane separations, cryogenics, and solid looping are some of the widely known techniques for carbon capture. Despite many established methods, the application and adoption of CCUS in industrial generator set system setup has not been widely investigated in the past.

The exhaust gases exit the tailpipe of the Genset(s) at normal atmospheric pressures of around 1.00-1.05

bar(a) and temperatures of around 439 °C for maximum power mode and around 415 °C for continuous power mode. Considering the operational constraints of the Scania genset, particularly the need to minimize additional exhaust backpressure imposed by the downstream Carbon Capture Container (CCC), a conventional energy-efficient amine-based Girbotol regenerative absorption process with packed columns was identified as the most suitable CO_2-capture technology for the proposed application. A more detailed reasons behind selecting this approach will be discussed in the *Literature Review* and *Methodology* sections of the report. Aspen Plus V14 Chemical Engineering Modelling software is used to model and the simulate the CCUS flowsheet for the primary Girbotol Carbon Capture circuit.

The assessment of Waste-Heat availability, Waste-Heat upgrading to desired temperature range and transfer to the Reboiler unit for CO_2 Stripping, and the Refrigeration Energy requirements for compressed CO_2 gas was performed separately using analytical equations and calculations with the help of Microsoft Excel and MATLAB R2025a softwares.

1.3 Scope and Limitations

The primary objective of the thesis is to develop a conceptual process design and perform a techno-economic assessment of the complete CCC system. The Carbon Capture process shall be based on chemical absorption of CO_2 into a liquid solvent, followed by thermal regeneration of the solvent in a CO_2 stripping unit. An extensive literature review for the choice of suitable energy-eficient solvent for the main Carbon Capture circuit, suitable Heat Pump technology to upgrade the available waste heat, choice of Multistage Compressor, Intercoolers and Chiller (Refrigeration Equipment) for post capture CO_2 conditioning before CO_2 transportation via Tank Trailer trucks were also conducted since they are an important part of this study.

The Scope of the project also covered Equipment sizing and Thermal Designing of the Absorber and Stripper units, Heat Exchangers, Pumps, Compressors and Chiller required to process and condition the Exhaust Gas, captured CO_2, Rich and Lean Solvent streams, Scania Genset Engine Coolant (50wt% ethylene glycol-water mixture), Process Water, Cooling Water and other process streams to their specified operating temperatures, pressures and flow rates. Dimension the Carbon Capture system to fit into one or two standard 45 foot container(s), cost estimation of the CCC unit from Genset tailpipe to Chiller outlet and quantifying the additional energy penalty on the Genset side (if any) for the the CCC unit are integral part of the thesis scope. A comparison of CO_2 Capture Efficiency for Ambient Environmental Conditions and Hot Environmental Conditions are presented in *subsection 5.1.2* under the Results and Discussion section.

This study does not include a detailed analysis of transient Cold-Start operation condition, during which partial release of CO_2 to the atmosphere is assumed. Long-term operational effects, including equipment corrosion and fouling (J. Gao et al., 2012; Soosaiprakasam & Veawab, 2008), solvent degradation, and the accumulation of insoluble carbamate salts, are also excluded from the scope of the study (Svensson et al., 2014; Karlsson et al., 2019). Furthermore, detailed analysis of exhaust-gas backpressure caused by piping, fittings, flanges, and other process components which could not be modelled is not evaluated in detail. Low-temperature CO_2 stripping is investigated in a parallel Master's thesis and is therefore outside the scope of the present work.

2 Literature Review

In this section various criteria and technologies are investigated in detail before moving forward with designing the CCC System. Since the project investigates the technical and economic feasibility of a containerised point-source Carbon-Capture system installed downstream of a stationary generator set, a detailed Literature study is required. The Carbon Capture Container (CCC) is intended to separate carbon dioxide gas from the generator unit exhaust, condition the captured CO_2, and deliver it in liquefied form at high pressure and low temperature for subsequent transportation and storage. Hence, a detailed literature review on the Climate Goals and Regulations, different CO_2 Capture techniques and solvents, various available Heat Pump technologies to upgrade the available Waste Heat from the Genset Exhaust and Coolant streams and additional fuel-to-power equipment such as Diesel Boilers and Oxy-Cat Unit are studied and compared to select the most optimized units for the final process flowsheet and design.

2.1 Climate Goals and Regulations

Anthropogenic CO_2 emissions from fossil-fuel combustion have been recognised for several decades as a principal driver of changes in the Earth's radiative balance and climate system. This is evident from the fact that environmental impact of fossil CO_2 as a potent greenhouse gas has been studied and discussed in the research community for a long time now (Manabe & Wetherald, 1967; Hansen et al., 1981). The continued dependence of the industrial and transport sectors on fossil fuels and its extensive use therefore represents a major challenge for climate-change mitigation. If things continue on the same pathway, it is inevitable that global warming effects will become more and more intense over the next few decades. In response to this, the Paris Agreement was enacted in 2015 to limit the global average temperature rise (Dessens & Weyant, 2016). The European Green Deal also reiterates the Paris Agreement and sets interim and long-term targets aiming for CO_2 Neutrality by 2050 in the EU (European Parliament and Council, 2021). Carbon taxes, emissions-trading systems, and sector-specific emission-performance standards are consequently being implemented across an increasing number of government bodies and local authorities around the globe to incentivise emission reductions and support national and international climate commitments. This also helps to abate direct release of CO_2 into the atmosphere.

The road-freight sector, especially the Long-Haul Heavy Duty Trucks constitutes an important area for decarbonisation. Studies suggest that transportation sector is still one of the most important yet neglected sectors for anthropogenic GHG emission reductions. To put the official data into perspective and highlight the numbers, the Heavy Duty Trucks are responsible for 27.60% and 4.07% of total CO_2 emissions from the transport sector in China and the United States, respectively (P. Gao et al., 2026; Su et al., 2025). In the EU, the HDVs (including Heavy Duty Trucks, Medium Lorries, City Buses, Coaches, and Trailers) also contribute to around 6% of the total EU GHG emissions (European Parliament and Council, 2024). The EU regulation and its subsequent amendment have thus set very strict intermediate and final emission reduction goals in its CO_2 emission standard regulation. Regulation (EU) 2019/1242 established along with its recent amendment Regulation (EU) 2024/1610 set the bar on tailpipe CO_2-emissions for the Heavy Duty Vehicles sector within the continent. For the main regulated groups of Heavy-Duty Vehicles (HDV), manufacturers such as Scania are required to reduce the fleet-average specific tailpipe CO_2 emissions of newly registered vehicles by 45% from 2030, 65% from 2035, and 90% from 2040, relative to the applicable reference-period emissions (European Parliament and Council, 2024). In accordance with the Environmental Laws and Climate Lesiglations, Scania decided to investigate into the techno-economic feasibility of integrating downstream CO_2-Capture equipment with its Induistrial Genset systems as an alternative LH HDV fleet charging solution before making a corporate investment decision regarding the proposed business case.

2.2 CO_2 Capture Technologies

The selection of an appropriate CO_2-capture technology is strongly dependent on the characteristics and conditions of the emission source and the exhaust flue gas stream. Important parameters include the CO_2 concentration and partial pressure, exhaust-gas temperature and pressure, volumetric flow rate, moisture content, and temporal variations in engine load. These parameters influence the thermodynamic driving force for separation, the required size of the capture equipment, and the energy needed to achieve the specified Capture Rate and product purity. Consequently, no single capture technology is optimal for

all stationary emission sources, and the capture process must be selected and designed according to the operating conditions and outlet CO_2 quality requirements of the specific application (Intergovernmental Panel on Climate Change (IPCC), 2005; Boot-Handford et al., 2014).

The composition of the exhaust gas plays an important role in CCUS process design. In addition to CO_2, combustion exhaust mix may contain significant amount of other gases like oxygen, water vapour, nitrogen, and traces of nitrogen oxides, sulphur compounds, carbon monoxide, unburned hydrocarbons, particulate matter, and trace contaminants. Depending on their concentrations, these components may react with or compete for active sites in the capture medium, reduce the CO_2-capture capacity, accelerate solvent or sorbent degradation, and contribute to corrosion, fouling, foaming, or the formation of solid deposits. Exhaust-gas pretreatment may therefore be required to reduce the concentration of particulates, sulphur compounds, nitous oxides, and other contaminants before the flue gas enters the capture unit. The external environmental conditions prevailing in the CCUS site must also be considered when selecting construction materials for absorbers, adsorbers, heat exchangers, piping, valves, and auxiliary equipment. The availability and quality of energy at the installation site represent additional criteria for process selection. Capture systems may require electricity for pumps, blowers, compressors, vacuum equipment, control systems, refrigeration, and CO_2 conditioning, while thermally regenerated processes require continuous supply of high grade heat energy at desired temperature level. Both the temperature and quantity of the available heat is as important, since low-grade waste heat cannot necessarily be transferred directly to a high-temperature regeneration process. For the proposed Carbon Capture Container, the potential integration of heat recovered from the genset exhaust and coolant circuits is therefore an important factor in the selection and design of the capture process. Effective heat integration into CCC Circuit helps to reduce the external energy demand and improve both the environmental and economic performance of the system.

There are several ways to capture CO_2 depending on the origin, exhaust flue gas composition, CO_2 concentration in the flue gas and exhaust flue gas temperature and pressure to name a few. The presence of impurities and potential toxic and corrosive components in the exhaust gas can be a vital factor to consider while choosing the material of construction and technology for the CCC. Also the amount and type of energy available for operating the CCUS can influence one's decision for choice and design of the Capture process.

Stationary point-source CO_2 capture can be implemented using several separation mechanisms, including absorption, adsorption, membrane separation, cryogenic separation, and looping processes.

Absorption based Carbon Capture methods consists of absorption and stripping processes and has been extensively studied and applied for fossil CO_2 Capture. It is widely used in industrial scale in fertiliser, soda ash, natural gas processing plants, etc. transfers CO_2 from the gas phase into a liquid solvent. Physical absorption relies primarily on the dissolution of CO_2 in the liquid and is generally more favourable at elevated CO_2 partial pressures. Chemical absorption employs a reversible reaction between CO_2 and an absorbent and is therefore more suitable for dilute, near-atmospheric-pressure post-combustion gases. Chemical solvent, such as amine-based solvent (Monoethanolamine (MEA), Diethanolamine (DEA), Methyldiethanolamine (MDEA), 2-Amino-2-methyl-1-propanol (AMP), Piperazine (PZ) or a blend of two or more amines in aqueous solution), is utilized to capture CO_2 from the flue gas stream in absorption column. CO_2 reacts with the solvent and absorbed to the solvent phase, hence separated from the other components of the flue gas. In the stripping column, CO_2 is separated by means of high temperature (typically 120-140 °C). Solvents are regenerated and recycled back to the absorber, while releasing a concentrated CO_2 gas stream from the stripping section Condenser unit. This method is very effective in capturing the CO_2 from flue gas and is so far the most commercially matured method in industrial scale. Typically, 30% MEA or MDEA can capture approximately 85-95% CO_2 from flue gas with purity above 99.5% (Madejski et al., 2022). However, this method is energy intensive due to its high temperature stripping process. The solvent regeneration step alone consumes 60% of the energy requirement of CO_2 capture (Yu et al., 2012). Therefore, adjustment of operating conditions or the solvent properties must be done to minimize the energy requirement. Another set of drawbacks for this method is solvent degradation, corrosion, solvent losses, and the need for cooling and heat-recovery equipment(Rochelle, 2009; Intergovernmental Panel on Climate Change (IPCC), 2005).

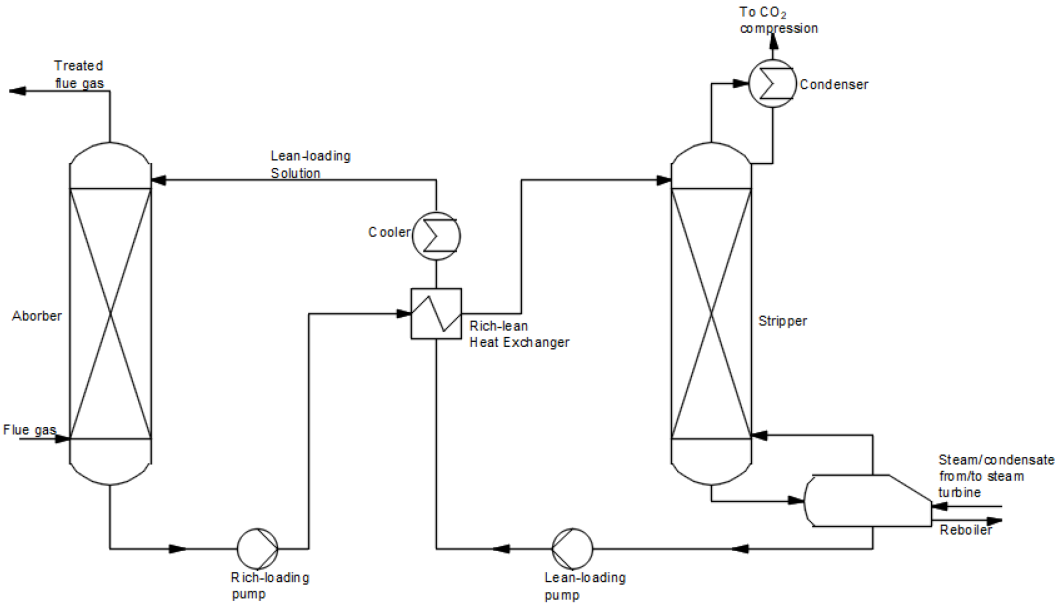


Figure 1: Schematic of Amine based CO_2 Absorption Process (Madejski et al., 2022)

Adsorption based Carbon Capture methods utilize the internal and external surfaces of a porous material to physically or chemically adsorb a compound. Potential adsorbents include activated carbon, zeolites, molecular sieves, metal–organic frameworks, and amine-functionalised solids (Barlow et al., 2025). In adsorption-based carbon capture, temperature-swing adsorption, pressure-swing adsorption, vacuum-swing adsorption, or combined pressure–temperature cycles are the normally deployed techniques. CO_2 is linked to the adsorbent through relatively weak physical Van der Waals interactions (physisorption) or covalent bonding (chemisorption), removing it from the influent gas mixture. When enough CO_2 has been adsorbed, the influent is closed off, normally switching to another unit, and the conditions in the saturated unit are then changed to release the CO_2. For physisorption processes the CO_2 is released by means of pressure swing, while for chemisorption processes it is normally released by a temperature swing. Their performance is governed by the equilibrium adsorption capacity, CO_2 selectivity, adsorption and desorption kinetics, cyclic working capacity, pressure drop, heat management, and resistance to moisture and exhaust-gas contaminants. Pressure swing adsorption has been implemented at an industrial scale in Port Arthur, Texas SMR-CCS, and several pilot and lab-scale projects for adsorption based Carbon Capture and Storage (CCS) are currently ongoing. Since CO_2 adsorbs to sites on a surface, the surface area available is the key aspect determining system capacity for adsorption processes. The higher the available surface area for a given adsorbent material, the more CO_2 that can be captured on it.

Membrane separation technique for CO_2 capture primarily uses differences in permeation rate across a semipermeable membrane to achieve separation (Barlow et al., 2025). A pressure difference between the feed side and permeate side of the membrane drives the separation. There are two main types of membrane technology for CCS: Gas separation membrane technology and gas absorption membrane technology. Both technologies utilizes membranes and benefit from membrane selectivity toward CO_2 and high CO_2 permeability. Membrane absorption processes, however, use the solubility of CO_2 in an absorbent on the permeate side rather than utilizing a pressure difference. This method can achieve a high removal rate of CO_2 while minimizing operational issues (Madejski et al., 2022). Membrane units will, however, require frequent maintenance as membrane fouling and pore clogging will eventually cause operational issues.

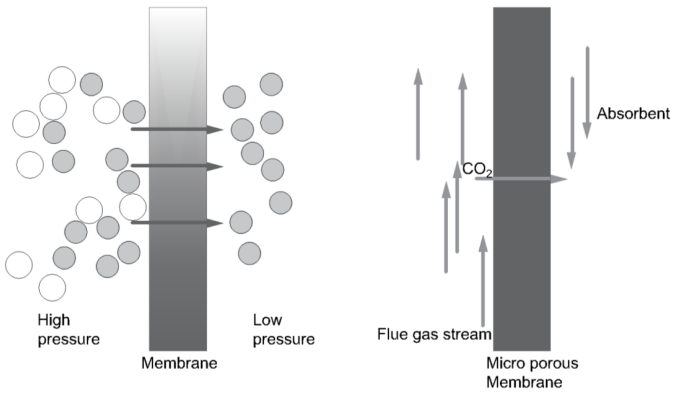


Figure 2: Schematic of CO_2 Membrane Separation Process (Madejski et al., 2022)

In the ***Cryogenics*** process, a series of steps, including compression, cooling, and condensation, is applied to achieve separation. The flue gas is exposed to a cold temperature environment (less than -56.6 °C). This process generates CO_2 in its liquid form, making it easier to separate it from other flue gas components. No solvent is required to generate a pure stream, which reduces the need for further processing. The product is high-purity CO_2 in liquid form. However, because of the large cooling demand needed to reach these extremely low temperatures, the technology can only be suitable for flue gases with a high CO_2 concentration. For flue gases with lower CO_2 concentrations, too much energy is wasted cooling the non-CO_2 components for this technology to be suitable temperature. Moreover, the compression also requires a lot of energy, making the process even more costly, especially if CO_2 concentrations are low (Fu et al., 2022).

Solid looping technology utilizes calcium oxide (CaO) to react with CO_2. The reaction consumes both reactants to produce calcium carbonate ($CaCO_3$) in a so called carbonation stage. The next stage which is calcination, $CaCO_3$ is heated, releasing the high purity CO_2 (up to 95%). The reactors used for these processes are coupled with a solid-material transport and separation unit as calcium oxide and calcium carbonate are solid materials (Madejski et al., 2022).

2.2.1 Solvent Absorption Based CO_2 Capture (Amine-Based Technologies)

The Girbotol process is around a century-old traditional gas sweetening process prevalent in the industrial sector dealing with foul-smelling corrosive and acidic gases such as Carbon Dioxide (CO_2), Hydrogen Sulphide (H_2S) and Mercaptans (R-SH). The process itself was developed by Robert Roger Bottoms at the Girdler Corporation during the late 1920s and was formally described in a United States of America (USA) patent document which was granted in 1930 (Bottoms, 1930). The Girbotol process uses basic compounds such as amines and selectively separates acidic gases via a regenerative chemical absorption mechanism from a mixture of acidic gas and other inerts. The acid gases were absorbed at relatively low temperatures, while subsequent heating of the solvent in a separate column at elevated temperatures and pressures released the absorbed gases and enabled solvent regeneration and recirculation.

Alternative chemical absorption methods include aqueous sodium hydroxide (NaOH) and potassium hydroxide (KOH), which react quickly with CO_2 to form carbonate and bicarbonate salts. Their main advantages include high absorption rates, effective CO_2 removal at low CO_2 partial pressures, low solvent volatility, and the absence of amine-related oxidative degradation. However, the resulting carbonate solutions cannot generally be regenerated through conventional low or medium-temperature stripping operations. Solvent recovery instead requires additional causticization, calcination, or electrochemical processing, thereby increasing the energy demand, equipment requirements, and operating complexity of a standalone CO_2 capture system (Keith et al., 2018; Medina-Martos et al., 2022).

Hot Potassium Carbonate (HPC) solutions offer relatively low toxicity, high chemical stability, low degradation rates, and lower regeneration energy requirements than conventional amines. Nevertheless, their relatively slow CO_2 absorption kinetics and requirement for elevated CO_2 partial pressures limit

with minimum p_{CO2} of 0.7 bar(a) which is not achievable in near-atmospheric standalone Genset Exhaust Gas outlet (Borhani et al., 2015; National Energy Technology Laboratory (NETL), 2024). Consequently, regenerative alkanolamine absorption was selected because it is a mature and robust technology capable of treating low-pressure flue-gas streams without prior compression (Rochelle, 2009).

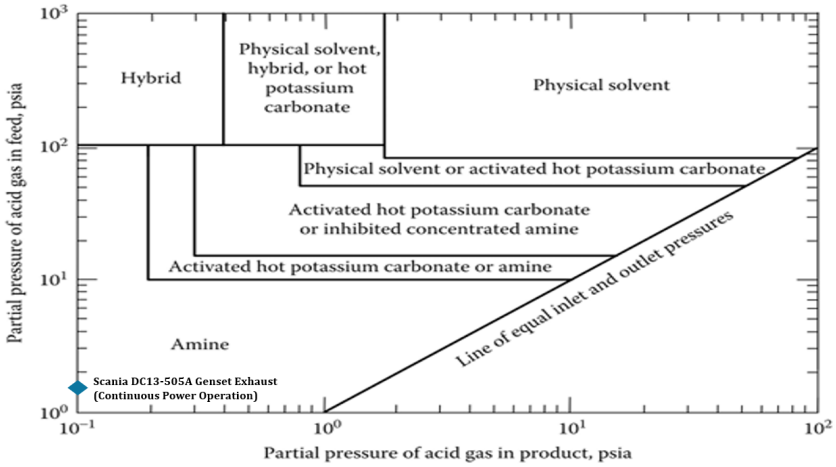


Figure 3: Different Carbon Capture Techniques based on CO_2 Partial Pressure (National Energy Technology Laboratory (NETL), 2024)

Monoethanolamine (MEA) is a primary alkanolamine containing one reactive amino group and is widely regarded as the benchmark solvent for post-combustion CO_2 capture. Its low molecular weight (61.08 g mol^{-1}), rapid reaction kinetics, and strong affinity for CO_2 enable high removal efficiencies from low-pressure and dilute flue-gas streams (Rochelle, 2009). MEA is typically applied as a 30 wt% aqueous solution and forms carbamate with a theoretical loading of approximately 0.5 mol CO_2/mol MEA. Pilot-scale studies using 30 wt% MEA have reported Specific Reboiler Duties of approximately 3.6–3.8 MJ $kg^{-1}CO_2$ at 90% capture rate (Moser et al., 2022). Its principal disadvantages include relatively high regeneration-energy demand, oxidative and thermal degradation, solvent loss, and equipment corrosion. These effects generally become more severe with increasing MEA concentration, CO_2 loading, oxygen exposure, and regeneration temperature (Davis & Rochelle, 2009; Rochelle, 2009). The MEA-induced corrosion issues starts to occur when the MEA concentration exceeds 20 wt% (N. Wang et al., 2023).

Methyldiethanolamine (MDEA) is a tertiary alkanolamine with a molecular weight of 119.16 g mol^{-1}. Unlike primary and secondary amines, MDEA does not contain an active hydrogen attached to the nitrogen atom and therefore does not directly form stable carbamate species. Instead, it promotes CO_2 hydration and bicarbonate formation, allowing a theoretical loading approaching 1 mol CO_2 per mol MDEA. Its major advantages include a relatively low heat of reaction, low regeneration-energy requirement, high thermal stability, lower toxicity, volatility and lower corrosiveness compared with MEA. However, the reaction between MDEA and CO_2 is comparatively slow, particularly at the low CO_2 partial pressures which is common in post-combustion flue gas from Gensets. Consequently, MDEA generally requires larger Absorber dimensions, higher solvent circulation rates or the addition of a kinetic promoter such as piperazine (PZ). MDEA is therefore more widely applied to high-pressure natural-gas sweetening (also because of its strong affinity towards H_2S) than to standalone post-combustion CO_2 capture (Rochelle, 2009; Closmann et al., 2009).

2-Amino-2-methyl-1-propanol (AMP) is a sterically hindered primary amine with a molecular weight of 89.14 g mol^{-1}. Steric hindrance destabilizes the carbamate formed during CO_2 absorption, favouring its conversion to bicarbonate and enabling a theoretical loading approaching 1 mol CO_2 per mol AMP. Consequently, AMP generally offers a higher cyclic capacity and lower heat of regeneration than conventional MEA (Sartori & Savage, 1983). However, steric hindrance also reduces its intrinsic reaction rate with CO_2, resulting in slower absorption kinetics and potentially larger absorber requirements when AMP is used independently. AMP additionally has a higher volatility than MEA, increasing the importance of an effective water-wash section and emission control. Although AMP demonstrates favourable regeneration characteristics and improved resistance to some degradation pathways, ther-

mal and oxidative degradation cannot be completely ignored. AMP is therefore commonly combined with a fast-reacting promoter, particularly PZ, to balance absorption kinetics, solvent capacity and regeneration-energy demand (Brúder et al., 2011; Morlando, Buvik, et al., 2024).

Piperazine (PZ) is a cyclic diamine with a molecular weight of 86.14 g mol^{-1} and two reactive nitrogen sites. Concentrated aqueous PZ exhibits very rapid CO_2 absorption kinetics; its absorption rate has been reported to be more than twice that of 7 M MEA under comparable conditions (Freeman et al., 2010). PZ also provides high cyclic capacity, good thermal stability and the possibility of regeneration at higher pressure, which may reduce downstream CO_2 compression requirements. However, concentrated PZ solutions have a restricted operating range because solid PZ or carbamate species may precipitate at low temperatures or high CO_2 loadings. Its relatively high viscosity may also increase pumping requirements and reduce mass-transfer performance. Furthermore, oxidative degradation and the potential formation of nitrosamine-related compounds require careful solvent management. For these reasons, PZ is frequently employed as a kinetic promoter in blended solvents rather than as the sole absorbent (Freeman et al., 2010; Rochelle, 2009).

The ***MDEA–PZ blend*** combines the high CO_2 capacity and relatively low regeneration energy of MDEA with the rapid reaction kinetics of PZ. No single composition is universally regarded as standard; reported aqueous formulations commonly contain approximately 20–35 wt% MDEA, 5–15 wt% PZ and the rest is water. For example, a blend containing 35 wt% MDEA, 15 wt% PZ and 50 wt% H_2O produced a simulated Specific Reboiler Duty of approximately 3.24 MJ $kg^{-1}CO_2$ under optimized conditions (X. Liu et al., 2020). A loaded 7 M MDEA/2 M PZ solution also demonstrated low measured degradation rates at 120,°C (Closmann et al., 2009). Nevertheless, the solvent remains limited by slower overall kinetics than concentrated PZ or MEA, particularly when the PZ fraction is low. Increasing the PZ concentration improves absorption but may increase viscosity, solvent cost and precipitation risk. The optimum blend composition is therefore strongly dependent on flue-gas composition, absorber dimensions, operating temperature and regeneration conditions.

CESAR-1 is a non-proprietary aqueous blend containing 3 M AMP and 1.5 M PZ, corresponding to approximately 27 wt% AMP, 13 wt% PZ and 60 wt% H_2O. The solvent combines the high cyclic capacity and relatively low regeneration energy of AMP with the rapid CO_2 absorption kinetics of PZ (Brúder et al., 2011; Morlando, Buvik, et al., 2024). Pilot-plant and modelling studies have demonstrated regeneration-energy requirements below approximately 3 MJ $kg^{-1}CO_2$ under optimized process conditions, compared with approximately 3.5–4.0 MJ $kg^{-1}CO_2$ for conventional MEA systems (Morlando, Y. Zhang, et al., 2025). CESAR-1 has also demonstrated lower degradation rates and improved energy performance relative to MEA, making it a promising second-generation benchmark solvent (Morlando, Buvik, et al., 2024). Its disadvantages include greater AMP volatility, potential aerosol and amine emissions, PZ-related precipitation constraints and incomplete long-term data concerning degradation products and environmental impacts. Nevertheless, its favourable balance of kinetics, capacity, stability and regeneration energy makes CESAR-1 particularly suitable for the present CCC application.

Following Genset engine Coolant and Exhaust Gas waste-heat extraction and upgrading, the thermal energy available for solvent regeneration was estimated at 3.4—3.6 MJ $kg^{-1}CO_2$ depending on the external environmental conditions and mode of Genset operation (CPO or MPO mode). CESAR-1 solvent was therefore selected as the working absorbent in the CCC circuit because its regeneration duty is compatible with this energy range. In addition, the CCC unit includes an Absorber gas water wash tower and CO_2-Lean Exhaust Gas cooler to recover entrained amine vapours, thereby reducing solvent losses and atmospheric amine emissions. These factors make CESAR-1 the most suitable solvent for the proposed system under the assessed operating conditions.

2.3 Waste Heat Upgradation and Heat Pumps

Industrial processes frequently reject large quantities of low-grade waste heat at temperatures below those required for direct energy-intensive process use. Heat pumps can upgrade this available low temperature thermal energy to a higher temperature level, thereby improving the energy efficiency and overall heat integration of the process, ultimately reducing fuel consumption and associated anthropogenic CO_2 emissions and GHG levels. The primary classifications of industrial heat pumps generally employed for large-scale thermal upgradation include Vapor Compression Heat Pumps (Closed Loop), Mechanical Vapor Recompression systems (Open Loop), Type I Absorption, Type II Absorption or Absorption Heat

Transformers, Adsorption, Chemical, Thermoacoustic, Cascade or Hybrid Heat Pumps, and Transcritical CO_2 Heat Pumps (Cudok et al., 2021). Industrial Vapour Compression Heat Pumps commonly achieve coefficients of performance for heating (COP_h) of approximately 2–6, although performance decreases as the required temperature lift increases (Schlosser et al., 2020).

In CCUS applications, heat pumps can recover heat from flue-gas cooling, stripper condensers, lean-solvent cooling and CO_2 compression intercoolers for solvent regeneration. For example, integration of a steam-generating heat pump into an amine-based capture process for steel-industry flue gas was estimated to reduce capture-process energy consumption by approximately 50% (Wilk et al., 2024). High-temperature heat pumps have also been investigated for MEA regeneration in cement plants using waste heat from the capture and production processes (Cremona et al., 2025). Outside CCUS, heat pumps are applied in food drying, pulp and paper production, chemical processing, district heating, wastewater treatment and Combined Heat and Power plants.

2.3.1 Compression Heat Pumps

A Compression Heat Pump (CHP) upgrades low-temperature heat through a mechanically driven vapour-compression cycle consisting primarily of an evaporator, compressor, condenser, and expansion valve. Industrial systems commonly achieve heating coefficients of performance, COP_h, of approximately 2–6, depending strongly on the required temperature lift; one large-scale installation achieved a COP of 6.67 at a heating capacity of approximately 9.5 MW (Schlosser et al., 2020; Hu et al., 2022). Compression Heat Pumps are commercially mature, compact, controllable, and capable of recovering low-grade heat from coolant, water, or low-temperature exhaust streams. However, the compressor constitutes a major rotational component and requires electrical power input for operation (cannot operate on waste heat), regular maintenance, lubrication, vibration control, and potentially expensive high-temperature refrigerants and materials. The COP also decreases as the temperature lift and compressor pressure ratio increase. In a Genset-based CHP system, the additional electrical consumption directly reduces the net electrical output available for vehicle charging, making this technology less favourable when abundant process waste heat but limited surplus electricity is available.

2.3.2 Absorption Heat Pumps

Absorption Heat Pumps are thermally driven heat-upgrading units which uses a Generator–Absorber pair to perform the function otherwise provided by the high-speed Mechanical Compressor unit. In a typical Absorption Heat Pump, a working fluid (eg. steam) is absorbed and desorbed within a liquid absorbent, replacing the mechanically driven compressor used in conventional vapour-compression cycles. As a result, their electrical demand is generally limited to small solution and circulation pumps, while the main energy input is supplied as waste heat. Common working pairs include H_2O–LiBr and NH_3–H_2O. Type I Absorption Heat Pumps typically deliver useful heat at an intermediate temperature with thermal COP values of approximately 1.4–1.8. Type II Absorption Heat Pumps, also known as Absorption Heat Transformers, upgrade part of a waste-heat stream to a higher temperature and generally achieve COP values of approximately 0.4–0.5 (Parham et al., 2014; Cudok et al., 2021). Here in Absorption Heat Pump systems, the COP_h is usually defined as the ratio of useful heat output at the Absorber and Condenser unit to the total driving heat input at the Generator unit for Type I. For Type II Absorption Heat Pumps, COP_h is defined as the ratio of useful heat output at the Absorber unit to the total driving heat input at the Generator and Evaporator unit.

2.3.2.1 Absorption Heat Pumps Type I

A Type I Absorption Heat Pump is a thermally driven system that uses high-temperature driving heat to recover additional heat from a lower-temperature source and deliver a larger quantity of heat at an intermediate temperature. Single-effect systems typically exhibit a thermal COP of approximately 1.4–1.8, meaning that the useful heat output exceeds the supplied driving heat because low-grade heat is simultaneously recovered (Z. Xu et al., 2022). Compared with Compression Heat Pumps, Type I systems require considerably less electricity because solution pumps replace the large mechanical compressor and contain fewer high-speed rotational components. They can therefore provide reliable operation with reduced noise and vibration. Nevertheless, a Type I Absorption Heat Pump requires a high quantity of separate high-temperature driving heat source which results in a lower COP when compared to Compression Heat Pump. Absorption Heat Pump generally delivers heat below the driving-source temperature. Hence, it

is consequently unsuitable when the principal objective is to upgrade a single available waste heat source to a temperature exceeding that of the source. Additional disadvantages include large heat-transfer areas, slower start-up, sensitivity to crystallization or corrosion, and reduced performance during transient Genset operation.

2.3.2.2 Absorption Heat Pumps Type II

A Type II Absorption Heat Pump, also known as an Absorption Heat Transformer, upgrades part of an intermediate-temperature waste-heat stream to a temperature above that of the original heat source. A conventional single-stage unit typically upgrades approximately 40–50% of the supplied waste heat, corresponding to a thermal COP of approximately 0.4–0.5, while rejecting the remaining heat at a lower temperature (Cudok et al., 2021; Parham et al., 2014). Its principal advantage is that the cycle is driven predominantly by waste heat and requires only a small quantity of electricity for solution and circulation pumps. The absence of a large refrigerant compressor reduces electricity consumption, vibration, noise, and the number of high-speed rotational components. This is particularly advantageous in a Genset-based CHP system containing substantial Exhaust Flue Gas and Engine Coolant waste heat, as more generated electricity remains available for vehicle charging. A Type II system can therefore upgrade otherwise underutilized heat for solvent regeneration. Its disadvantages include a COP below unity, rejection of approximately half of the input heat, larger equipment size, limited commercial deployment, working-pair constraints, corrosion or crystallization risks, and reduced flexibility under rapidly varying loads.

For the Scania Genset CCC unit design, the Type II Absorption Heat Pump or Absorption Heat Transformer was identified as the most suitable waste heat-upgrading technique. The basis for this selection is discussed in greater detail in the *Methodology* section under *Heat Pump Selection and Waste Heat Integration* (Section 3.3.2).

2.4 Diesel Boilers and Oxy-Cat Units

A ***Diesel Boiler*** is a combustion unit which produces hot water or steam through the direct combustion of input diesel fuel. Fossil diesel fuel typically has a Lower Heating Value (LHV) of approximately 42.80 MJ kg^{-1}. Modern oil-fired boilers typically achieve thermal efficiencies of approximately 85–95%, providing reliable, controllable heat independently of Genset operating conditions (Heywood, 2018; Council of Industrial Boiler Owners, 1997). However, the system requires additional fossil fuel, increases fossil CO_2 and NO_x emissions, and introduces fuel-storage, burner-maintenance, ventilation, and fire-safety requirements. It is therefore more suitable as an auxiliary or backup heat source when the primary Genset coolant and Exhaust Gas upgraded heat energy supply is not enough to meet the CCC unit thermal energy demand.

An ***Oxygen Catalytic or Oxy-Cat Unit*** generates heat by injecting a controlled quantity of diesel or another hydrocarbon into the oxygen-containing Genset Exhaust Flue Gas upstream of a Diesel Oxidation Catalyst. The fuel is exothermically combusted to produce CO_2 and H_2O and a lot of thermal energy, increasing the exhaust-gas temperature without requiring a separate combustion-air system. Diesel-hydrocarbon light-off may occur at approximately 180–250,°C, depending on catalyst formulation and operating conditions (Luo et al., 2019; Y. Wang et al., 2024). Advantages include compactness, rapid heat generation, and utilization of residual exhaust O_2. Limitations include additional fuel consumption, catalyst aging, hydrocarbon slip, additional exhaust gas treatment steps, backpressure, and possible thermal-control difficulties.

A literature review of these systems was conducted to identify a suitable option for on-site thermal-energy supply to the reboiler of the standalone Genset CCC unit, particularly if the process heat demand increases or an alternative solvent with a higher Specific Reboiler Duty (SRD), such as MEA, is selected. However, both systems would require dedicated exhaust after-treatment and regular maintenance, thereby increasing operational complexity, equipment requirements, and overall process costs.

2.5 CO_2 Post-Capture Processing

Post-capture processing is an essential stage of the CCUS value chain and prepares the separated CO_2 stream for transport, permanent storage, or as a raw material for industrial utilization. The process

generally includes cooling and water separation, purification, dehydration, multistage compression, and, where required, liquefaction using chillers. For dense-phase pipeline transport and geological injection, CO_2 is commonly compressed to pressures above approximately 80–110 bar, depending on the transport network, impurity composition, ambient conditions, and required injection pressure. For ship transport, the CO_2 is generally liquefied, with pre-defined transport conditions commonly located near 7–15 bar and sub-zero temperatures (Wetenhall et al., 2014; International Organization for Standardization, 2024).

The conditioned CO_2 must satisfy the composition and operating specifications imposed by the transport and storage infrastructure regulations. Excess water must be removed to prevent hydrate formation, two-phase flow, carbonic acid formation and corrosion, particularly in the presence of impurities such as SO_x, NO_x, O_2, and H_2S. Recommended water-content limits for carbon-steel pipelines are commonly in the range of approximately 50–500 ppmv (volume basis), although the accepted specification remains project-specific (National Energy Technology Laboratory, 2019). Non-condensable gases also affect phase behaviour, compression duty, liquefaction temperature, pipeline capacity, and storage efficiency. Consequently, CO_2 purity requirements are not universal and must be agreed between the capture facility and the transport, storage, or utilization operator (Wetenhall et al., 2014; International Organization for Standardization, 2024).

The ***CO_2 Quality Recommendations*** are based on several considerations. Pipeline transport is mainly concerned about the corrosion and hydrate formation, as both will damage the pipes leading to repair costs and potential safety hazards. The presence of other gases in the final stream also affect the compression process, mainly by increasing power demand. It is recommended to have H_2S and CO less than 200 ppm and 2000 ppm respectively due to health and safety conditions in case of a leakage (Aspelund et al., 2010; De Visser et al., 2008). If the goal is to use it for sustainable urea production, the CO_2 stream should be very clean and free of water, as water shifts the reaction equilibrium backwards (reducing yield) and can lead to corrosion of the equipment (Santos Mesquita et al., 2025). Therefore, clean CO_2 is essential because urea synthesis involves high-pressure carbamate formation and is highly sensitive to contaminants that interfere with both reaction equilibrium and equipment durability. Figure 4 below show a simplify flow diagram of the CO_2 capture and purification process.

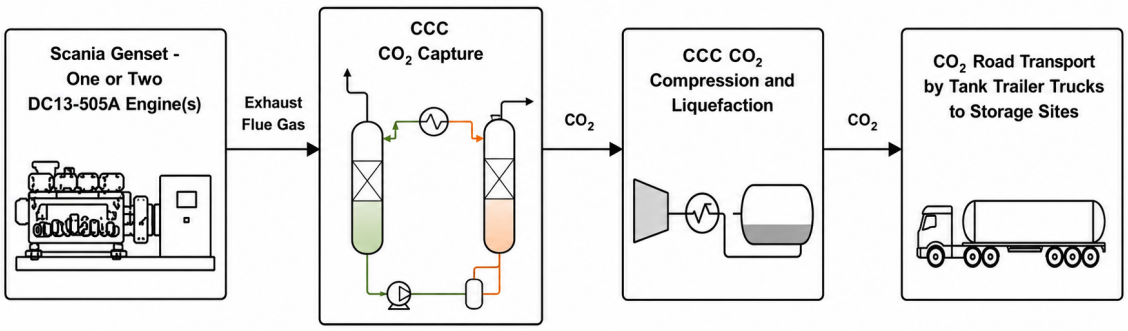


Figure 4: Simplified Flow Diagram of CO_2 Capture and Post-Capture Conditioning Process

2.5.1 CO_2 Quality Recommendations

Table 6 below represents the CO_2 quality recommendations according to the DYNAMIS (i.e., the European 6th Framework Programme DYNAMIS) and the European project Enhanced Capture of CO_2 (ENCAP) (De Visser et al., 2008).

Table 6: DYNAMIS CO_2 Quality Recommendation

Component	Concentration	Limitation
H_2O	500 ppm	Design and operational considerations
H_2S	200 ppm	Health and safety considerations
CO	2000 ppm	Health and safety considerations
CH_4	Aquifer < 4 vol.%, EOR < 2 vol.%	As proposed in ENCAP project
N_2	< 4 vol.% (all non-condensable gases)	As proposed in ENCAP project
Ar	< 4 vol.% (all non-condensable gases)	As proposed in ENCAP project
H_2	< 4 vol.% (all non-condensable gases)	Further reduction of H_2 recommended due to energy content
CO_2	> 95.5 vol.%	Balanced with other compounds in CO_2

2.5.2 CO_2 Compression

CO_2 Compression using a Multistage Compressor system is a Standard Operating Procedure (SOP) followed by almost every CCUS plant as a post-capture CO_2 conditioning step. Following solvent regeneration and preliminary cooling via condenser and CO_2 stream dehydration and purification, the CO_2-rich product from an amine- or CESAR-1-based capture process must be compressed to the pressure required for transport, liquefaction, or geological injection for storage. Multistage compression with intercooling is commonly employed to limit discharge temperatures and reduce compressor work. For dense-phase pipeline transport, the final pressure is typically approximately 100–150 bar(a), while the captured gas may leave the stripper condenser near 1.5–2 bar(a) (Modekurti et al., 2017; National Energy Technology Laboratory, 2026). The main advantages of compression are reduced CO_2 volume, continuous pipeline transport, and compatibility with high-pressure storage injection. Interstage cooling and knockout vessels can additionally remove condensed water. Intercoolers are placed between successive stages of a Multistage CO_2 Compression system to reduce the gas temperature before it enters the next compressor stage. This improves material and lubricant safety, limits thermal stresses, and can extend compressor service life. Intercooling also reduces the inlet specific volume of the CO_2, thereby lowering the compression work required in the subsequent stage and improving the overall efficiency of the Compression Train. However, compression represents a substantial auxiliary electrical load, commonly requiring approximately 80–120 $\mathrm{kWh}_{elec}\, tonne_{\mathrm{CO_2}}^{-1}$, depending on inlet pressure, purity, efficiency, and final pressure. Crucial issues include multiple rotating stages, vibration, maintenance requirements, cooling-water demand, and sensitivity to water and non-condensable impurities (Wetenhall et al., 2014).

2.5.3 CO_2 Liquefaction

CO_2 liquefaction converts the compressed gas into a dense liquid suitable for temporary storage and transport by ship, road tanker, or rail. For ship-based Carbon Capture and Storage (CCS), representative conditions include approximately 6.5–7 bar(a) at about -50 to $-52\,^{\circ}\mathrm{C}$, or intermediate-pressure conditions near 15 bar(a) and -28 to $-30\,^{\circ}\mathrm{C}$ (Seo et al., 2015; Wetenhall et al., 2014). Liquefaction substantially reduces the storage volume and offers greater transport flexibility than dedicated pipelines, particularly for dispersed capture sites, moderate flow rates, offshore storage, or routes with variable destinations. Nevertheless, refrigeration and compression impose a considerable electricity demand, often approximately 90–140 $\mathrm{kWh}_{elec}\, tonne_{\mathrm{CO_2}}^{-1}$, depending on feed pressure, cooling technology, impurities, and selected transport condition (Øi et al., 2016). Water and non-condensable gases must be removed because they can increase refrigeration duty, alter phase behaviour, reduce CO_2 recovery, and promote freezing or hydrate formation. Liquefaction also requires insulated pressure vessels, boil-off management, intermediate storage, and more complex loading infrastructure.

CO_2 cooling and liquefaction systems can be broadly classified as open-loop, closed-loop, or hybrid refrigeration processes. In an open-loop or self-refrigerated process, the captured CO_2 itself acts as the refrigerant. The stream is compressed, cooled, and expanded through a Joule–Thomson valve or expansion device, producing partial liquefaction without requiring a separate refrigerant circuit. This configuration is relatively simple and compact but may suffer from lower CO_2 recovery, flash-gas recompression requirements, and sensitivity to feed impurities (Engel & Kather, 2017; Alabdulkarem et al., 2012). In a closed-loop process, an external refrigerant circulates independently of the CO_2 stream through the Chiller unit's compressor, condenser, expansion valve, and evaporator. Common refrigerants include ammonia (R717), propane (R290), CO_2 (R744), and single- or mixed-refrigerant systems. Closed-loop systems provide greater operational control and CO_2 recovery but require additional rotating

equipment, refrigerant inventory, and heat exchangers. Hybrid systems combine external refrigeration with CO_2 self-expansion and can reduce energy consumption while maintaining high recovery (Gong et al., 2022; López et al., 2024).

For this project, an open-loop CO_2 liquefaction design scheme was studied and rejected because the available cooling medium limited the Chiller unit's condenser outlet to approximately 30 °C due to the available cooling medium and the performance of the refrigeration cycle. This temperature is close to the CO_2 critical temperature of 31 °C, preventing CO_2 condensation at 20 bar(a) as CO_2 is still a vapour at these conditions. Consequently, a conventional closed-loop R290 (propane) vapour-compression cycle was selected to achieve -25 °C and 20 bar(a), which is 5 °C lower than CO_2 road transport conditions of -20 °C and 20 bar(a) to ensure a stable liquid product for tank-trailer transport (Span & Wagner, 1996; Chen & Morosuk, 2021).

2.6 CO_2 Transport and Sequestration

CO_2 transportation is a critical step of the CCUS value chain because capture facilities and suitable utilization or geological storage sites are often geographically separated. The principal transport options are high-pressure Pipeline Transport, Road Transport using Pressurized or Refrigerated Tank Trailers, and Ship Transport of liquefied CO_2. The preferred mode depends on the transported volume, distance, infrastructure availability, delivery flexibility, and required CO_2 storage conditions.

In the case of ***Pipeline Transport***, CO_2 conditions are chosen as high pressure in the range of 100–150 bar(a), a pressure well above the critical point at ambient temperatures. At ambient temperatures at these pressures, CO_2 is in a supercritical phase, or dense-phase liquid, but still compressible. In contrast, ***Ship Transport*** requires CO_2 to be carried in a liquid state at much lower pressures, which is achieved by cooling it to approximately -50 °C. This approach reduces the pressure requirement and enables safe and economical large-scale marine transport (Anheden et al., 2005).

For ***Road Transport***, CO_2 is commonly processed as a refrigerated liquid at approximately -20 °C and 20 bar(a) conditions already used by commercial CO_2 tankers serving industries such as food and beverage production (UK Research and Innovation, 2024). The liquid is transported in insulated, pressure-rated trailer tanks equipped with transfer pumps, pressure-relief devices, valves, and dedicated loading and unloading connections (European Industrial Gases Association, 2021). A typical semi-trailer can transport approximately 20–27 tonne of liquid CO_2 per journey. Road transport offers considerable routing flexibility and requires less fixed infrastructure than pipelines, making it suitable for small or geographically dispersed capture facilities and shorter transport distances. However, its limitations include relatively low payload per trip, additional fuel consumption and driver and vehicle requirements, which ultimately accounts for increased transport costs and emissions as the CO_2 flow rate or delivery distance increases.

For this study, the plan is to transport the captured and conditioned CO_2 as a refrigerated liquid at approximately -20 °C and 20 bar(a) conditions by Tank Trailer Trucks to the nearest shared Regional CO_2 collection hub. At the receiving facility, the CO_2 may be temporarily stored, recompressed if necessary, and subsequently transported via Pipeline Transport to be injected into a suitable geological formation (North Sea for EU's case) for permanent sequestration.

2.7 Carbon Capture Container Process Description

Figure 5 presents a schematic Process Flow Diagram (PFD) of the Carbon Capture process considered in this study. The process begins with the *Heat-Recovered-Genset Exhaust Gas* stream leaving the Scania Genset's exhaust after-treatment system and Waste Heat Extraction cum Integration Circuit [stream 2] and entering the *Exhaust Gas Cooler* unit. This unit primarily functions as a flue gas condenser, with the purpose of cooling the exhaust gas from approximately 110 °C and 1.02 bar(a) to 40 - 50 °C and approximately 1.013 bar(a), depending on the surrounding atmospheric temperature and the temperature of the cooling water available in the CCC unit. Another purpose of the *Exhaust Gas Cooler* unit is to condense the excess moisture present in the exhaust gas (due to lower vapour pressure of water at 40 °C or 50 °C compared to at 110 °C) and produce an outlet flue gas that is saturated with water vapour at 40 - 50 °C.

For most parts of this study, the temperature of the flue gas leaving the *Exhaust Gas Cooler* unit [stream 3] and the temperature of the lean solvent stream [stream 1] are assumed to be 40 °C. This represents operation under typical EU ambient climatic conditions, with cooling water available at 25 °C. An additional case study presented at the end of the project considers hot European summer conditions, for which the cooling water is assumed to be available at 40 °C. In this case, the Exhaust Flue Gas and Lean Liquid Mix streams [stream 3 and stream 1, respectively] are assumed to enter the process at 50 °C.

The cooled exhaust gas [stream 3] subsequently enters the packed *Absorber Unit*, which operates at approximately 1.01 bar(a), through the bottom of the column. The lean solvent [stream 1] enters from the top and flows downward through the column. This flow arrangement is commonly referred to as counter-current flow. The temperatures and pressures of the lean solvent stream and the cooled exhaust gas stream are assumed to be equal in this study. In actual Carbon Capture plants and pilot-scale facilities, these streams are also generally operated at similar temperatures and pressures.

The exhaust gas and lean solvent streams undergo intimate contact within the *Absorber Unit*, facilitated by the high specific surface area of the packing material. CO_2 is transferred from the gas phase into the liquid phase due to the difference between the operating and equilibrium partial pressures of CO_2 at the relatively low operating temperature of the *Absorber Unit*. The reactive absorption of CO_2 into the Liquid Mix taking place in the Absorber tower is exothermic and therefore results in heat generation. Since the process equipment is assumed to operate adiabatically, a steep temperature bulge is observed at an intermediate position, generally within the upper half of the tower.

The CO_2-lean flue gas, referred to as *Absorber Gas Out*, leaves the upper section of the *Absorber Unit* [stream 6] at a higher temperature than the inlet exhaust gas due to partial absorption of the heat released by the absorption reactions. The gas stream also contains a considerable amount of moisture transferred from the solvent within the *Absorber Unit* due to higher vapour pressure of water at elevated temperatures. The CO_2-lean exhaust gas [stream 6] is subsequently directed to another packed column, referred to as the *Amine Wash Tower*. This unit operates at approximately 1.0 bar(a), and its primary function is to remove entrained amine droplets and vapours from the flue gas through counter-current contact with process water [stream 7] supplied at 40 °C.

The final *CO_2 Lean Exhaust Gas* [stream 9] enters the *CO_2 Lean Gas Cooler* unit, where it is cooled to conditions close to those at the Absorber column inlet, namely approximately 40 °C and 1.00 bar(a). This cooling step removes excess moisture from the gas stream before it is released into the atmosphere [stream 10]. The condensed excess moisture, together with trace amounts of amines [stream 11], is returned to the *Process Water Storage Tank*. The *Amine Wash Water* [stream 8] exits from the bottom of the *Amine Wash Tower* and enters the *Amine Wash Water Pump*, where its pressure is increased from approximately 1.0 bar(a) to 1.84 bar(a) before entering the *Stripper Unit 1 Amine Wash Section* [stream 12].

The CO_2 rich solvent stream leaving the bottom of the *Absorber Unit* is referred to as the Rich Liquid Mix [stream 5] because of its high CO_2 loading. This stream is also at a higher temperature than the inlet exhaust gas due to partial absorption of the heat released by the absorption reactions and contains less water because of moisture transfer to the exhaust gas. The rich solvent stream passes through the *Rich Liquid Mix Pump*, where its pressure is increased from approximately 1.01 bar(a) to 1.84 bar(a).

The pressurized Rich Liquid Mix [stream 13] is then heated to approximately 112 °C by recovering heat

from the hot CO_2 lean solvent stream or Lean Liquid Mix [stream 25] in the *Recuperator Unit*, which is a plate-type heat exchanger. The heated CO_2 rich solvent subsequently enters the upper section of the packed *Stripper Unit 1* [stream 14]. This unit can be described as an upper half-stripper equipped with a top *Condenser Unit* but without a Reboiler unit.

Within *Stripper Unit 1*, the hot rich solvent comes into intimate contact with the hot stripping vapour mixture, referred to as *Stripper 2 Gas* [stream 20], entering from *Stripper Unit 2* top. This vapour stream removes CO_2 from the rich solvent and consists primarily of water vapour, which acts as the stripping medium, together with CO_2 gas. The hot *Stripper 1 Liquid* [stream 21] exits from the bottom of *Stripper Unit 1* and enters the top of the packed *Stripper Unit 2*. This unit can be described as a lower half-stripper equipped with a bottom *Reboiler Unit*. The remaining CO_2 is stripped from the solvent by the counter-current stream of stripping vapours generated at the *Reboiler Unit* [stream 23].

Both Stripper units operate at approximately 1.84 bar(a). Mass transfer occurs because of the difference between the operating and equilibrium partial pressures of CO_2 at the elevated temperatures of *Stripper Unit 1* and *Stripper Unit 2*. In contrast to the reactions occurring in the *Absorber Unit*, the CO_2 stripping reactions are endothermic. Since the system is assumed to operate adiabatically, a gradual decrease in temperature can be observed from the bottom of *Stripper Unit 2* towards the top of *Stripper Unit 1*.

The *Condenser Unit* operates at approximately 40 °C and is primarily used to cool the incoming *CO_2 + Steam* mixture [stream 15]. It also separates CO_2 from the water vapour, as the later is condensed and divided into the *Excess Water to Storage* [stream 18] and *Reflux Water* [stream 19] streams. The equal flow rate of process water as *Excess Water to Storage* [stream 18] is later used in [stream 7] for removing the entrained amine droplets and vapours in the *Amine Wash Tower*, which then enters the *Stripper Unit 1 Amine Wash Section* as [stream 12]. The highly concentrated CO_2 stream, with a concentration of approximately 95 - 99 wt%, leaves the *Condenser Unit* and is directed to the downstream Compression and Liquefaction processes.

The CO_2 lean solvent or Lean Liquid Mix [stream 22] leaving the bottom section of *Stripper Unit 2* enters the kettle-type *Reboiler Unit*, whose main purpose is to provide the thermal energy required for solvent regeneration in the Stripper units by generating high-temperature steam. The Reboiler Duty is met using hot pressurized process water at 3.5 bar(a) and 130 °C (maintained in the liquid phase under elevated pressure). This heat-transfer medium recovers sensible heat from the flue gas exhaust (439–135 °C or 415–135 °C) combined with upgraded coolant waste heat supplied at 128 °C by the Heat Integration Circuit. The Pressurized Process Water leaves the cold side of the Reboiler at 123 °C. The *Reboiler Unit* supplies this thermal energy input as sensible heat required to increase the solvent temperature, the latent heat required for the partial vaporization of the water contained in the solvent, and the reaction enthalpy required for CO_2 desorption, as shown in Equation 1. Previous studies have reported that approximately 45 - 50% of the total regeneration energy may be associated with Q_{reaction}, while Q_{latent} may account for approximately 35% of the total heat demand (Mohd Rozaiddin & Lau, 2022).

$$Q_{\text{regeneration}} = Q_{\text{sensible}} + Q_{\text{latent}} + Q_{\text{reaction}} \tag{1}$$

The concentrated CO_2 gas [stream 16] from the outlet of *Condenser Unit* is send to Multi-stage *Compressor Train* unit with Intercoolers to pressurize it to 25 bar(a) at 40 or 50 °C depending on the available cooling water temperature. This pressurized CO_2 gas stream is then sent to the Refrigeration Cycle (*CO_2 Chiller Unit*). After this step, the outlet CO_2 stream is a liquid at 20 bar(a) and -25 °C, ready for road transportation via Tank Trailer Trucks.

The hot Lean Liquid Mix [stream 24] exits the *Reboiler Unit* at approximately 120 °C and passes through the *Lean Liquid Mix Pump* to compensate for the pressure losses encountered in the Stripper Columns. The stream leaves the Pump at approximately 1.84 bar(a). The Lean Liquid Mix [stream 25] then enters the *Recuperator Unit*, where it transfers heat to the incoming CO_2 rich solvent stream, subsequently cooling it down to approximately 53 °C [stream 26].

The CO_2 lean solvent subsequently enters the *Lean Liquid Mix Cooler*, where it is further cooled to approximately 40 °C and depressurized to approximately 1.013 bar(a), before entering the *Lean Liquid Mix Storage Tank* [stream 27] as the *Recycle Lean Liquid Mix* stream.

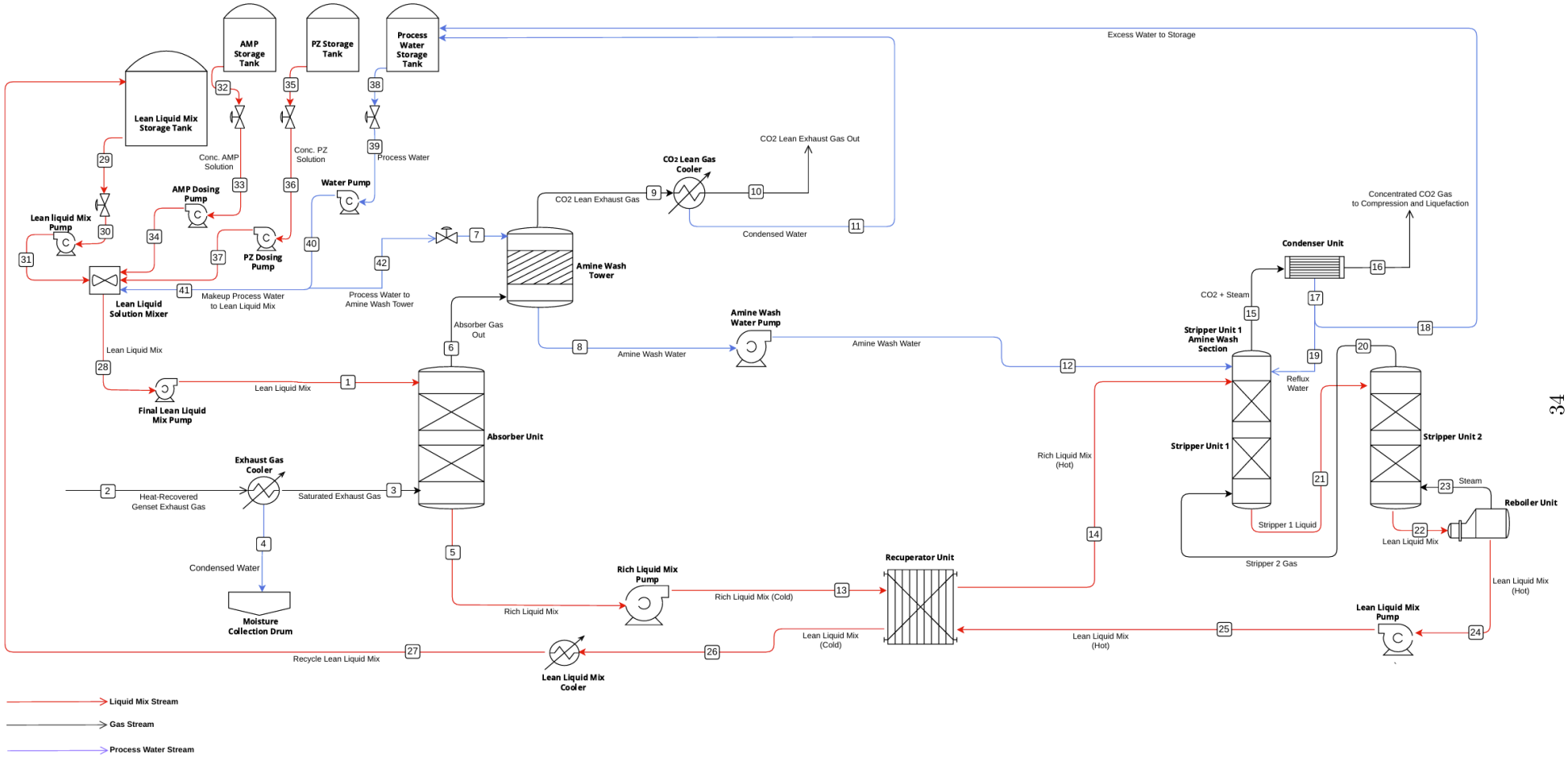


Figure 5: Process Flow Diagram for Amine Solvent based CO_2 Absorption process in Carbon Capture Container

3 Methodology

This section describes the methodology adopted for the design and evaluation of the proposed Carbon Capture system for the Scania Genset(s). The overall methodology involved breaking down the primary objective of the study into a set of well-defined research questions, which collectively guided the scope and direction of the work. This was followed by an extensive literature review to identify the most suitable Carbon Capture Technology, Solvent System, Process Configuration, Equipment, and Operating Conditions for the proposed application.

The selected process was subsequently implemented (modelled and executed) and analysed with the help of process simulations in Aspen Plus V14, supported by calculations and data analysis in Microsoft Excel and MATLAB. Finally, a techno-economic assessment was performed by estimating the Capital Cost of the individual Process Equipments and the overall operating costs of the Carbon Capture plant. These cost estimates were then used to evaluate the economic feasibility of the proposed process through the calculation of the specific CO_2 capture and CO_2 abatement costs.

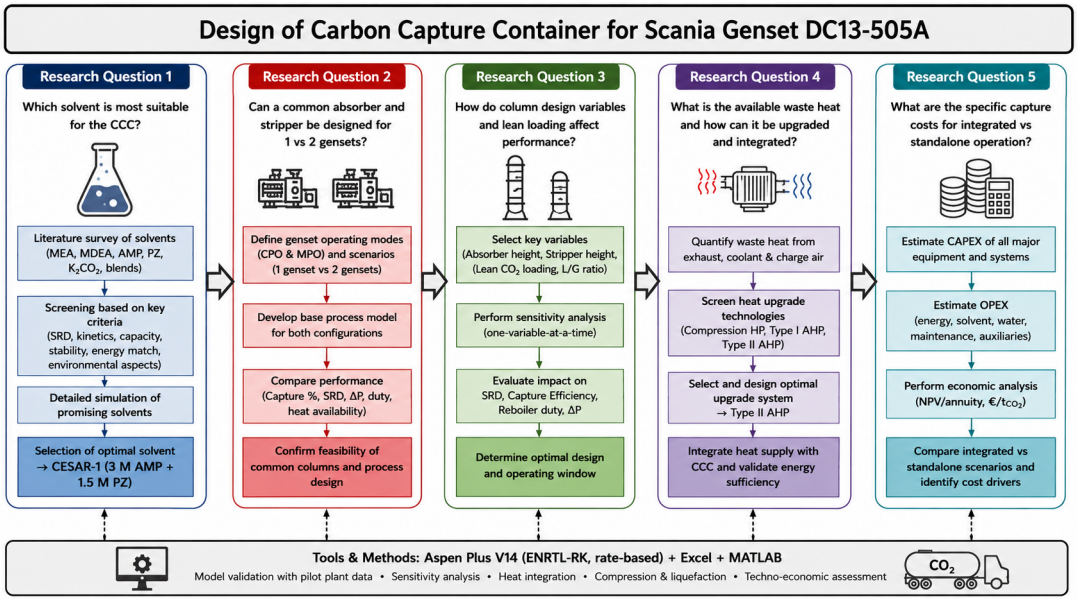


Figure 6: Methodology Workflow and CCC Unit Research Roadmap

The first research question addressed in this study was the selection of the most suitable solvent for the Carbon Capture Container (CCC) system. The selected solvent was required to show a high CO_2 Absorption Capacity, low Regeneration Energy requirement at the Stripping Section, and good Chemical Stability, particularly with respect to Thermal and Oxidative Degradation under the Process Operating Conditions. An extensive literature review was therefore conducted to evaluate the performance of commercially established and emerging solvent systems. Based on the findings, the popular CESAR-1 solvent, an aqueous blend consisting of 3 (M) 2-amino-2-methyl-1-propanol (AMP) and 1.5 (M) piperazine (PZ), which corresponds to approximately 27 wt% AMP, 13 wt% PZ, and 60 wt% H_2O, was selected as the most suitable solvent for the proposed carbon capture process.

The remaining research questions—including the possibility of using a common Absorber and Stripper Column design for both Single- and Dual-Genset operation under CPO and MPO modes, the optimization of column dimensions and lean-solvent CO_2 loading, the availability of engine and exhaust gas waste heat, and the estimation of the specific CO_2 capture cost were investigated thoroughly and answered systematically one by one in the later sections of the report.

Before discussing the process design and simulation methodology, it is useful to establish a consistent terminology for the liquid-phase solvent components used throughout this thesis. The following naming convention is adopted to improve clarity and readability for the readers.

Solvent: The solvent refers to the chemically active component responsible for absorbing CO_2 within the Carbon Capture process. It may consist of a single substance, such as MEA, or a blend of multiple substances, such as MDEA/PZ or AMP/PZ (this study).

Diluent: The diluent is the primary liquid medium in which the amine solvent is dissolved. Water is the most commonly used diluent, although other liquids such as methanol or ethanol may be used for specific applications. The diluent reduces the viscosity of the solvent (amine), improves its pumpability in the Carbon Capture circuit, reduces toxicity and corrosiveness, and facilitates easier heat and mass transfer within the Carbon Capture process.

Liquid Mix (LM): The Liquid Mix (LM) denotes the complete circulating liquid absorbent, consisting of the solvent(s) dissolved in the diluent. This is the working fluid that continuously circulates through the Absorber and Stripper columns to capture and subsequently release the CO_2 from the flue gas. For the present study, the Liquid Mix comprises 27 wt% AMP, 13 wt% PZ, and 60 wt% H_2O (CESAR–1).

3.1 Modelling Methodology Using Aspen Plus

Similar to any other engineering challenge, modelling the Carbon Capture process is the most essential part and the first step forward. Process modelling plays an important role from both economic and technical feasibility points of view as it enables preliminary evaluation of technical performance and economic feasibility before investment in laboratory- or pilot-scale facilities. Building a pilot plant or an industrial-scale prototype could be a costly affair. Hence, such models provide estimates of process behaviour, process dynamics, energy requirements, equipment performance, and overall capture efficiency, thereby supporting design optimisation and subsequent scale-up. The Aspen Plus V14 software was selected as the simulation software for the CCC unit design and simulation. Aspen Plus is a widely used Chemical Engineering simulation software at both Industrial and Academic levels due to its vast and extensive thermophysical-property database of nearly each and every Chemical Compound present, robust mass and heat transfer models and accurate kinetic predictions. These parameters makes Aspen Plus capable of modelling complex phase equilibria, chemical reactions, ionic interactions and heat- and mass-transfer processes. All these features made it an obvious choice for modelling the CCC unit for Scania Genset system and suitable for representing solvent-based CO_2 regenerative absorption. Many studies show successful validation of CO_2 absorption models in Aspen Plus with available pilot plant data by adjusting the thermodynamic and kinetic parameters and transport properties to closely replicate the test run results from real operational Carbon Capture plants (Garcia Ortega et al., 2017).

The Aspen Plus modelling methodology followed a systematic validation and application procedure. Initially, the modelling framework was validated using monoethanolamine (MEA) solvent, as this solvent has been extensively studied at both pilot-plant and industrial scales and is supported by robust experimental data. This step was used to verify the selected Thermodynamic Model, Reaction Scheme, Column Configuration, and Rate-Based Simulation approach.

The validated framework was then extended to the CESAR–1 solvent and compared with available pilot-plant data to assess the accuracy of predicted CO_2 Capture Performance, Lean and Rich Solvent Loading, Temperature Profiles, and Stripper Regeneration Energy Demand. Subsequently, the CESAR–1 model was adapted to represent the Carbon Capture Container (CCC) integrated with Scania Genset(s). Simulations were performed for single- and dual-Genset configurations under Continuous Power Operation (CPO) and Maximum Power Operation (MPO) conditions. Finally, Key Performance Indicators (KPIs), including Capture Rate (CR), Specific Reboiler Duty (SRD) and Capture Efficiency (CE), were evaluated against expected literature and pilot-scale ranges.

$$\text{Capture Rate (CR)} = \left(1 - \frac{\dot{m}_{\mathrm{CO_2,out,Absorber}}}{\dot{m}_{\mathrm{CO_2,in,Absorber}}}\right) \times 100\% \tag{2}$$

$$\text{Specific Reboiler Duty (SRD)} = \frac{Q_{\mathrm{reboiler}}}{\dot{m}_{\mathrm{CO_2,captured}}} \left[\frac{\mathrm{MW_{th}}}{\mathrm{kg\,CO_2/s}}\right] = \left[\frac{\mathrm{MJ_{th}}}{\mathrm{kg\,CO_2}}\right] \tag{3}$$

$$\text{Capture Efficiency (CE)} = \left(\frac{\dot{m}_{CO_{2,\mathrm{in,CCC}}} - \dot{m}_{CO_{2,\mathrm{out,CCC}}}}{\dot{m}_{CO_{2,\mathrm{in,CCC}}}}\right) \times 100\% \tag{4}$$

3.1.1 Aspen Plus Software Framework and Simulation Environment

The Aspen Plus simulation software operates with two main environments namely ***Properties*** (compound properties) and ***Simulation***. In the properties environment, users define the different chemical species and an appropriate thermodynamic property method is selected. Relevant equations of state, activity-coefficient models, and component interaction parameters are specified to describe the thermodynamic behaviour, reaction equilibria, and vapour–liquid equilibria associated with the CO_2 absorption process.

In the simulation environment of Aspen Plus software, one can construct and simulate the process flow diagram and use the previously defined chemical components and property methods. The required Process Units, and Material or Energy Streams are added to the flowsheet, after which their operating conditions, design specifications, and model parameters are specified. The final flowsheet is then solved via iterative numerical approximations using inbuilt Aspen Plus model equations to evaluate the mass

and energy balances, reaction kinetics, equipment performance, and overall behaviour of the Carbon Capture process.

3.1.2 Thermodynamics and Kinetics Study

To begin with defining the thermodynamics and kinetics for the Aspen Plus software, one needs to start with the *Properties* environment. For aqueous amine systems employed in Girbotol-type CO_2 capture processes, the electrolyte Non-Random Two-Liquid (e-NRTL) model is widely used to describe the liquid-phase thermodynamic behaviour for Aspen Plus simulations. This activity-coefficient model accounts for the strong non-idealities arising from molecular and ionic interactions and therefore enables the calculation of component activities, phase equilibria, CO_2 solubility, and chemical speciation within the solvent system (Y. Liu et al., 1999; Sanku & Svensson, 2019). The e-NRTL model is an extension of the conventional NRTL model, originally developed by Renon and Prausnitz in 1968, and is specifically formulated for electrolyte-containing systems (Prausnitz et al., 1998). It is particularly suitable for reactive amine–CO_2 systems, in which dissolved molecular species coexist with ionic species formed through acid–base and carbamate-forming reactions. The model is therefore capable of representing the complex chemical equilibria and strongly non-ideal liquid-phase behaviour characteristic of aqueous amine solvents (J. Wang et al., 2020; Esmaeili et al., 2020).

In Aspen Plus V14 the *ENRTL-RK* equation of state or property method which can be found in the *Properties* environment combines the e-NRTL model for the liquid phase with the Redlich–Kwong equation of state for the vapour phase. The latter is used to account for vapour-phase non-idealities, while Henry's law is applied to describe the dissolution of sparingly soluble gases, including CO_2, where applicable. The property method requires several model parameters, including Henry's law constants, binary interaction parameters, equilibrium constants, and thermodynamic data for the molecular and ionic species involved (Aspen Technology, Inc., 2001).

For the validation of the MEA-based system, the required solvent property data and model parameters were obtained from the standard Aspen Plus example model database for CCUS applications. For the CESAR–1 solvent system, the required thermodynamic parameters, supporting property data, and the Aspen Plus Solvent Chemistry simulation model was developed, validated and kindly provided by Morlando and Knuutila (Morlando, Y. Zhang, et al., 2025). The use of this validated model facilitated the implementation of the solvent system and improved the reliability of the predicted process performance.

The packed Absorber and Stripper columns were modelled in Aspen Plus using the *RadFrac* unit-operation block. *RadFrac* provides a rigorous framework for simulating multistage separation processes and is capable of representing both vapour–liquid equilibrium and reactive mass-transfer phenomena. The block may be configured using either an equilibrium-stage approach (Riva et al., 2018), which assumes that the exiting vapour and liquid phases reach thermodynamic equilibrium at each stage or at a particular packing height, or a rate-based approach, which explicitly accounts for mass-transfer resistances, reaction kinetics, interfacial area, and column hydrodynamics closer to a dynamic and steady-state process like CO_2 Capture (Morlando, Y. Zhang, et al., 2025).

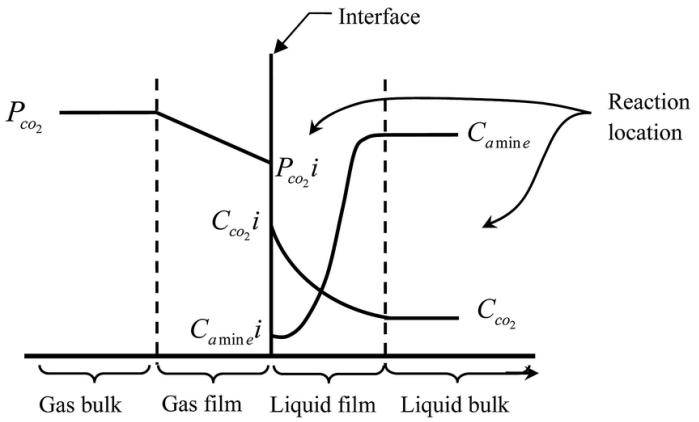


Figure 7: Schematic Representation of the Two-Film Theory for Rate Based CO_2 Capture (Afkhamipour & Mofarahi, 2017)

The rate-based approach is based on the Lewis–Whitman two-film theory for describing mass transfer across the gas–liquid interface. As shown in Figure 7, CO_2 is transferred from the bulk gas phase to the interface, where the principal resistance to mass transfer is assumed to be concentrated within a thin stagnant layer, or film, adjacent to the interface. Separate films are considered for the gas and liquid phases. Depending on the diffusion rate and the associated resistance in each film, CO_2 then passes from the interface into the bulk Lean Liquid Mix (Afkhamipour & Mofarahi, 2017). For the CESAR–1–CO_2 system, the rate-based formulation was selected because the absorption and desorption processes are governed by finite-rate chemical reactions and mass-transfer limitations. Under these conditions, the assumption of complete phase equilibrium at each theoretical stage or packing height may not adequately represent the behaviour of an industrial packed column (Morlando, Y. Zhang, et al., 2025).

3.1.3 Absorbent Analysis and Implementation in Aspen Plus Model

This section describes the main reactions involving aqueous-phase CO_2 in two different solvent systems. It also outlines the methodology used to implement these reactions in the Aspen Plus model, including the types of the relevant chemical species, reaction mechanisms, equilibrium or kinetic expressions, and corresponding reaction constants. Particular focus is given to how the selected reaction framework represents CO_2 absorption, solvent protonation, and the formation of ionic and carbamate species within each solvent system.

Beginning with the aqueous CO_2–MEA system, for which several model validation studies are presented in the subsequent sections of this report, the chemical reaction framework is defined by seven different reactions. Of these, three are equilibrium reactions (instantaneous and non-kinetically controlled), while the remaining four are reversible kinetic expressions. Together, these reactions describe the principal acid–base, bicarbonate, and carbamate formation pathways governing CO_2 absorption in aqueous MEA solutions (Errico et al., 2016). The complete reaction set and the corresponding reaction types are presented in Table 7. The Reaction Rate Constant and Activation Energy values are presented in Table 8.

Table 7: Absorber and Stripper model Reactions for the MEA–CO_2–H_2O system (Errico et al., 2016; Aspen Technology, Inc., 2012)

No.	**Stoichiometry**	**Reaction type**
1	$2H_2O \leftrightharpoons OH^- + H_3O^+$	Equilibrium
2	$MEAH^+ + 2H_2O \leftrightharpoons H_3O^+ + OH^- + MEA$	Equilibrium
3	$HCO_3^- + H_2O \leftrightharpoons CO_3^{2-} + H_3O^+$	Equilibrium
4	$OH^- + CO_2 \rightarrow HCO_3^-$	Kinetic
5	$HCO_3^- \rightarrow OH^- + CO_2$	Kinetic
6	$MEACOO^- + H_3O^+ \rightarrow MEA + CO_2 + H_2O$	Kinetic
7	$MEA + CO_2 + H_2O \rightarrow MEACOO^- + H_3O^+$	Kinetic

Table 8: Rate constants (k_0) and activation energies (E, kJ/mol) for the kinetic reactions in the MEA system (Errico et al., 2016; Aspen Technology, Inc., 2012)

No.	k_0	E **[kJ/mol]**
4	1.33×10^{17}	5.55×10^{1}
5	6.63×10^{16}	1.07×10^{2}
6 (Absorber)	5.52×10^{23}	6.92×10^{1}
6 (Stripper)	6.50×10^{27}	6.92×10^{1}
7	3.02×10^{14}	4.12×10^{1}

The aqueous CO_2–AMP–PZ system was considered next. Model validation using available pilot-plant data, followed by the development and application of the model under the Scania Genset Exhaust Flue Gas and CCC operating conditions, is presented in the subsequent sections of this report. The chemical reaction framework for this solvent system is defined by ten different reactions. Of these, six reactions namely the protonation and de-protonation reactions are equilibrium reactions and non-kinetics controlled, while the remaining four are reversible kinetic expressions. Together, these reactions describe the principal acid–base, bicarbonate, carbamate and dicarbamate formation pathways governing CO_2 absorption in aqueous AMP-PZ solutions (Morlando, Y. Zhang, et al., 2025; Spek et al., 2016). The complete reaction set and the corresponding reaction types are presented in Table 9. The Reaction Rate Constant and Activation Energy values are presented in Table 10.

Table 9: Absorber and Stripper model reactions for the AMP–PZ–CO_2–H_2O system (Morlando, Y. Zhang, et al., 2025; Spek et al., 2016)

No.	**Stoichiometry**	**Reaction type**
1	$2H_2O \leftrightharpoons H_3O^+ + OH^-$	Equilibrium
2	$2H_2O + CO_2 \rightarrow H_3O^+ + HCO_3^-$	Kinetic
3	$H_2O + HCO_3^- \leftrightharpoons H_3O^+ + CO_3^{2-}$	Equilibrium
4	$AMP + HCO_3^- \rightarrow AMPCOO^- + H_2O$	Kinetic
5	$H_2O + AMPH^+ \leftrightharpoons AMP + H_3O^+$	Equilibrium
6	$H_2O + PZH^+ \leftrightharpoons PZ + H_3O^+$	Equilibrium
7	$H_2O + PZH_2^{2+} \leftrightharpoons PZH^+ + H_3O^+$	Equilibrium
8	$PZ + HCO_3^- \rightarrow PZCOO^- + H_2O$	Kinetic
9	$PZCOO^- + HCO_3^- \rightarrow PZ(COO^-)_2 + H_2O$	Kinetic
10	$H_2O + PZH^+CO_2^- \leftrightharpoons PZCOO^- + H_3O^+$	Equilibrium

Table 10: Rate constants (k_0) and activation energies (E, kJ/mol) for the kinetic reactions in the AMP–PZ system (Morlando, Y. Zhang, et al., 2025; Spek et al., 2016)

No.	**Reaction**	**Concentration basis**	**Rate constant k_0**	**E [kJ/mol]**
2.1	$CO_2 + OH^- \rightarrow HCO_3^-$	Activity	1.33×10^{17}	5.55×10^{1}
2.2	$HCO_3^- \rightarrow CO_2 + OH^-$	Activity	6.63×10^{16}	1.07×10^{2}
4.1	$AMP + CO_2 + H_2O \rightarrow AMPCOO^- + H_3O^+$	Activity	1.38×10^{10}	2.04×10^{1}
4.2	$AMPCOO^- + H_3O^+ \rightarrow AMP + CO_2 + H_2O$	Activity	6.14×10^{24}	6.17×10^{1}
8.1	$PZ + CO_2 + H_2O \rightarrow PZCOO^- + H_3O^+$	Activity	3.99×10^{10}	2.53×10^{0}
8.2	$PZCOO^- + H_3O^+ \rightarrow PZ + CO_2 + H_2O$	Activity	1.26×10^{24}	6.17×10^{1}
9.1	$PZCOO^- + CO_2 + H_2O \rightarrow PZ(COO^-)_2 + H_3O^+$	Activity	3.21×10^{14}	3.52×10^{1}
9.2	$PZ(COO^-)_2 + H_3O^+ \rightarrow PZCOO^- + CO_2 + H_2O$	Activity	8.53×10^{17}	1.05×10^{1}

Since the equilibrium constants for Reactions 1–3 for aqueous CO_2–MEA system and Reactions 1, 3, 5–7, and 10 for aqueous CO_2–AMP-PZ system are strongly temperature-dependent, they were calculated using a rigorous thermodynamic formulation based on the standard Gibbs free-energy change, ΔG_j°. The required ΔG_j° values and associated thermodynamic data were obtained from the Aspen Plus database (Madeddu et al., 2019). Here, j denotes the individual equilibrium reaction, ΔG_j° is its standard Gibbs free energy change, and $(K_{eq})_j$ is the corresponding equilibrium constant of the equilibrium reaction j.

$$(K_{eq})_j = \exp\left(-\frac{\Delta G_j^\circ}{RT}\right) \tag{5}$$

For the reversible kinetic reactions (Reactions 4–7 for aqueous CO_2–MEA system and Reactions 2, 4, 8 and 9 for aqueous CO_2–AMP-PZ system), Aspen Plus applies a power-law rate expression based on the Arrhenius equation (Madeddu et al., 2019). In this formulation, T is the absolute temperature, n is the temperature exponent, which is equal to 1 when no reference temperature T_0 is specified, R is the universal gas constant, and N is the number of components involved in the reaction. Furthermore, χ_i represents the mole fraction of component i, γ_i its activity coefficient, and α_i its reaction-order coefficient. The pre-exponential factor, k_0, and activation energy, E, were specified by the user using values obtained from the literature and the Aspen Plus database for CO_2–MEA, as summarized in Table 8 (Aspen Technology, Inc., 2024; Zacchello et al., 2017) and from literature alone for CO_2–AMP-PZ as summarized in Table 10 (Morlando, Y. Zhang, et al., 2025).

$$r = k_0 T^n \exp\left(-\frac{E}{RT}\right) \prod_{i=1}^{N} (x_i \gamma_i)^{\alpha_i} \tag{6}$$

3.1.4 MEA Based Models

Validating the Process Simulation model in Aspen Plus is necessary to ensure the reliability of the CCC system design, especially before and if Scania decides to move forward with the idea of building a prototype plant. Validation studies for the MEA-based capture system were performed using Aspen Plus V14 process modelling and simulation software to establish the modelling methodology, develop familiarity with the software and the CCUS process, and get preliminary estimates of the design parameters and dimensions of amine-based CO_2 capture systems with similar flue gas or CO_2 handling capacities. The validation procedure was conducted in a stepwise manner, beginning with validation of the packed absorber model, followed by validation of the packed stripper and the associated liquid- and gas-phase streams. Although several experimental and simulation studies are available in the literature, only a limited number of well-established studies containing extensive and reliable datasets were selected. These reference cases, described in the subsequent sections, were used to assess the accuracy of the thermodynamic framework, rate-based column model, mass and energy balances, and predicted process-performance indicators.

As a common rule of thumb, a model validation result is considered near accurate when the percentage of error or the deviation of the simulation output from the experimental data is less than or around 5% and maximum 10%.

3.1.4.1 Model Validation Study for Absorber

For validation of the *Absorber Unit* model, two Pilot Plant data from two different sources, representing different operating conditions and column dimensions, were simulated and the model predictions (results) were compared with the corresponding experimental data. The same underlying thermodynamic framework and kinetic parameters were applied consistently across all two cases.

The National Carbon Capture Center (NCCC) Pilot Plant: The National Carbon Capture Center (NCCC), which is located in Alabama, USA, operates a Pilot Scale Solvent Test Unit with a CO_2 capture capacity of approximately 10 tonne day^{-1}. The Absorber unit comprises of a packed column containing three beds, each with 6.1 m of Sulzer MellapakPlus 252Y structured packing, and has an internal diameter of 0.64 m. Intercoolers are installed between the packing beds and were considered in the present model to generate close to real plant conditions during the validation procedure. The Stripper unit comprises of a packed column containing two beds, each with 6.1 m of Sulzer MellapakPlus 252Y structured packing, and has an internal diameter of 0.59 m (Akula et al., 2021). Absorber unit pressure was maintained around 1.07 bar(a) and Stripper unit pressure was maintained around 1.83 bar(a). A comparison between the reported experimental data and the corresponding Aspen Plus simulation results is presented in Table 11. Here the Liquid–to–Gas or *L/G ratio* is represented in molar ratio format (moles of Lean Liquid Mix (LLM)/moles of Flue Gas (FG)).

Table 11: Comparison of Experimental Data with Absorber Simulation Results for NCCC Pilot Plant (Akula et al., 2021)

Independent Variable	Variable Value	Modelled Variable	Variable value (Literature)	Variable value (This Work)	Modelled Variable	Variable value (Literature)	Variable value (This Work)
L/G ratio–Low (mol LLM/mol FG)	1.75	CO_2 Capture rate –Absorber (%)	78.57	79.45	Absorber Rich Liquid Mix flow rate (kg/hr)	3343	3272.40
L/G ratio–Medium (mol LLM/mol FG)	3.75	CO_2 Capture rate –Absorber (%)	99.91	99.97	Absorber Rich Liquid Mix flow rate (kg/hr)	7242	7224.25
L/G ratio–High (mol LLM/mol FG)	6.67	CO_2 Capture rate –Absorber (%)	98.07	99.96	Absorber Rich Liquid Mix flow rate (kg/hr)	12043	11983.86

As evident from the simulation results, the error margin between the experimental data and the simulation results is well within the 5% error range, successfully validating the Absorber model.

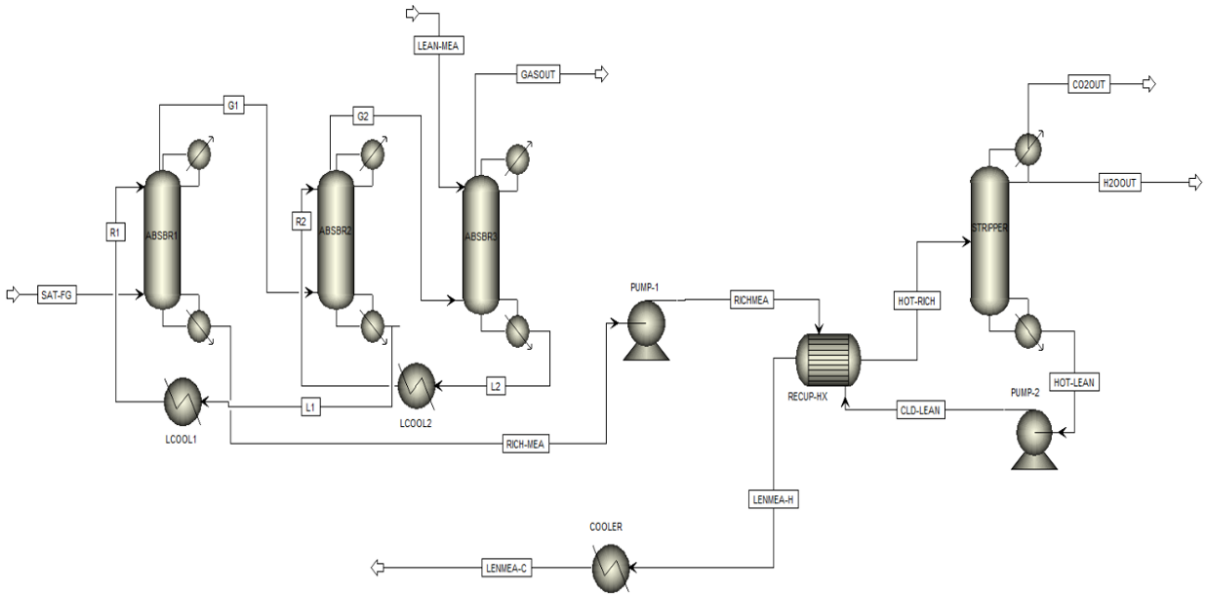


Figure 8: Aspen Plus Process Flow Diagram for NCCC Pilot Plant

The ***CESAR Pilot Plant*** located in Esbjerg, Denmark, is a post-combustion Carbon Capture facility designed to evaluate both conventional and advanced amine solvents. The absorber column is packed with Sulzer Mellapak 2X structured packing, with an approximate packed height of 17 m and an internal diameter of 1.1 m. The entering lean MEA solution is at $40\,°C$ and the overall CO_2 Removal Efficiency is around 87–90% (Razi et al., 2013). The plant provides experimental temperature measurements at several locations along the absorber height. These data were used to construct an experimental temperature profile and compare it with the corresponding Aspen Plus simulation results, as shown in Table 12. The comparison indicates that both the exact location and magnitude of the temperature bulge are predicted with considerable accuracy, which is consistent with the relatively medium to low Liquid-to-Gas ratio (*L/G ratio*) which is presented as moles of lean Liquid Mix (LM)/moles of Flue Gas (FG).

This validation study was interesting and particularly informative because it shows how the assumptions for ideal conditions used in Aspen Plus may deviate from real life operating behaviour of the Capture Plant as we go to lower Liquid–to–Gas ratios. At an L/G ratio of approximately 2.9 (Test 1A–4), the experimental and simulated temperature profiles differed by more than 19% at intermediate Absorber heights, close to the temperature-bulge peak. In contrast, at an increased L/G ratio of approximately 4.7 (Test 1A–1), the deviation between the experimental and simulated temperature profiles were found to be below 6.5%. One possible explanation is that the model may overestimate gas–liquid contact, effective interfacial area due to ideal packing, or ideal wetting of the structured packing. This would enhance the predicted mass-transfer rate and consequently increase the calculated heat release from the exothermic CO_2–MEA absorption reactions relative to actual Pilot Plant operation. This hypothesis could have been verified by comparing the experimental rich-solvent CO_2 loading at the Absorber outlet

with the corresponding simulation value. However, the experimental rich-solvent loading at the bottom of the Absorber was not explicitly reported for different L/G ratios in the referenced study. In addition, the common Aspen Plus assumption of adiabatic column operation, with all the generated heat being taken up by the mixing streams and no heat loss to the surroundings, may further contribute to the larger temperature deviations observed at lower L/G ratios.

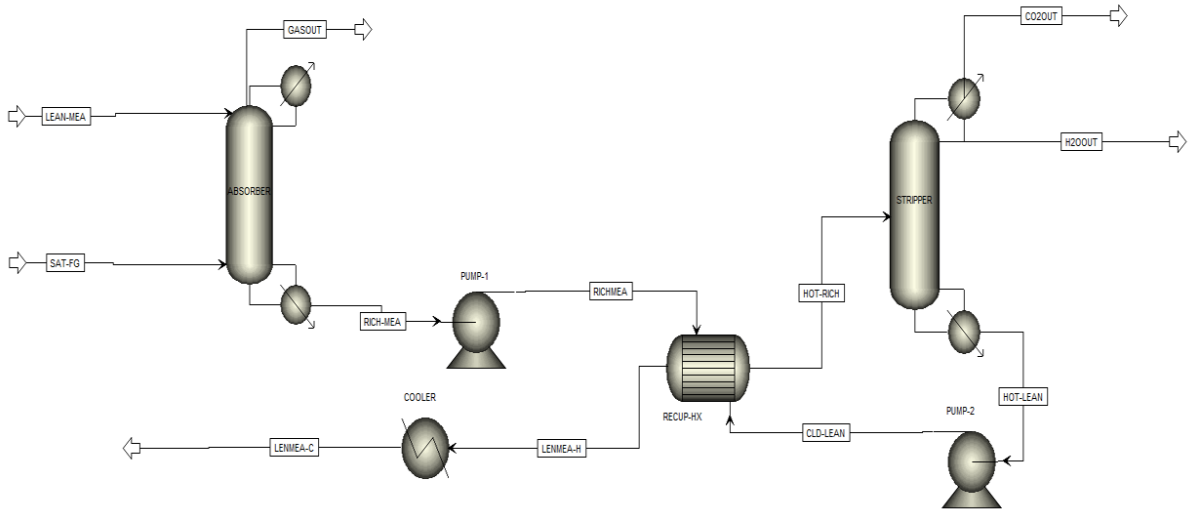


Figure 9: Aspen Plus Process Flow Diagram for CESAR Pilot Plant

Table 12: Comparison of Experimental and Simulated Absorber Temperature Profiles for CESAR Pilot Plant (Razi et al., 2013)

(a) Test 1A–1: $L/G = 4.70$ mol LM/mol FG

Height from bottom (m)	Temperature Literature (°C)	Temperature This Work (°C)	Error (%)
17.00	56.43	54.95	2.62
12.75	74.45	72.31	2.87
8.50	69.52	71.23	2.46
4.25	62.45	63.94	2.39
0.00	55.60	52.03	6.42

(b) Test 1A–2: $L/G = 4.10$ mol LM/mol FG

Height from bottom (m)	Temperature Literature (°C)	Temperature This Work (°C)	Error (%)
17.00	59.42	57.14	3.84
12.75	73.40	72.55	1.16
8.50	66.80	72.08	7.90
4.25	60.32	64.68	7.23
0.00	53.77	51.24	4.71

(c) Test 1A–3: $L/G = 3.49$ mol LM/mol FG

Height from bottom (m)	Temperature Literature (°C)	Temperature This Work (°C)	Error (%)
17.00	61.06	59.74	2.16
12.75	70.32	72.52	3.13
8.50	62.70	72.13	15.04
4.25	57.92	64.13	10.72
0.00	52.62	50.31	4.39

(d) Test 1A–4: $L/G = 2.92$ mol LM/mol FG

Height from bottom (m)	Temperature Literature (°C)	Temperature This Work (°C)	Error (%)
17.00	62.03	62.40	0.60
12.75	67.00	72.57	8.31
8.50	59.65	71.44	19.77
4.25	57.40	60.20	4.88
0.00	52.08	48.87	6.16

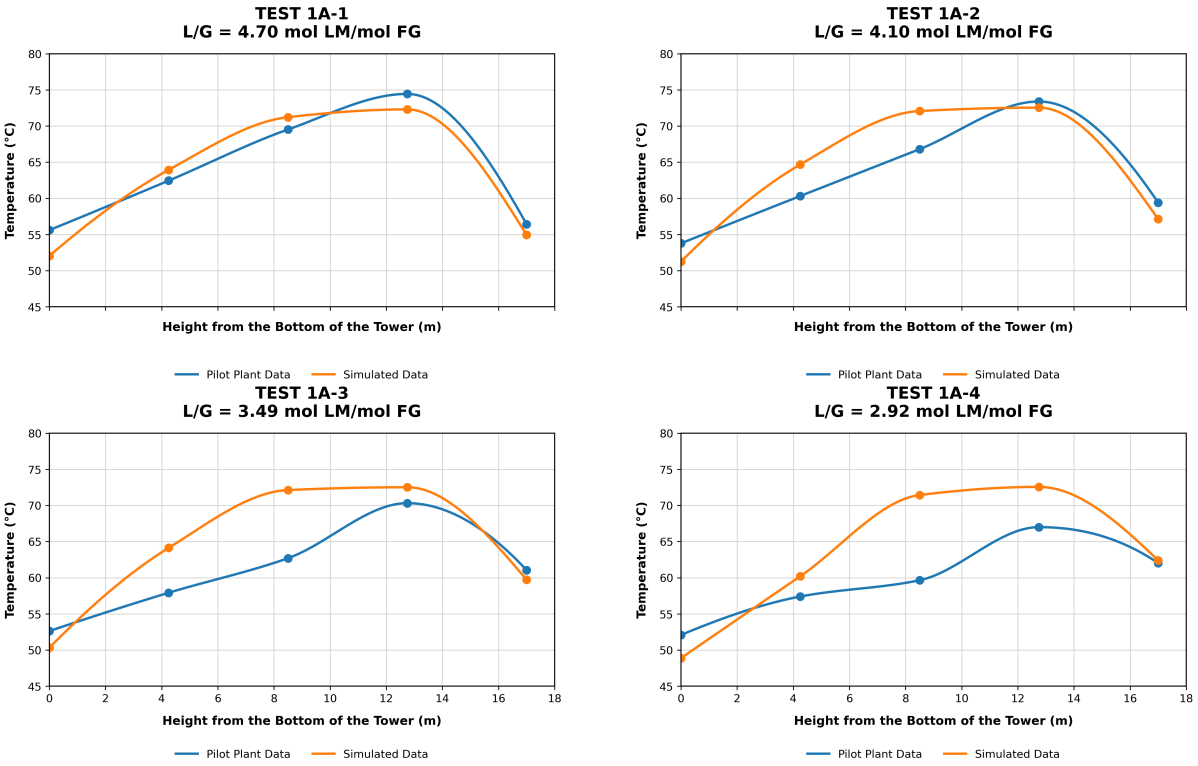


Figure 10: Comparison Plot for Experimental and Simulated Absorber Temperature profiles of CESAR Pilot Plant

3.1.4.2 Model Validation Study for Stripper

The Stripper unit model requires two degrees of freedom to be defined unlike the Absorber unit because of the presence of both a Condenser unit and a Reboiler unit. The operating specifications for the SINTEF Plant was therefore selected for the validation study.

The ***SINTEF*** CO_2 ***Plant*** (also known as the Tiller CO_2 Capture Pilot Plant or Tiller CO_2 Laboratory) is located at Tiller, Norway. For validation of the SINTEF Plant Stripper model, the same degrees of freedom reported in the literature were used in the model, namely the Condenser temperature of 15 $°C$ and the Reboiler Heat Duty of 11.60 kW (Madeddu et al., 2019). The same Liquid-to-Gas ratio of around 10.88 (moles of Lean Liquid Mix (LLM)/moles of Flue Gas (FG)) and Stripper pressure of 1.97 bar(a) reported in the literature were used in the Absorber and Stripper models respectively (Tobiesen et al., 2008). The experimentally reported column temperature profile was compared with the liquid-phase temperature profile predicted by the Aspen Plus model. The study results are shown in Table 13. Furthermore, the temperature profile deviations at the selected Stripper heights remain below 5 °C, indicating satisfactory agreement between the experimental data and the simulation results with less than 1.5% error margin.

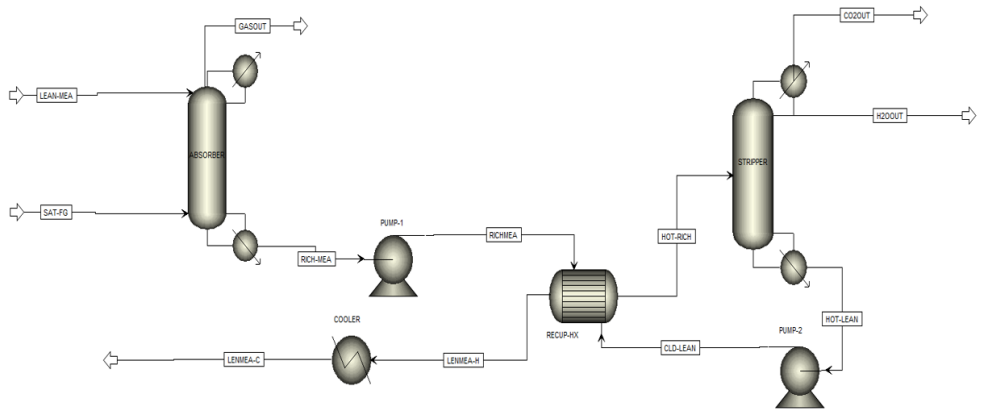


Figure 11: Aspen Plus Process Flow Diagram for SINTEF Pilot Plant

Table 13: Comparison of Experimental and Simulated Stripper Liquid-Temperature Profiles from the SINTEF CO_2 Capture Plant (Madeddu et al., 2019)

Height from Stripper Bottom (m)	**Liquid Temperature from Literature (K)**	**Liquid Temperature This Work (K)**	**Error (%)**
0.01	393.59	396.06	0.63
1.05	392.41	396.06	0.93
2.15	391.50	396.06	1.16
3.25	391.00	396.05	1.29
4.35	389.81	388.32	0.38

The comparison results show that the simulated temperature is generally higher than the corresponding experimental value along the Stripper height, except near the top of the column, where the hot steam–CO_2 mixture encounters the relatively cooler rich-amine stream. This result further suggests that the Aspen Plus model simulates ideal gas–liquid contacting in packed towers and perfect mixing while neglecting heat losses to the surroundings under the assumed adiabatic operating conditions. The trend observed in Figure 12 also suggests relatively higher CO_2 desorption and mass transfer rates from liquid amine phase to the gaseous steam phase in the upper section of the column. Owing to the endothermic nature of the desorption reactions, this increased reaction rate may explain the more pronounced temperature decrease over the upper 1.5–1.8 m of the tower compared to the remaining 2.55–2.85 m of packed height.

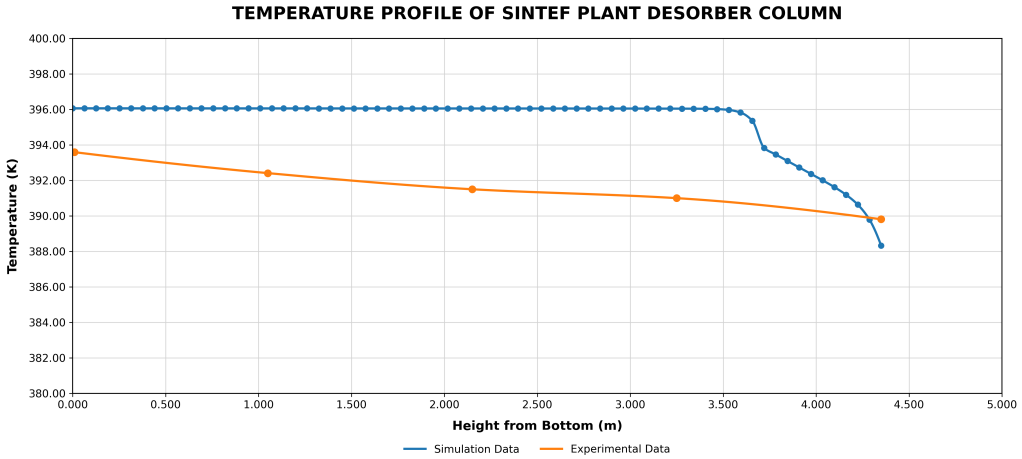


Figure 12: Comparison Plot for Experimental and Simulated Stripper Liquid-Temperature Profiles from the SINTEF CO_2 Capture Plant

3.1.5 CESAR-1 Based Models

The CESAR–1 model validation was conducted to evaluate the reliability and accuracy of the solvent-chemistry file obtained from the literature and to confirm its suitability for subsequent process-design studies (Morlando, Y. Zhang, et al., 2025; Spek et al., 2016). This validation also served as an important preliminary verification step before applying the model to the Scania Genset Exhaust Gas compositions, Gas and Solvent Flow Rates, and CCC Operating conditions. Owing to the limited availability of comprehensive, publicly accessible and robust CESAR–1 based Pilot Plant datasets, two widely reported test runs from the University of Kaiserslautern (UKL) Pilot Plant in Kaiserslautern, Germany, were selected as the simulation reference cases. The corresponding operating conditions, solvent properties, and column dimensions and specifications were implemented in Aspen Plus. The main model inputs and important simulation outputs are summarized in Tables 14 and 15, together with comparisons of key performance indicators (KPIs) such as Overall CO_2 Removal Efficiency, the Specific Reboiler Heat Duty (SRD), and Rich Liquid Mix CO_2 loading.

Table 14: Operating and Design Parameters for Test Runs G11 and G13 for UKL Pilot Plant (Morlando, Y. Zhang, et al., 2025; Spek et al., 2016)

Parameter	**Run G11**	**Run G13**
Absorber Packing type	Mellapak BX 500	Mellapak BX 500
Stripper Packing type	Mellapak BX 500	Mellapak BX 500
Absorber pressure (bar(a))	1.00	1.00
Stripper pressure (bar(a))	2.00	2.00
Absorber diameter (m)	0.125	0.125
Stripper diameter (m)	0.125	0.125
Absorber height (m)	4.25	4.25
Stripper height (m)	2.55	2.55
Reboiler Heat Duty (W)	9858.50	11170.80
Lean Liquid Mix (LLM) loading (mol CO_2/mol AMP+PZ)	0.13	0.21
L/G ratio (mol LLM/mol FG)	1.52	2.41

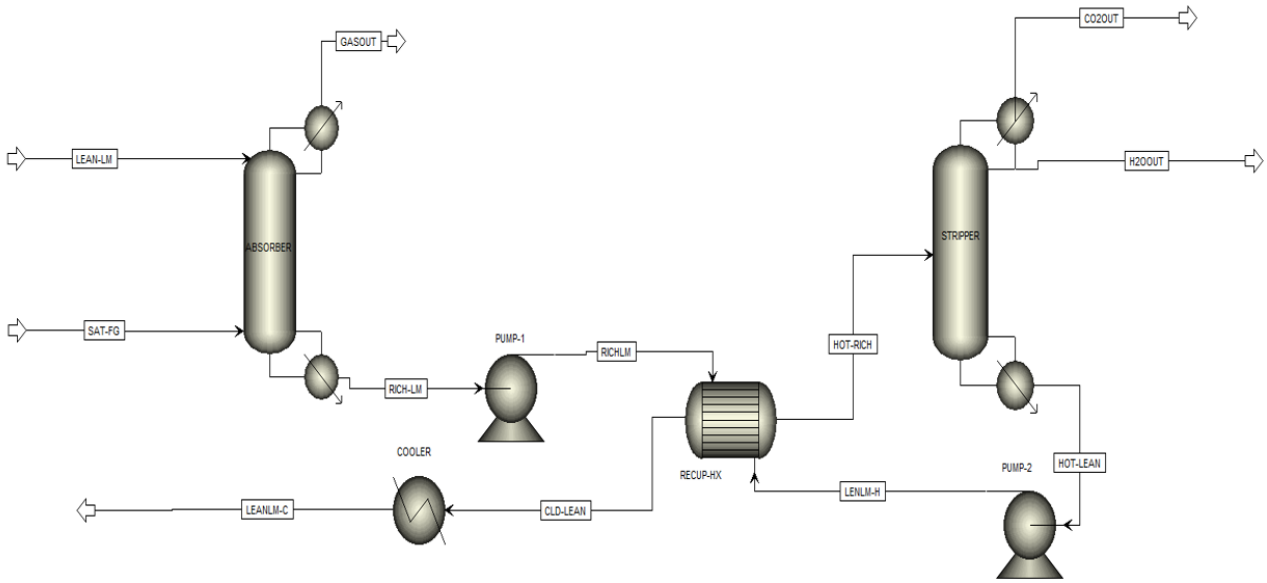


Figure 13: Aspen Plus Process Flow Diagram for UKL Pilot Plant (Morlando, Y. Zhang, et al., 2025; Spek et al., 2016)

Table 15: Comparison of Aspen Model Results with UKL Pilot Plant data for Test Runs G11 and G13

Parameter	**Run G11**	**Run G13**
Captured CO_2 in Absorber Unit (kg/hr)	11.58	11.43
Desorbed CO_2 in Stripper Unit (kg/hr)	10.13	10.23
Absorber CO_2 Capture Efficiency – Aspen model (%)	98.05	96.86
Overall CO_2 Removal Efficiency – Aspen model (%)	85.78	86.69
Overall CO_2 Removal Efficiency – Pilot Plant data (%)	90.51	89.33
Error in Overall CO_2 Removal Efficiency (%)	4.73	2.64
Specific Reboiler Duty – Aspen model (MJ/kg CO_2)	3.50	3.93
Specific Reboiler Duty – Pilot Plant data (MJ/kg CO_2)	3.32	3.83
Error in Specific Reboiler Duty (%)	5.42	2.61
AMP concentration in Absorber gas outlet (ppmv)	1661	1214
PZ concentration in Absorber gas outlet (ppmv)	136	57
Rich Amine loading – Aspen model (mol CO_2/mol AMP+PZ)	0.56	0.47
Rich Amine loading – Pilot Plant data (mol CO_2/mol AMP+PZ)	0.52	0.48
Error in Rich Amine loading (%)	7.70	1.72

As shown by the simulation results, the deviations between the experimental data and the model predictions were within approximately 1.7–8.0% for the principal KPIs, thereby providing satisfactory validation of the Carbon Capture plant model. Additional simulations were performed for the same pilot-plant test runs by incorporating an Amine Wash section in the top part of the Absorber unit, using fresh water at 40 °C and flow rates of 2–3.8 kg hr^{-1}. This section was included to capture entrained AMP and PZ vapours from the CO_2-lean flue gas and return them to the Stripper Unit as a solvent-recovery

measure via adding the water from the bottom of the Amine Water Wash section into the top part of the Stripper column after its pressure has been increased using the Amine Wash Water Pump. Under these conditions, the Specific Reboiler Duty (SRD) increased by approximately 3–7% for the same overall CO_2 Removal Efficiency.

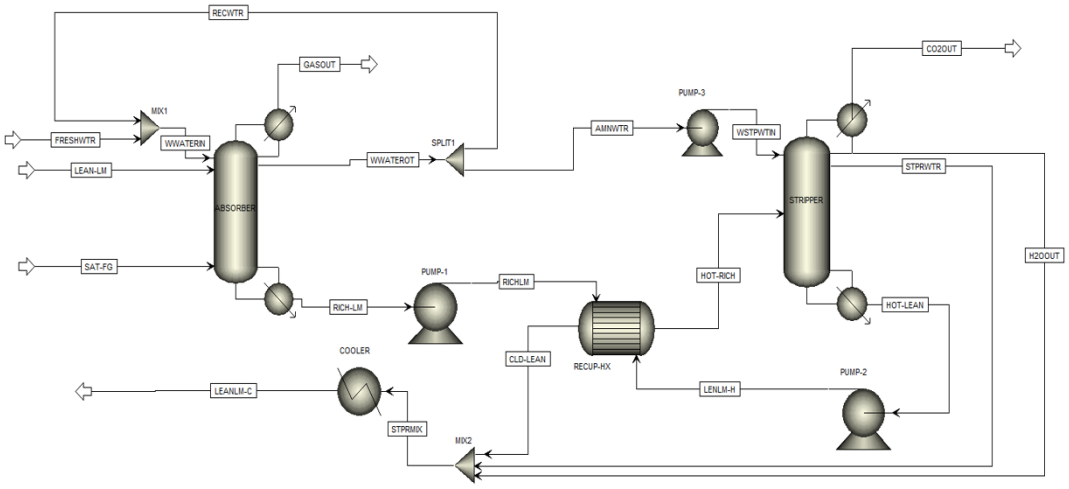


Figure 14: Aspen Plus Process Flow Diagram for Modified UKL Pilot Plant

3.2 Genset CCC System Modelling

This section describes the design dimensions and process parameters of the various components and equipment included in the Genset CCC Unit model developed in Aspen Plus V14. As mentioned earlier, the Genset Engine(s) operate under steady-state conditions, resulting in a constant Exhaust Gas flow rate for each operating mode (CPO and MPO modes). In accordance with the widely accepted design philosophy, the design point for the CCC components was selected at the Maximum Power mode of operation, corresponding to the Rated Power condition of one or two 13-Liter DC13-505A Genset Engine(s). The flow rate, composition, and temperature of the Exhaust Gas downstream of the *Exhaust Gas Cooler* are presented in Table 16. The Exhaust Gas leaving the *Exhaust Gas Cooler* unit and entering the *Absorber Unit* was assumed to be saturated with water vapour. The fresh CESAR–1 Liquid Mix was specified as 27 wt% AMP, 13 wt% PZ, and 60 wt% H_2O, which is a commonly reported composition that provides favourable capture performance without imposing excessive corrosion concerns (Morlando, Buvik, et al., 2024).

Table 16: Exhaust Gas CCC Inlet Conditions for Single and Two Genset Scenarios

Parameter	**Single Genset**		**Two Gensets**	
	Maximum Power Operation	**Continuous Power Operation**	**Maximum Power Operation**	**Continuous Power Operation**
Gas Flow rate (kg/hr)	1928.58	1507.82	3857.17	3015.64
CO_2 Conc. (wt%)	14.96	13.16	14.96	13.16
H_2O Conc. (wt%)	4.33	4.36	4.33	4.36
N_2 Conc. (wt%)	71.94	72.22	71.94	72.22
O_2 Conc. (wt%)	8.77	10.26	8.77	10.26
Gas Inlet Temperature (°C)	40	40	40	40
Gas Inlet Pressure (bar(a))	1.013	1.013	1.013	1.013

During steady-state operation, the Lean Liquid Mix retains a certain amount of CO_2, since complete regeneration in the *Stripper Units* is neither technically practical nor energetically feasible or desirable. Reported lean loadings for AMP–PZ systems generally range from approximately 0.02 to 0.22 moles of CO_2 per mole of AMP+PZ (Morlando, Y. Zhang, et al., 2025). For the present study, the optimum Lean Liquid Mix CO_2 loading was selected as 0.10 mol CO_2 per mol AMP+PZ (Morlando, Y. Zhang, et al., 2025). Although lower Lean Liquid Mix loadings may improve the driving force for CO_2 absorption into the CESAR–1 stream, it also increases the Reboiler duty because a larger quantity of stripping steam is now required to achieve higher solvent regeneration. Further insights of this sensitivity analysis are provided in the *Lean Amine CO_2 Loading Optimization* subsection of the *Results and Discussion* section.

For all process columns, namely the *Absorber Unit*, *Stripper 1*, and *Stripper 2*, a packed-tower configuration was identified as the most suitable design option. Structured packing was selected because it generally provides a high specific surface area, low pressure drop, low liquid holdup, and favourable mass-transfer performance compared with random packing. It also offers relatively simple installation and replacement during maintenance (T. G. Walmsley et al., 2018). These advantages are consistent with the findings of Correa Bernal's thesis, who compared packed-bed and tray columns and reported lower complexity, pressure drop, and liquid holdup for packed systems, which is particularly beneficial for thermally sensitive solvents such as MEA and CESAR–1, together with improved cost-effectiveness (Correa Bernal, 2024).

Sulzer Mellapak 250Y structured packing was selected for the *Absorber Unit*, *Stripper 1 Unit*, *Stripper 2 Unit*, and the *Amine Wash Section* located at the top of *Stripper 1 Unit*, because of its widespread use in amine-based CO_2 capture studies (Park & Øi, 2017). For the Absorber Gas *Amine Wash Tower*, Flexipac 700Y structured packing was used to provide a higher specific surface area and improve the transfer of entrained amine species from the gas phase to the water stream within a comparatively short packed height. A *Rate-based RadFrac* model was used to simulate the packed towers with the *mixed* flow model, *discretized film reactions* for the liquid phase, *consider film* for the vapor phase and reaction condition and transfer condition factors were chosen to be 0.5 each. *Brf-85* correlation was used for the mass transfer coefficient and interfacial area models, *Chilton and Colburn analogy* was used to model the heat transfer coefficient and *Billet-93* was used to simulate holdup correlation (Morlando, Y. Zhang, et al., 2025; Spek et al., 2016).

To avoid column flooding, packed columns are commonly designed to operate at approximately 70% of the flooding capacity, as recommended by Sherwood correlations (*Fortran Programs for Chemical Process Design, Analysis, and Simulation* 1995). Flooding occurs when the upward gas velocity is very high which ultimately results in the gas flow restricting the downward liquid flow, causing excessive liquid accumulation within the packing. To maintain the flooding percentage within the design limit of approximately 80%, the pressure drop along the packed section height was generally maintained within the range of 1.5–5 mbar(a) m^{-1} (Coulson & Richardson, 1979). These hydraulic constraints were considered throughout the column-sizing procedure.

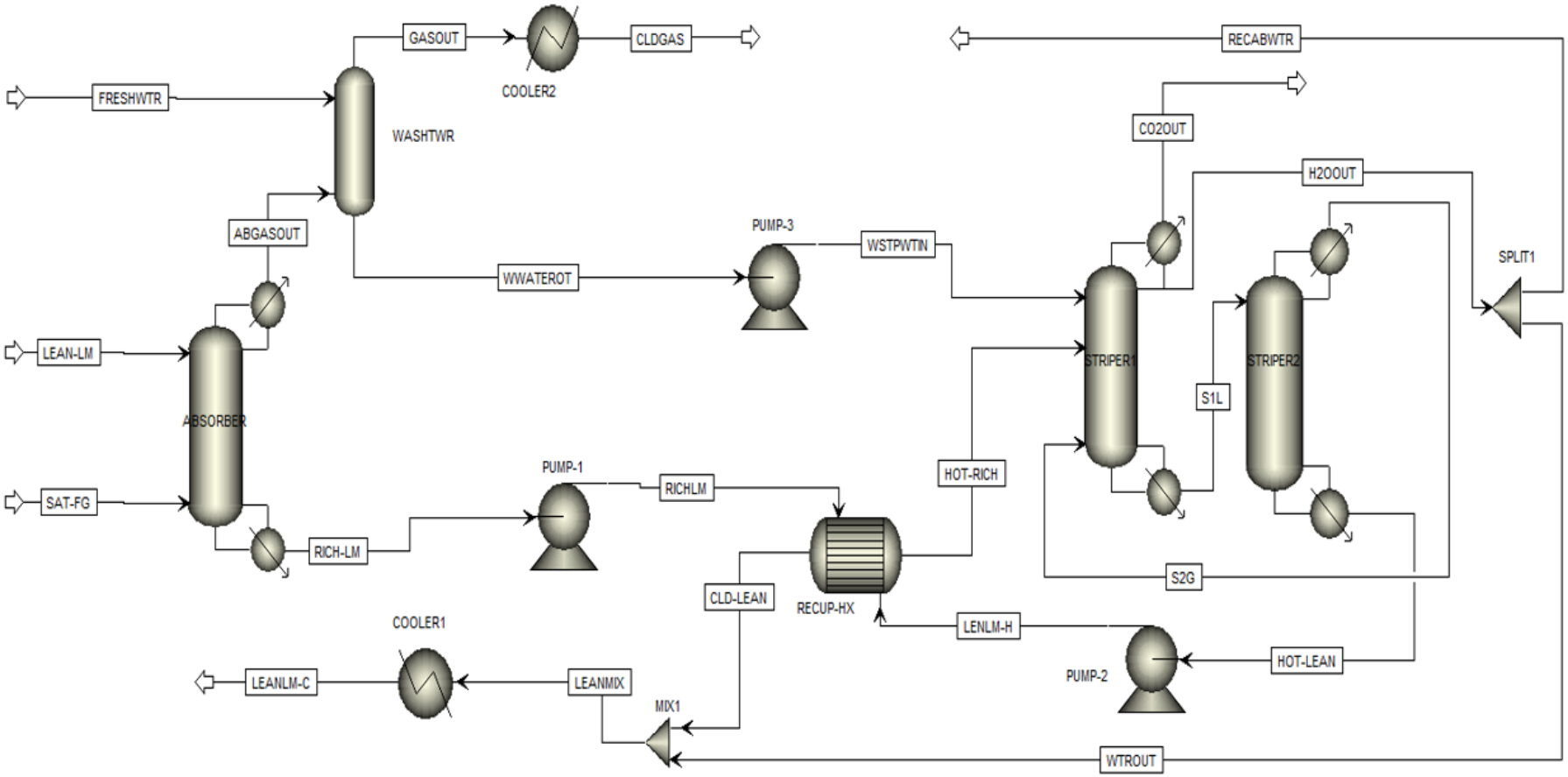


Figure 15: Aspen Plus Process Flow Diagram for Amine Solvent based CO_2 Absorption process in Carbon Capture Container

3.2.1 Design Configuration for Steady-State Point Source CO_2 Capture

This section includes the design of the *Absorber Unit* and *Stripper Units*, followed by the *Absorber Gas Amine Wash Tower* and the *Recuperator Unit*. A common column design was targeted for all operating scenarios, including One Genset and Two Gensets under both Maximum Power Operation (MPO) and Continuous Power Operation (CPO) conditions. This approach was adopted to simplify CCC Unit manufacturing, reduce the number of column-specific spare parts, improve design standardization, and lower the overall system cost while maintaining comparable process performance and operational efficiency. The feasibility of this common design approach also constitutes one of the main research questions of the study.

Other Process Equipments, including the *Recuperator*, Heat Exchangers, Coolers, Pumps, Compressors, and Chiller, were sized according to the operating requirements of the One Genset or Two Gensets configurations. Where appropriate, adjustments were made to the equipment capacity factors to account for the corresponding differences in flow rate, heat duty, and power demand.

3.2.1.1 Absorber Design

The first step in sizing the *Absorber Unit* was to determine the minimum Lean Liquid Mix flow rate, $L_{\min}$. This was estimated by modelling an absorber with an effectively infinite packed height, approximated in the present study as 100 m, such that the solvent requirement approaches its theoretical minimum value. The maximum allowable Absorber internal diameter was constrained to 10 m to avoid unrealistic column dimensions for a full-scale CESAR–1-based CO_2 Capture Plant. Under these assumptions, the minimum Lean Liquid Mix flow rate required to achieve 90% CO_2 Capture Efficiency was calculated as 4685 kg h^{-1} for the Two Gensets Maximum Power Operation mode scenario. This case was intentionally chosen as Two Gensets Maximum Power Operation (MPO) produces the highest amount of exhaust gas, which corresponds to the highest probability for column flooding. After establishing $L_{\min}$, the Absorber diameter was recalculated using the relationship between the pressure drop, the flow parameter F_{LV}, and the modified gas-load factor $K_{4,gas}$, which are defined as follows (Coulson & Richardson, 1979; Perry et al., 1997):

$$F_{LV} = \frac{L_w^*}{V_w^*}\sqrt{\frac{\rho_V}{\rho_L}} \tag{7}$$

$$K_{4,gas} = \frac{13.1\,(V_w^*)\,.\,F_p.\,(\mu_L/\rho_L)^{0.1}}{\rho_V\,.\,(\rho_L - \rho_V)} \tag{8}$$

Here, ρ_L represents the density of the liquid and gas phases, while μ_L denotes the liquid viscosity. L/G is the liquid-to-gas molar flow ratio, and L_w^* and V_w^* represent the liquid and gas mass flow rates per unit column cross-sectional area, respectively (kg/m^2s). The packing factor of the structured packing (Mellapak 250Y in this case) is denoted by F_p, whereas F_{LV} and $K_{4,gas}$ represent the flow parameter and pressure drop parameter, respectively. Furthermore, P denotes the operating pressure, T is the operating temperature, and ΔP_c represents the pressure drop across the packing (Canepa et al., 2013). The calculated flow parameter, F_{LV}, is 0.045, which supports the selection of structured packing. Within the approximate range $0.02 \leq F_{LV} \leq 0.10$, tray columns and random packing generally exhibit similar hydraulic performance, while structured packing can provide approximately 1.5 times higher separation efficiency because of its larger effective interfacial area and lower pressure drop (Perry et al., 1997).

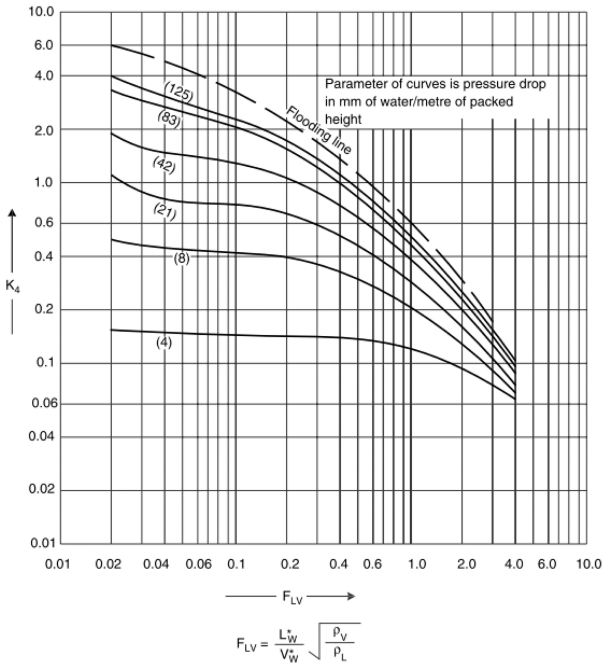


Figure 16: Generalised Pressure Drop Correlation from Sinnott (Sinnott, 2005)

For sizing the *Absorber Unit*, a pressure-drop curve of 42 mm H_2O m^{-1}, equivalent to approximately 4.12 mbar(a) m^{-1}, was selected (Canepa et al., 2013). This value lies within the previously defined acceptable pressure-drop range and provides a suitable basis for column operation. The viscosity and density of the Lean Liquid Mix and Exhaust Gas were obtained from Aspen Plus simulation. Using the calculated F_{LV} value and the selected pressure drop, the modified gas-load factor, $K_{4,gas}$, was determined from Figure 16. The gas mass flux, V_w^*, was then calculated using Equation 8. Combined with the Exhaust Gas mass flow rate entering the absorber, V_w^* was used to determine the required column cross-sectional area and, subsequently, the Absorber diameter which was found to be 0.95 m. The calculated Absorber diameter was subsequently used as the initial guess value to estimate the actual Absorber diameter via iterative simulations in Aspen Plus. Following hydraulic evaluation and adjustments of the column dimensions, the final *Absorber Unit* diameter was determined to be 0.77 m, corresponding to the prescribed value of 80% of the maximum flooding limit.

Following the determination of $L_{\min}$, a sensitivity analysis was performed by varying the Lean Liquid Mix flow rate to identify the optimum packed height required to achieve 90% CO_2 capture. The details of this sensitivity analysis are provided in the *Absorber Column Height Optimization* subsection of the *Results and Discussion* section. After performing the sensitivity analysis, the packed height was of the Absorber Column was found to be 1.95 m.

3.2.1.2 Stripper Unit Design

After determining the packed height of the Absorber Column and fixing the Lean Liquid Mix CO_2 loading, a sensitivity analysis was performed to identify the optimum packed height of the Stripper column required to achieve 90% CO_2 desorption. The details of this sensitivity analysis are provided in the *Stripper Column Height Optimization* subsection of the *Results and Discussion* section. Following the sensitivity analysis, the total packed height required for the Stripper tower was found to be 4 m. Since the *Stripper Unit* operates at an elevated temperature and pressure of approximately 120 °C and 1.84 bar(a), respectively, the required packed height was divided between two separate 2 m units. This configuration allows both *Stripper 1 Unit* and *Stripper 2 Unit* to be installed inside a 45-foot container, which provides an additional protective layer of insulation and reduces heat losses to the surroundings.

Based on the column hydraulic evaluation and subsequent adjustment of the dimensions to satisfy the prescribed maximum flooding limit, the diameters of *Stripper 1 Unit* and *Stripper 2 Unit* were determined to be 0.38 m and 0.55 m, respectively from Aspen Simulation. Analytical estimation of Stripper Diameter

using the same equations 7 and 8 can be difficult because we only have the value of the Rich Liquid Mix CO_2 loading and need to account for the Stripping Vapor flow rate produced by the Reboiler. So, the diameter was adjusted iteratively until the pressure drop and flooding parameters were found within specified ranges. Also, since the Reboiler unit is attached to *Stripper 2 Unit* which therefore handles a higher stripping vapour flow rate, this may explain why *Stripper 2 Unit* is thicker than *Stripper 1 Unit*.

Though this kind of split *Stripper Units* design is new, it is not unconventional. Studies have shown that splitting a single Stripper unit into two Half-Stripper sections with proper vapour connections can work effectively. Research has also shown that a water-saturated warm air-driven secondary Stripping unit inserted into the amine regeneration loop can operate more efficiently and effectively (Thompson et al., 2017). In addition, replacing the conventional Single Stripper tower with a multi-section arrangement operating at different pressures can help reduce operational cost and lower the overall Energy consumption (Oyenekan & Rochelle, 2007; Rochelle & Oyenekan, 2011). The main safety and operational requirement is that the vapour line between the two Stripper units must be short and properly connected to ensure stable operating conditions. The pipes must also be insulated properly to avoid condensation of stripping vapour (steam) and unwanted heat loss. Although some condensation is inevitable, Condensate Drains, Steam Traps, and flow control can be used as mitigation measures. The two Stripper arrangement also provides the advantages of reduced material packing volume requirements and a lower total Stripper volume compared to a single 4 m tall Stripper column having a uniform diameter of 0.55 m.

Another unconventional aspect of this design is the water management as the flowsheet uses of a part of the condensing steam (Stripper Reflux line) from the *Stripper 1 Condenser Unit* as washing water in the *Amine Wash Tower Unit* downstream of the *Absorber Unit*, followed by recycling it back to the *Stripper 1 Unit* through the *Stripper 1 Unit Amine Wash* packed section. This approach is used instead of directly sending the excess water flow back to the *Stripper 1 Unit* with the main Reflux stream for three important purposes: entrained amine recovery from the vapour phase, after-treatment of the Absorber and Stripper outlet gas, and reduction in fresh process-water consumption in the CCC circuit.

3.2.1.3 Recuperator Unit Design

The *Recuperator Unit*, denoted as *RECUP-HX*, is a plate-type heat exchanger used to recover heat from the Hot Lean Liquid Mix leaving the Stripper-section *Reboiler Unit* and transfer it to the cold Rich Liquid Mix leaving the bottom of the *Absorber Unit*. For a clean plate-type heat exchanger, the overall heat-transfer coefficient was assumed to be approximately 950–1000 W m^{-2} K^{-1} (Jensen et al., 2024; Aromada, Eldrup, Normann, et al., 2020). To account conservatively for fouling and operational uncertainties, an over-design allowance of approximately 10–15% was incorporated to calculate the final heat exchanger area and a final heat-transfer coefficient value of 850 W m^{-2} K^{-1} was selected (Aromada, Eldrup & Øi, 2021). A sensitivity analysis, presented in the subsequent chapters, was performed to determine the optimum Heat Transfer area of the *Recuperator Unit*. The details of this sensitivity analysis are provided in the *Sensitivity Study for Recuperator Unit* subsection of the *Results and Discussion* section. Based on the resulting thermal performance and approach-temperature requirements, the optimum overall Heat Transfer area was found to be approximately 65.0 m^2 for Two Gensets CCC design.

Table 17 gives a detailed description of the design and operating parameters of the Carbon Capture unit design. The Liquid-to-Gas (L/G) ratio is taken for the total Exhaust Gas flow (including the inlet CO_2 flow rate) or on a non-inert gas basis.

Table 17: Design and Operating parameters for Single and Two Genset(s) CCC Unit under Normal Temperature Conditions determined in this Work

Parameter	**Single Genset**	**Two Gensets**
Packing Type – Absorber	Mellapak 250Y	Mellapak 250Y
Packing Type – Stripper	Mellapak 250Y	Mellapak 250Y
Void Fraction – Mellapak 250Y	0.987	0.987
Absorber Pressure (bar(a))	1.00	1.00
Stripper Pressure (bar(a))	1.84	1.84
Absorber Internal Diameter (m)	0.77	0.77
Stripper 1 Internal Diameter (m)	0.38	0.38
Stripper 2 Internal Diameter (m)	0.55	0.55
Packed Absorber Height (m)	1.95	1.95
Packed Stripper 1 Height (m)	2.00	2.00
Packed Stripper 2 Height (m)	2.00	2.00
Net Absorber Volume (m^3)	0.896	0.896
Net Stripper Volume (Stripper 1 + 2) (m^3)	0.693	0.693
Absorber Height-to-Diameter Ratio (H/D)	2.532	2.532
Stripper 1 Height-to-Diameter Ratio (H/D)	5.263	5.263
Stripper 2 Height-to-Diameter Ratio (H/D)	3.636	3.636
Recuperator unit Heat Transfer Coefficient (W/m^2K)	850	850
Recuperator unit Heat Transfer Area (m^2)	31.0	65.0
Flue Gas (FG) Flow rate (kg/hr)	1928.58 (MPO) 1507.82 (CPO)	3857.17 (MPO) 3015.64 (CPO)
Flue Gas (FG) Inlet Temperature (°C)	40	40
Lean Liquid Mix (LLM) loading (mol CO_2/mol AMP+PZ)	0.10	0.10
Lean Liquid Mix (LLM) loading (% CO_2/AMP+PZ)	10.00	10.00
Lean Liquid Mix (LLM) Flow rate (kg/hr)	2655 (MPO) 1785 (CPO)	5750 (MPO) 3785 (CPO)
Lean Liquid Mix (LLM) Inlet Temperature (°C)	40	40
Absorber L/G Ratio (kg LLM/kg FG)	1.38 (MPO) 1.18 (CPO)	1.49 (MPO) 1.26 (CPO)
Absorber L/G Ratio (mol LLM/mol FG)	1.52 (MPO) 1.30 (CPO)	1.64 (MPO) 1.38 (CPO)
Packing type – Amine Wash Tower	Flexipac 700Y	Flexipac 700Y
Packing type – Stripper Amine Wash Section	Mellapak 250Y	Mellapak 250Y
Amine Wash Tower Height (m)	3 × 0.30	3 × 0.30
Amine Wash Tower Internal Diameter (m)	0.77	0.77
Void Fraction – Flexipac 700Y	0.920	0.920
Stripper 1 Amine Wash Section Height (m)	0.20	0.20
Stripper 1 Amine Wash Section Internal Diameter (m)	0.38	0.38
Net Amine Wash Tower Volume (m^3)	0.385	0.385
Amine Wash Tower Height-to-Diameter Ratio (H/D)	1.169	1.169
Net Stripper 1 Amine Wash Section Volume (m^3)	0.007	0.007
Stripper 1 Amine Wash Section Height-to-Diameter Ratio (H/D)	0.526	0.526
Amine Wash Tower Fresh Wash Water Flow rate (kg/hr)	23 (MPO) 16 (CPO)	46 (MPO) 33 (CPO)
Fresh Wash Water Inlet Temperature (°C)	40	40

3.2.2 Simulation Scope and Model Assumptions

To simplify the process model, several process units were placed outside the defined system boundary and were therefore excluded from the Aspen Plus simulation. These units were modelled separately using Microsoft Excel or MATLAB 2025a. The excluded equipment included the Type II Absorption Heat Pump, the Exhaust Flue Gas-to-Pressurized Process Water Heat Exchanger, the Charge Air-to-Engine Coolant Heat Exchanger, Exhaust Flue Gas-to-Engine Coolant Heat Exchanger, Expansion Valves, Storage Tanks, and the CO_2 Chiller. The multistage CO_2 Compressor and Intercooler system was also modelled separately.

The Lean Liquid Mix circuit was not implemented as a closed recycle loop in Aspen Plus. Instead, the inlet and outlet Lean Liquid Mix streams were specified to have identical compositions and operating conditions to mimic a closed loop system. This was done to ensure numerical convergence of the overall mass and energy balances. Minimal pressure drops were assumed across simple process equipment, such as Coolers and Heat Exchangers, while pressure losses in pipelines were neglected because the piping network was not explicitly modelled.

All simulations were performed under steady-state conditions and represented continuous point-source CO_2 capture. The columns and Heat Exchangers were assumed to operate adiabatically, with no heat loss to the surroundings. The minimum temperature (pinch point) approach was maintained at 5 °C in liquid–liquid Heat Exchangers and at 10 °C in gas–liquid Heat Exchangers. In the *Recuperator Unit*, the minimum approach temperature, $\Delta T_{\min}$, was maintained at 5.3 °C.

The engine exhaust gas stream was assumed to contain only N_2, H_2O, O_2, and CO_2. Trace impurities, including SO_x, NO_x, particulate matter, hydrocarbons, and other combustion products, were therefore neglected. One of the primary focus of this study was to investigate the moderate–temperature and ambient–pressure desorption of CO_2 inside the Stripper column. Hence, the maximum desorber temperature was limited to 120 °C, representing a typical upper operating range for the regeneration of aqueous amine solvents.

Solvent degradation was not included in the model; consequently, solvent make-up, reclaiming, and degradation-product removal units were excluded. No mass losses, fugitive emissions, or process leaks were considered. Furthermore, long-term effects associated with corrosion, thermal degradation, mechanical vibration, fouling, erosion, and equipment ageing were outside the scope of the simulation. Aspen Plus represents steady-state process behaviour rather than long-term plant operation; therefore, the model does not account for cumulative changes that may occur over extended operating periods. These assumptions should be considered when interpreting the predicted process performance and techno-economic results.

3.3 Heat Pump Selection and Waste Heat Integration

The selection of a suitable heat pump is a crucial step in any engineering process involving waste-heat upgrading and energy integration. In the present study, it is particularly important to ensure efficient recovery of waste heat from the Genset Engine Coolant, Exhaust Gas, and Charge Air Intercooler, so that sufficient thermal energy can be supplied to the hot side of the Reboiler unit for CO_2 stripping.

3.3.1 Waste Heat Availability Study and Quality Analysis

The Waste Heat data obtained from Scania as provided by our Industrial Supervisor, Antonius Kies, are shown in Table 18 and Table 19. The relevant thermophysical properties of the Exhaust Gas and Charge Air components were also cross-checked using the NIST Chemistry WebBook and, where applicable, the NIST REFPROP database (National Institute of Standards and Technology (NIST), 2026).

Table 18: Data Table for Waste Heat as Reboiler heat supply at CCC Design Point: 1500 rpm and 437 kW-mech (from Scania)

T_{Reboiler} **Hot side (°C)**	**Heat Duty (kW-th)**				**Specific Heat Duty 90% CO_2 Capture (MJ/kg CO_2)**			
	$\mathbf{Q}_{\text{Exhaust(E)}}$	$\mathbf{Q}_{\text{ChargeAirCooler(CAC)}}$	$\mathbf{Q}_{\text{Coolant(C)}}$	$\mathbf{Q}_{\text{E+CAC+C}}$	$\mathbf{Q}_{\text{Exhaust(E)}}$	$\mathbf{Q}_{\text{ChargeAirCooler(CAC)}}$	$\mathbf{Q}_{\text{Coolant(C)}}$	$\mathbf{Q}_{\text{E+CAC+C}}$
90	214	58	96	368	2.96	0.80	1.34	5.10
100	208	52	89	350	2.88	0.72	1.24	4.80
110	202	47	82	331	2.80	0.65	1.14	4.60
120	196	41	75	313	2.72	0.57	1.04	4.30
130	190	36	68	294	2.64	0.50	0.94	4.10

Table 19: Data Table for Waste Heat as Reboiler heat supply at CCC Operation Point: 1500 rpm and 312 kW-mech (from Scania)

T_{Reboiler} **Hot side (°C)**	**Heat Duty (kW-th)**				**Specific Heat Duty 90% CO_2 Capture (MJ/kg CO_2)**			
	$\mathbf{Q}_{\text{Exhaust(E)}}$	$\mathbf{Q}_{\text{ChargeAirCooler(CAC)}}$	$\mathbf{Q}_{\text{Coolant(C)}}$	$\mathbf{Q}_{\text{E+CAC+C}}$	$\mathbf{Q}_{\text{Exhaust(E)}}$	$\mathbf{Q}_{\text{ChargeAirCooler(CAC)}}$	$\mathbf{Q}_{\text{Coolant(C)}}$	$\mathbf{Q}_{\text{E+CAC+C}}$
90	152	24	73	249	3.07	0.48	1.48	5.00
100	148	20	67	234	2.98	0.39	1.36	4.70
110	143	15	61	219	2.89	0.31	1.23	4.40
120	138	11	55	205	2.80	0.22	1.11	4.10
130	134	7	49	190	2.71	0.14	0.99	3.80

Although the available Waste Heat may appear sufficient at 130 °C to meet the Reboiler unit thermal energy demand for CESAR-1 System (Morlando, Y. Zhang, et al., 2025), a conservative design approach is still essential because Aspen Plus simulation results are idealized which always considers perfectly insulated and steady state system and do not include real-plant heat losses, fouling, or off-design effects. In preliminary design to build process, it is therefore reasonable to have a conservative margin of roughly 15–20% to reduce the risk of underestimating actual duty requirements and to absorb design uncertainty which might affect the overall energy balance of the Capture plant (Agbonghae et al., 2014).

In real practice, the Engine Coolant cannot be assumed to deliver Waste Heat at 130 °C directly, since the maximum Coolant Outlet temperature from the engine side is typically closer to 100–110 °C, with only a 2–5 K allowable temperature drop for the coolant loop across the Genset Engine without engine modification. This makes direct Waste Heat supply to the Reboiler less feasible and creates a trade-off between losing a substantial fraction of usable waste heat from the Coolant circuit and upgrading the available Waste Heat from the Coolant and Exhaust Gases from 130–110 °C through a suitable Heat Pump. Waste heat recovery from the Charge Air Intercooler is also difficult than it may initially appear on paper because it may require additional Genset Intercooler unit modifications. For these reasons, Waste Heat upgradation through a Heat Pump remains the desirable option to reduce opportunity losses and improve the overall feasibility of Energy Integration into the CCC System.

3.3.2 Design Selection – Compression Heat Pump versus Absorption Heat Pumps

Compression Heat Pumps are the most mature and widely available option for low temperature heat upgrading, but they are also electricity-intensive because they require an external driving mechanical

or electrical power rather than using waste heat as the primary energy input. Since the Coefficient of Performance (COP) is defined as useful heat output divided by electrical power input, a COP of 4 corresponds to an electricity demand of about 0.25 kW_{el} per 1 kW of upgraded Waste Heat, which is still a significant electricity demand for a stand-alone Genset-based Carbon Capture system handling roughly 70–200 kW of recoverable waste heat depending on the mode of operation and number of Genset units. Compression Heat Pump systems also rely on rotating components that introduce additional efficiency drop due to frictional losses, maintenance needs, and sensitivity to rotational speed matching and pressure ratio. High Temperature Heat Pump compressors often require high rotational speeds and careful design, which can add complexity, cost, and failure risk. Driving the Compressor shaft by connecting it to the Genset Engine shaft is also not a feasible option due to the large difference in rotational speed of the two shafts. The rotational speed of the Compressor is at least around *15000 rpm*, which is almost 10 times the rotational speed of of the Genset (*1500 rpm*), so the use of a gearbox is essential, which has its own efficiency losses and failure issues due to the large difference in rotational speeds. For these reasons, the compression option was not preferred for the present CCC application.

On the other hand, Absorption heat pumps are thermally driven heat-upgrading systems and are therefore much better aligned with waste-heat integration. It is especially useful where low-grade heat is available in abundance, because part of the input heat is recovered and upgraded rather than discarded. In particular, a Type II Absorption Heat Pump, also known as an Absorption Heat Transformer, can upgrade an intermediate-temperature waste-heat stream to a higher useful temperature using only a negligible amount of electrical power (less than 1% of the total upgraded Waste Heat output) in the solvent pumps. Absorption Heat Transformer intakes a part of the intermediate-temperature waste-heat stream at the Evaporator unit and upgrades it to a temperature above that of the original heat source and releases it at the Absorber unit. The other part of the intermediate-temperature waste-heat stream is absorbed at the Generator unit and given out at the Sink or Condenser Unit. A Type I Absorption Heat Pump can also be useful as they have a higher COP (more than 1.0) compared to Type II. But Type I Absorption Heat Pumps generally requires a high temperature driving waste heat source at the Generator unit and does not suit the present study conditions to lift available low grade Waste Heat to the Reboiler's Hot Side without the availabiltiy of any high grade Waste Heat and limited electrical power supply. After considering all these factors, a Type II Absorption Heat Pump or an Absorption Heat Transformer was selected as the Heat Pump for Waste Heat upgrading.

3.3.3 Absorption Heat Pump Type II Design Selection and Analysis

The main operating principle of a Type II Absorption Heat Pump, or Absorption Heat Transformer, is that it upgrades a part of the Waste Heat at the Evaporator–Absorber system while utilizing another part at the Generator–Condenser system to sustain the cycle. The ratio between the upgraded Waste Heat and the supplied Waste Heat depends primarily on the coefficient of performance (COP) of the Heat Pump, which is typically less than 1.00 for a Type II system. The COP value depends on the temperature lift required, the temperatures of the low-grade and upgraded Waste Heat streams, the Sink temperature, and other operating conditions. The formula for the COP of an Absorption Heat Transformer is given as follows by equation 9. For design considerations, a pinch temperature of 5 K is maintained across all the counter-current heat exchangers of the Absorption Heat Transformer.

$$COP_{AHT,\max} = \frac{T_2\,(T_1 - T_0)}{T_1\,(T_2 - T_0)} \tag{9}$$

Here, $COP_{AHT,\max}$ represents the maximum theoretical COP of the Absorption Heat Transformer. T_0 is the absolute temperature of the Heat Sink (cooling medium), T_1 is the absolute temperature of the low-grade Waste Heat source supplied to the Generator or Evaporator unit, and T_2 is the absolute temperature of the upgraded useful heat delivered by the Absorber unit. All temperatures are expressed in Kelvin (K). The actual efficiency for an Absorption Heat Transformer operating with 75–110 °C driving Waste Heat and delivering 110–145 °C upgraded Waste Heat is around 65% of the $COP_{AHT,\max}$ used for this study (Ayou et al., 2025).

As evident from the COP Expression, equation 9, the lower the Source temperature is kept while maintaining a constant Output temperature, the lower the COP becomes. However, a lower source temperature also means that more waste heat can be extracted from the Coolant stream, Exhaust Gas stream,

and Charge Air Intercooler, which gives rise to a classical Optimization Problem. Therefore, an Optimization study for Single Genset Continuous Power Operation (CPO) mode was conducted to identify the most suitable Operating Point of the Type II Absorption Heat Pump. Since the Coolant waste heat is available at around 100 °C, and an additional small portion of Waste Heat from the Charge Air Intercooler and Exhaust Flue Gases, as shown in Figure 18, is also available together with the Coolant Waste Heat, the COP equation is used to estimate the upgraded waste-heat duty.

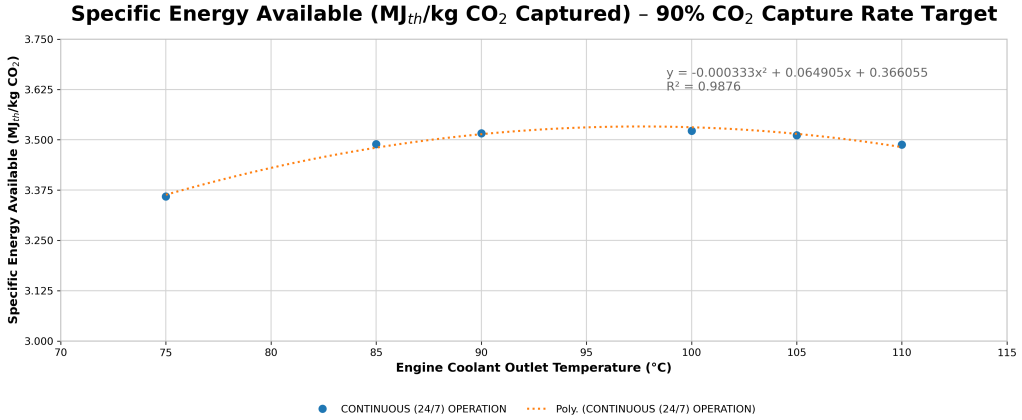


Figure 17: Total Available Thermal Energy Optimization Study at different Coolant Outlet Temperatures for Single Genset Continuous Power Operation (CPO) Mode

Figure 17 gives the total upgraded Specific Waste Heat available per kg of CO_2 for 90% CO_2 Capture Target. This includes the Waste Heat from Coolant, Hot Exhaust Gas and Charge Air Intercooler from the Heat Intergration Circuit (Figure 18) that includes both the Absorption Heat Transfromer and Process-Process Heat Exchanger Network. As evident from Figure 17, the Outlet temperature of Coolant is chosen as 100 °C which also agrees with the maximum recommended Coolant Outlet temperature from the Genset Engine outlet.

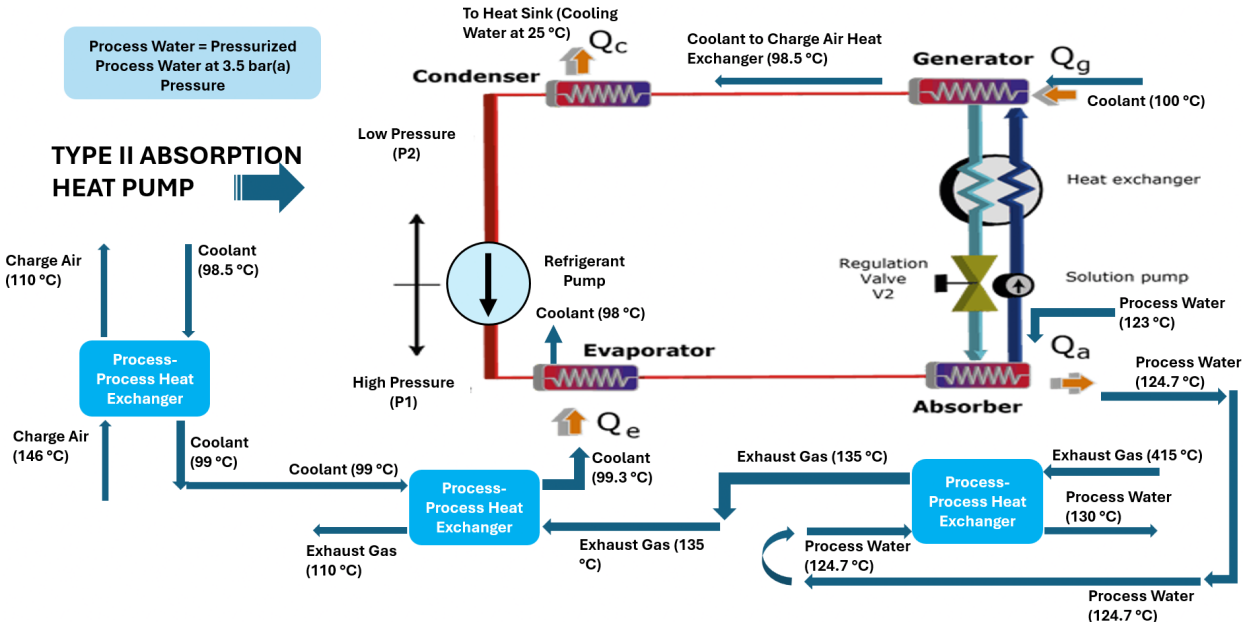


Figure 18: Heat Integration Circuit Flowsheet depicting Waste Heat Transfer and Upgradation via Absorption Heat Transformer for the CCC Unit

Figure 18 helps to visualize and better understand the Waste Heat upgradation and Table 20 provides the

values of the different parameters for low-grade Thermal Energy upgradation for Single Genset Continuous Power Operation (CPO) Mode via Absorption Heat Transformer at 100 °C Coolant Genset Outlet Temperature. The above equipment setup is chosen for the Heat Integration Circuit design because it ensures that the heat load on the *Evaporator* side of the *Absorption Heat Transformer Unit* is always minimum and the high grade Waste Heat from the hot Engine Exhaust Gases are transferred to Pressurized Process Water at 3.5 bar(a) and 124.7 °C from the *Absorber Unit* via *Exhaust-to-WaterProcess–Process Heat Exchanger* before the Exhaust Gas at 135 °C continues to the *Exhaust-to-Coolant Process–Process Heat Exchanger*.

Table 20: Absorption Heat Transformer Waste Heat Upgradation and Heat Integration Data

Parameter	**Value**	**Unit**
Genset Coolant Entry Temperature	98	°C
Genset Coolant Exit Temperature	100	°C
Thermal Power from Genset Coolant	67.0	kW-th
Coolant Mass Flow rate	32289	kg/hr
Specific Heat Capacity of the Coolant	3.735	kJ/kg K
Initial Temperature of Exhaust Gas	415	°C
Intermediate Temperature of Exhaust Gas	135	°C
Final Temperature of Exhaust Gas	110	°C
Thermal Power from Exhaust Gas to Water	131.75	kW-th
Thermal Power from Exhaust Gas to Coolant	11.0	kW-th
Initial Temperature of Charge Air	146	°C
Final Temperature of Charge Air	110	°C
Thermal Power from Charge Air tp Coolant	16.0	kW-th
Heat Pump Evaporator Heat Intake	94.0	kW-th
COP_{Ideal} (or $COP_{\text{AHT,max}}$)	0.703	–
COP_{Real} (65% of $COP_{\text{AHT,max}}$)	0.456	–
Evaporator Temperature	95	°C
Heat Pump Condenser Power	51.23	kW-th
Upgraded Heat at the Absorber Unit	43.02	kW-th
Pressurized Process Water Mass Flow rate	21400	kg/hr
Initial Temperature of Pressurized Process Water	123	°C
Final Temperature of Pressurized Process Water	130	°C
Specific Heat Capacity of Water	4.20	kJ/kg K

3.4 Compressor Train Design

A Multi-stage Compressor or *Compressor Train* is an important component of post-capture CO_2 processing and conditioning prior to CO_2 transportation. In this study, the Multi-stage Compressor was used to process the 90% captured CO_2 from the Carbon Capture unit. The CO_2 entering the *Compressor Train* at the Stripper section *Condenser Unit* outlet conditions (40 °C and 1.84 bar(a)) is processed to 40 °C and 25 bar(a), considering a maximum pressure drop of 5 bar(a) across the CO_2 Refrigeration or Chiller unit. The compressed CO_2 gas will then be sent to the Chiller unit to be processed as Liquid CO_2 at -25 °C and 20 bar(a).

To determine the optimum number of Compressor and Intercooler Units in the *Compressor Train*, a

sensitivity analysis was performed. The details of this sensitivity analysis are provided in the *Sensitivity Study for Compressor Train* subsection of the *Results and Discussion* section. Following the sensitivity analysis, the optimum number of Compressor and Intercooler Units in the *Compressor Train* was found to be 3.

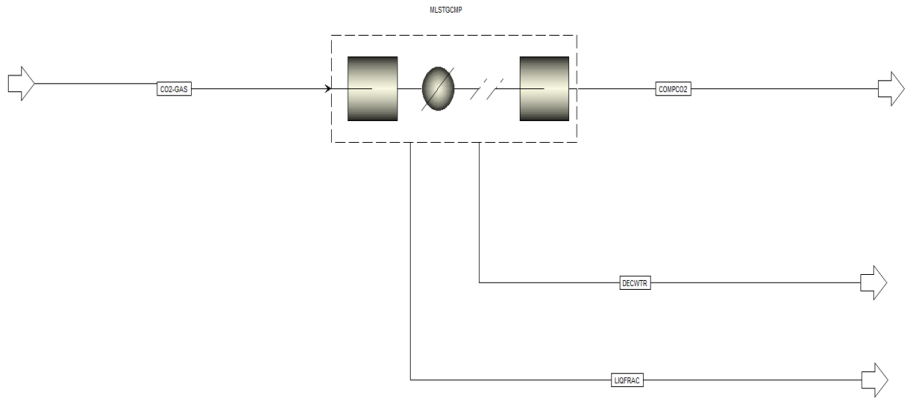


Figure 19: Aspen Plus Process Flow Diagram for Compressor Train Unit

3.5 Heat Exchanger Selection Suggestions

Heat exchangers are an integral part of any chemical-engineering process flowsheet, especially in systems involving heat transfer and thermal-energy integration. Based on general recommendations from the literature, a Kettle-Type Reboiler is selected for the *Reboiler Unit* because of its high heat-transfer coefficient and its ability to accommodate phase change effectively (Aromada, Eldrup, Normann, et al., 2020; Amine Best Practices Group (ABPG), 2003). In this setup, hot pressurized water at 130 °C and 3.5 bar(a) was supplied as the heating medium on the tube side, while the Lean Liquid Mix flows through the shell side to enable boiling and steam generation. For the *Condenser Unit*, a Fixed-Tube Shell-and-Tube heat exchanger with a single pass on both the shell and tube sides can be used (Aromada, Eldrup, Normann, et al., 2020; Amine Best Practices Group (ABPG), 2003). The hot vapour stream consisting of Steam and CO_2 is passed through the shell side, while cooling water flows through the tube side to promote partial condensation of water on the shell side. For *Exhaust Gas Cooler* and heat exchange between hot Exhaust Gas or Charge Air and Pressurized Process Water or Coolant, a Duct-Mounted Tubular Flue-Gas Condenser or Condensing Economizer can be employed (Aromada, Eldrup, Normann, et al., 2020; Qian et al., 2021). In this arrangement, the hot Exhaust Gas or Charge Air flows through the shell side, while Coolant, Cooling Water, or Pressurized Process Water passes through the tube side, allowing higher heat-transfer coefficients due to gas-side turbulence and facilitating partial condensation of water from the Exhaust Gas for *Exhaust Gas Cooler*. For Lean Liquid Mix stream Coolers, a U-Tube Shell-and-Tube heat exchanger can be used, with Lean Liquid Mix stream on the tube side and Cooling Water on the shell side, to reduce the risk of amine-related corrosion and deposition caused by local hot and cold spots on the shell side (Amine Best Practices Group (ABPG), 2003). Finally, for cooling the Cooling Water stream with Ambient Air, a Fin-Type heat exchanger can be considered, as it provides a large surface area and a high heat-transfer coefficient for effective cooling.

4 Economic Assessment

The Economic Assessment is an important part of any Techno-Economic Ananlysis of a project. For a Carbon Capture process, the economic analysis helps to determine the levelized cost of CO_2 capture, usually expressed in €/tonne CO_2 or \$/tonne CO_2, based on the Total Capital Investment (TCI), Total Production Cost (TPC) or total Operational Expenditure (OPEX), and Total Revenue (TR), if any, generated over the project lifetime. In the present study, the different process elements and equipment items are considered individually, and their capital costs are estimated using suitable cost correlations and scientific references. These equipment-level costs are then combined with appropriate economic factors to build up the overall cost structure of the Capture system. On this basis, the Capital Expenditure (CAPEX) and Operational Expenditure (OPEX) of the Carbon Capture Container (CCC) unit are calculated and presented. This approach provides a structured economic framework for comparing the feasibility of the proposed CCC design under different operating scenarios.

As an important consideration, it should be kept in mind that in this study, the CAPEX is always calculated for the equipment at the CCS design point, i.e., under Single or Two Genset Maximum Power Operation (MPO) conditions, whereas the OPEX is always calculated for the CCC system at the operating point, i.e., under Single or Two Genset Continuous Power Operation (CPO) conditions.

4.1 Capital Expenditure (CAPEX)

The costs of the main process equipment, initial chemical inventory, process water, cooling water, and other relevant units were considered in the economic assessment. The Genset cost was excluded, along with the cost of the empty 45-foot container, because the Genset is to be provided by Scania and is not inherently part of the capture process economics.

The total capital investment (TCI) consists of the fixed capital investment (FCI), Contingencies and the working capital (WC). The FCI includes both direct and indirect costs. Direct costs primarily account for equipment purchase, delivery, installation, and associated auxiliary items, whereas indirect costs cover engineering and construction overhead and other plant-related expenditures. The proportions of direct costs, indirect costs, and working capital contributing to the TCI for a typical chemical plant are outlined in Table 22 later in this section. The TCI was estimated using standard cost correlations and factors commonly applied to large-scale process plants, as reported in the literature.

The total delivered cost of the major equipment was estimated using the four main economic correlations. They are the Power–Law Scaling correlation (Eq. 10), the Quadratic Logarithmic Cost Correlation (or Log—Quadratic Cost Correlation) (Eq. 11), the Polynomial Cost Correlation (Eq. 12) and the Power Function Cost Correlation (or Nonlinear Power Cost Correlation) (Eq. 13). The equations and the variables used to represent different economic and capacity parameters are elaborated as follows.

$$C_p^0 = C_p^1 \left(\frac{A^0}{A^1} \right)^n \tag{10}$$

$$\log_{10} C_p^0 = K_1 + K_2 \log_{10}(A^0) + K_3 \left[\log_{10}(A^0) \right]^2 \tag{11}$$

$$C_p^0 = \log_{10}(A^0) + K_4 (A^0)^2 + K_5 (A^0) + K_6 \tag{12}$$

$$C_p^0 = K_7 + K_8 (A^0)^m \tag{13}$$

Here, C_p^0 represents the manufacturer's cost of the required equipment (required equipment cost), while C_p^1 denotes the manufacturer's cost of the given equipment from literature data (given equipment cost). The parameter A^0 is the size parameter or cost attribute of the required equipment, whereas A^1 represents the corresponding size parameter or cost attribute of the given equipment. The constants K_1 – K_8 are empirical coefficients that depend on the type of equipment and the selected cost correlation. The exponents n and m are the equipment-specific cost exponents used to account for economies of scale in

equipment cost estimation. Table 21 summarizes the cost exponent (n) or capacity scaling factor values for common process equipments obtained from the literature (Turton et al., 2012).

Table 21: Parameters to estimate the Capacity Scaling Factor of Major Process Equipments (Turton et al., 2012)

Equipment Type	**Capacity Range**	**Units of Capacity**	**Cost Exponent (n)**
Reciprocating Compressor with Motor Drive	0.75 – 1490	kW	0.84
Shell-and-Tube Heat Exchanger (Carbon Steel)	1.9 – 1860	m^2	0.59
Vertical Tank (Carbon Steel)	0.4 – 76	m^3	0.30
Centrifugal Blower	0.24 – 71	std m^3/s	0.60
Jacketed Kettle (Glass Lined)	0.2 – 3.8	m^3	0.48

Another factor that may increase the equipment cost is the thickness of the vessel depending on the operating pressure. The American Society of Mechanical Engineers (ASME) Code for Pressure Vessel Design is given by Equation 14 as follows (Turton et al., 2012). Here the vessel diameter (D) is in meters and Operating Pressure (P) is in bar(g).

$$F_{P,\text{vessel}} = \frac{\dfrac{(P+1) \times D}{2 \times [850 - 0.6(P+1)]} + 0.00315}{0.0063} \tag{14}$$

In this study the operating pressure range for most of the process units (towers, vessels, pumps, and heat exchangers) except for Compressor Train, Chiller or Absorption Heat Transformer equipments (to be purchased as pre-designed modules and packs) is around 1.00–1.84 bar(a) or 0.00–0.83 bar(g). Hence from Equation 14, the Pressure Factor for the vessel ($F_{P,\text{vessel}}$) is calculated to be less than 1.0 so $F_{P,\text{vessel}}$ is taken as 1.0 and the vessel thickness (t_{vessel}) is considered to be 0.0063 m. Accordingly, the relevant Capacity and Scaling-Factor Correlation constants were used as recommended in literature (Turton et al., 2012).

Equipment cost reported in literature can be an important source of information for cost calculation. But because the cost data in the literature were not reported consistently, a normalization step was necessary. Some sources reported purchased equipment cost (PEC), while others reported the total direct cost (TDC). In the latter case, the TDC values were converted to PEC using the standard relationship:

$$TDC = 2.202 \times PEC \tag{15}$$

This is as per the recommended values in chemical process design literature (Guthrie, 1974). The resulting PEC values were then scaled using the correlation above, after which the total capital investment (TCI) was calculated using the appropriate installation and indirect cost factors.

The cost-inflation index used to update literature-reported equipment costs to current costs was calculated using the standard Chemical Engineering Plant Cost Index (CEPCI) for the relevant year and then converted to €$_{2024}$ using the equation given below (Eq. 16).

$$C_{2024}^{Euro} = C_{\text{lit,year}}^{\$} \left(\frac{\text{CEPCI}_{2024}}{\text{CEPCI}_{\text{lit,year}}} \right) \left(\frac{Euro}{\$} \right)_{2024} \tag{16}$$

Here, $C_{\text{lit,year}}$ is the literature cost (in \$), $\text{CEPCI}_{\text{lit,year}}$ is the original year's index value for the reference cost, and CEPCI_{2024} is the index value for year 2024.

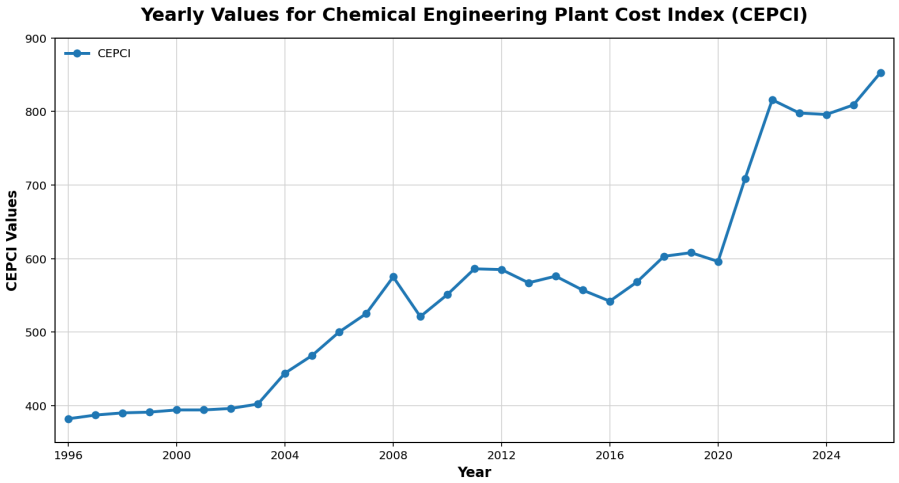


Figure 20: Historical CEPCI Index Value (Yearly) (Chemical Engineering, 2026)

Table 22 summarizes the commonly used values and cost-scaling factors for estimating the Total Capital Investment (TCI) from the Purchased Equipment Cost (PEC). Although these factors are suitable for a typical large chemical plant with 15–20 years lifetime which is first-of-its-kind (FOAK), the final factor value of 400.4% is likely too high for a small containerized Carbon Capture system, particularly one that may be produced in batches of 20–200 units per year rather than as a single large stationary CCUS installation. For small-scale capture systems, the capital cost is more strongly affected by modularity, conditioning equipment, and shared infrastructure. Recent techno-economic studies suggests that the initial capital investment is highly sensitive to scale effects and unit production quantity with Cost Factor in between 1.47–1.93 for small scale Carbon Capture systems (IEAGHG, 2024; Sorrels et al., 2018). Also, a recent study for a small-scale Waste Biomass gasification plant to be used for power production uses a TCI factor of 1.67 to estimate the TCI from PEC (Safarian et al., 2020). Based on this reasoning, a lower mid-range value of 1.75 was chosen here as the TCI factor for scaling PEC to TCI.

Table 22: Factors for estimation of Total Capital Investment (TCI) Cost as a multiple of the Purchased Equipment Cost (PEC)

Components	**Factor (%) (Guthrie, 1974; Peters et al., 2003)**	**Factor (%) (This Work)**
Direct Costs:		
Purchased Equipment Cost	100.0	–
Installation Cost, including Insulation and Painting	4.0	–
Instrumentation and Controls (Installed)	7.3	–
Piping (Installed)	32.0	–
Electrical Systems (Installed)	8.3	–
Concrete and Steel	10.6	–
Labour	58.0	–
Total Direct Plant Cost	**220.2**	–
Indirect Costs:		
Engineering, Supervision and Construction	74.9	–
Contingency and Fees	53.1	–
Total Indirect Plant Cost	**128.0**	–
Fixed Capital Investment (FCI) = Direct Costs + Indirect Costs	**348.2**	–
Working Capital Investment (WCI) (15% of FCI)	**52.2**	–
Total Capital Investment (TCI)	**400.4**	**175.0**

4.2 Operational Expenditure (OPEX)

The operational expenditure (OPEX) includes all recurring costs associated with operating and maintaining the system, including labor, electricity and other essential utilities, maintenance, solvent replacement and refill, operating overhead, property taxes and insurance, and other consumables.

To ensure consistency, all operational costs were estimated following common factors and correlations

obtained from literature. These are standard factors and correlations and are widely used for any typical chemical process. Adjustments were also applied to these factors to meet CCC process specific requirements and operating conditions.

4.2.1 Carbon Capture Container OPEX

Utility consumption constitutes a significant component of the operating expenditure (OPEX) of the proposed CCC system. The major utility requirements arise from the electricity demand of the CO_2 Post Capture Processing Unit, including the Multi-stage CO_2 Compression system, and the Refrigeration Cycle employed for CO_2 Liquefaction prior to transport. Among these, the CO_2 Compressor Train represents the largest electricity consumer due to the substantial electrical energy required for CO_2 compression to elevated pressures, followed by CO_2 Refrigeration, and Liquid Mix and Process Water Pumps. The electricity consumption of auxiliary equipment, such as pumps and control systems, was also included where applicable. The electricity demand of the CCC Unit was assumed to be fulfilled by burning extra fossil diesel in the Genset unit(s), and hence the amount and price of electricity were calculated based on price values provided by Scania AB to account for the electrical energy demand of the CCS system. The thermal energy conversion factor or Lower Heating Value (LHV) for fossil diesel is 42.80 MJ_{th} kg_{Diesel}^{-1}, and 1 DLE_{Diesel} is equivalent to 0.84 kg_{Diesel}. Given an average fuel-to-mechanical efficiency of 45.70% and an average mechanical-to-electrical efficiency of 93.50%, the overall thermal-to-electrical efficiency is 42.73%, yielding a total electricity output of 4.27 kWh_{elec} DLE_{Diesel}^{-1} (or 5.08 kWh_{elec} kg_{Diesel}^{-1})

In contrast, makeup materials such as Cooling Water, Process Water, and Amine Chemicals (AMP and PZ) contribute to a smaller share of the overall OPEX, but their cost was also accounted for where applicable. The unit costs of feedstock and utilities and the cost percentages (of *Total Capital Investment (TCI)*) for other overheads used in the Economic Analysis are summarized in Table 23.

Table 23: List of Operating Cost Assumptions for Carbon Capture Container Unit

Parameter	Value	Unit	Reference
Diesel	0.851	€$_{2024}$/DLE	(European Commission, 2026)
Electricity	0.216	€$_{2024}$/kWh	(Scania AB, 2024)
AMP	10.42	€$_{2024}$/kg	(Manzolini et al., 2015)
PZ	7.81	€$_{2024}$/kg	(Manzolini et al., 2015)
Process Water	7.81	€$_{2024}$/m^3	(Manzolini et al., 2015)
Cooling Water	0.34	€$_{2024}$/m^3	(Nazir et al., 2020)
Discount Rate	8.00	%	(International Energy Agency, 2020)
Plant Working Hours	2000	hours/year	–
Plant Lifetime	12	years	–
Maintenance and Repair	8.00	%TCI	(Peters et al., 2003; Turton et al., 2012)
Operating Supplies	0.90	%TCI	(Peters et al., 2003; Turton et al., 2012)
Local Taxes and Insurance	3.20	%TCI	(Peters et al., 2003; Turton et al., 2012)

Finally, the specific CO_2 abatement cost was calculated using the following equations.

$$\text{Annualized CAPEX} = \text{Total CAPEX} \times \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right) \tag{17}$$

$$C_{\text{CO}_2,\text{Capture}}^{Euros,2024} = \frac{\text{Annualized CAPEX} + \text{Total Annual OPEX}}{m_{\text{CO}_2,\text{Captured}}} \tag{18}$$

The total Capital Expenditure (CAPEX) (equal to the *Total Capital Investment (TCI)*) was annualized using the Capital Recovery Factor (CRF), as given in Eq. 17, to account for the time value of money over the project lifetime. Here, i is the Annual Discount Rate and n is the economic lifetime of the carbon capture system in years. The annual operating expenditure (OPEX) was then added to the annualized capital cost to obtain the total annualized cost of operating the system. Finally, the specific

CO_2 capture cost was calculated using Eq. 18 by dividing the total annualized cost by the annual mass of CO_2 captured (in tonnes). Consequently, the capture cost is expressed in $€_{2024}\, tonne_{CO_2}^{-1}$ and represents the average cost of capturing one tonne of CO_2 over the lifetime of the plant.

4.2.2 CO_2 Transportation Cost

Road transport of CO_2 via Tank Trailer Trucks is the last step of the CO_2 Capture Process and can potentially increase the operational expenditure (OPEX) of the overall CCUS chain depending on the Charging Station CCC unit and Storage hub locations. This is because it requires Tank Trailer Truck fuel, driver labour, loading and unloading operations, trailer maintenance, and maybe an intermediate storage infrastructure as well. In addition, transport distance, delivery frequency, and handling losses can further raise the cost per tonne of CO_2 delivered. For liquefied or refrigerated road transport, additional costs may also arise from insulation, pressure control, safety systems, and temporary buffer storage at the capture and receiving sites. These factors can become significant when the capture plant operates at small to medium scale or when the storage site is located far from the source. However, a detailed economic assessment of CO_2 road transport was not included in this thesis due to the limited scope of the study, and the transport cost is therefore discussed only qualitatively.

4.3 Economic Feasibility Assessment

Economic Feasibility is always evaluated to asses whether the proposed system is feasible and profitable to build, based on time criteria and cost criteria. Time criteria such as payback period, and cost criteria such as Net Present Value (NPV) can be used in the calculation referring to Turton et al., 2012. Some parameters listed in the Table 24 could be assumed according to current year economic situation.

Annual profit in the future can be determined based on the difference between the predicted future values of yearly CO_2 Tax (Income) and net of yearly operating expenditures (OPEX), with both components adjusted to account for an inflation rate of 3.30%, which is the 10 years average for EU (Eurostat, 2026). The Net Present Value (NPV) is subsequently calculated according to the Equation 19 defined below, which discounts the projected cash flows to their present value and deducts by the initial investment. A project is considered financially viable if the NPV is positive, whereas a negative NPV indicates that the project is not economically feasible.

$$NPV = \sum_{n=1}^{20} \frac{\text{Income}_n - \text{OPEX}_n}{(1+r)^n} - \text{Fixed Capital Investment (FCI)} \tag{19}$$

Table 24: Parameter for NPV Economic Analysis

Parameter	**Value**	**Unit**
Discount Rate (r)	8	%
Inflation	3.30	%
Average Carbon Tax	65.0	Euro_{2024}/tonne CO_2

5 Results and Discussion

5.1 Technical Study Results

5.1.1 Main Carbon Capture Container System Results

In this section, the results from the main Carbon capture Circuit is presented and discussed. With the Simulation design, Model parameters and Equipment dimensions as stated previously in the *Methodology* section, we simulate the Aspen Plus Simulation flowsheet and obtain the following results as presented in Table 25 for Single Genset System and Two Genset System.

Table 25: Comparison of Carbon capture Circuit Results and Performance for Single and Two Genset(s) System Operation

Parameter	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
Captured CO_2 – Single Genset (kg/hr)	259	179
Captured CO_2 – Two Gensets (kg/hr)	519	357
CO_2 Capture Efficiency – Single Genset (%)	∼90.00	∼90.00
CO_2 Capture Efficiency – Two Gensets (%)	∼90.00	∼90.00
Reboiler Thermal Energy Demand – Single Genset (kW-th)	217	147
Reboiler Thermal Energy Demand – Two Gensets (kW-th)	462	307
Specific Reboiler Heat Duty – Single Genset (MJ-th/kg CO_2)	3.02	2.97
Specific Reboiler Heat Duty – Two Gensets (MJ-th/kg CO_2)	3.20	3.09
AMP Concentration in Absorber Gas out – Single Genset (ppmv)	914.30	831.30
AMP Concentration in Absorber Gas out – Two Gensets (ppmv)	846.00	783.50
PZ Concentration in Absorber Gas out – Single Genset (ppmv)	27.90	25.50
PZ Concentration in Absorber Gas out – Two Gensets (ppmv)	26.00	24.40

As evident from the data presented, it is quite possible and feasible to use a single *Absorber Unit*, *Stripper Units*, and *Amine Wash Tower* design for both CPO and MPO operation in the Single and Two Genset Systems. The Specific Reboiler Energy demand for the Two Genset System is slightly higher than that of the corresponding Single Genset System cases, primarily because the required Lean Liquid Mix-to-Exhaust Gas flowrate Ratio (L/G ratio) in the *Absorber Unit* is slightly higher for Two Genset System scenario compared to Single Genset System scenario. This is due to the larger exhaust-gas flow rate in the Two Genset case, which leads to less favourable column hydrodynamics as the Absorber approaches its design flooding limit. The values for amine entrainment (AMP and PZ) are also shown in PPMV or Parts Per Million (Volume basis) in Table 25.

Backpressure is an important design constraint for Capture systems installed downstream of the Genset units. For Scania Gensets, a rule of thumb provided is that a pressure increase of 100 mbar(a) corresponds to approximately 1.5% Brake Efficiency (BE) loss cost, or about 3 g kWh^{-1} of additional diesel consumption. As stated by Scania, the total backpressure introduced by the CCC unit should therefore be limited to a maximum of 50 mbar(a) in the Exhaust Gas Stream, which the Genset Engine can handle easily without the need for any additional modifications. The pressure-drop values for the flowsheet Process units and Components were obtained from the Aspen Plus simulation results. However, additional pressure losses in pipelines, valves, and the Exhaust Gas-to-Water or Coolant Heat Exchagers in the Waste Heat extraction and upgradation circuit are more difficult to estimate and are outside the scope of this present thesis. The simulated backpressure results can be found in the *Appendix B* as Table 35 for the Single Genset System and Table 36 for the Two Gensets System.

The pressure drop for the Exhaust Gas Stream is within the prescribed limit of 50 mbar(a) for both Single and Two Genset Operations. This data can be found in Tables 35 and 36 in the *Appendix B* section. The simulation results show that the backpressure is less than 12 mbar(a) for all scenarios for both exhaust gas and CO_2 streams presenting favourable operating conditions for Genset engines without additional BE losses.

Another important aspect of truly Environment-Friendly, or Climate-Friendly Genset operation is compliance with emission limits especially those for entrained Amines vapours in the outlet CO_2-lean Exhaust Gas. CESAR–1 in particular is a highly volatile solvent with AMP being more volatile than PZ in the blend and hence shows a higher Amine-slip behaviour. More about the permissible Amine–slip limits and Environmental Regulations are discussed in the *Air Emissions and Water Usage* part (*section 6.1*) under *Environmental Aspect* Chapter.

The authors of the solvent database file used to build this simulation flowsheet (Morlando, Y. Zhang, et al., 2025) confirmed the robustness of the vapour–liquid equilibrium data for both the fresh CESAR–1 solvent and the CO_2-loaded CESAR–1 Liquid Mix. The amine–slip results obtained from the Aspen Plus simulation can be found in the *Appendix B* as Table 37 for the Single Genset System and Table 38 for the Two Genset System. Here PPBV means Parts Per Billion (Volume basis). The amine concentrations in the final CO_2-Lean Exhaust Gas stream leaving *Cooler–2* are low (below 12 PPBV for AMP and negligible for PZ) and remain within the applicable environmental limits. However, further investigation and Pilot-scale model validation are still recommended to check the accuracy and build trust in the predicted Amine-slip behaviour.

5.1.2 Waste Heat Integration and Post Capture Process Unit Results

In this section, the results from the Heat Integration Circuit (Figure 18) are discussed, together with the results from the Post-Capture Process Study, CO_2 Compression and Liquefaction. For the Waste Heat analysis, an additional case corresponding to hot European summer conditions was also considered, in which the ambient temperature is higher and the cooling water is now available at 40 °C instead of the previously studied 25 °C case. This was included to evaluate the sensitivity of the Heat Demand and Heat-Integration concept under less favourable operating conditions. The investigation also helps to better understand possible effects of seasonal variations in real life operation conditions. The available Waste Heat quantities, Reboiler Heat Duty, and other relevant process-parameter values obtained from the Aspen Plus Simulation and Analytical calculations are presented here. However, due to the limited time and scope of the Masters Thesis, a detailed process-design study of these downstream units was not carried out, and the discussion is therefore focused on the main results and their implications on the CCC operation.

5.1.2.1 Waste Heat Upgradation via Heat Integration Circuit Results

The Waste Heat Upgradation and Heat Integration Circuit results are an important aspect of the Standalone CCC unit, since they are essential for meeting the *Reboiler Unit* Thermal-Duty demand in the CO_2 Stripping section. No Heat loss is considered during the Waste Heat calculations and all the Equipments, Pipelines and Processes were assumed to be perfectly insulated with thick layers of insulator and operating fully under Adiabatic conditions. The results from the Waste Heat calculations are presented in Table 26 where Cases I and II means Single Genset and Two Gensets and subparts A and B means Maximum Power Operation (MPO) and Continuous Power Operation (CPO) modes respectively.

Table 26: Thermal Energy Availability and Heat Demand Analysis

PARAMETER		CASE IA	CASE IB	CASE IIA	CASE IIB
Reboiler Unit Thermal Energy Required (kW-th)	Normal Condition (25 °C)	217.0	147.0	462.0	307.0
	Hot Summer (40 °C)	233.0	158.0	508.0	335.0
CO_2 Captured (kg/hr) – 90% Capture Efficiency		259.65	178.64	519.29	357.28
Specific Reboiler Duty (MJ-th/kg CO_2)	Normal Condition (25 °C)	3.02	2.97	3.20	3.09
	Hot Summer (40 °C)	3.23	3.18	3.52	3.38
Available Heat from Exhaust Gas Cooled to 135 °C (kW-th)		187.25	131.75	374.50	263.50
Available Heat from Engine Coolant at 100 °C (WH1) (kW-th)		89.0	67.0	178.0	134.0
Additional Heat from Exhaust Gas Cooled to 110 °C (WH2) (kW-th)		15.0	11.0	30.0	22.0
Available Heat from Charge Air Cooled to 110 °C (WH3) (kW-th)		47.0	15.0	94.0	30.0
Upgraded Thermal Energy at 130 °C (kW-th) ($COP_{AHT} \times (WH_1 + WH_2 + WH_3)$)	Normal Condition (25 °C)	68.9	42.4	137.7	84.8
	Hot Summer (40 °C)	61.6	37.9	123.2	75.9
Net Available Thermal Energy at 130 °C (kW-th)	Normal Condition (25 °C)	256.10	174.20	512.20	348.30
	Hot Summer (40 °C)	248.90	169.70	497.70	339.40
Specific Thermal Energy Available at 130 °C (MJ-th/kg CO_2)	Normal Condition (25 °C)	3.55	3.51	3.55	3.51
	Hot Summer (40 °C)	3.45	3.42	3.45	3.42
Net Difference (Available – Required) (kW-th)	Normal Condition (25 °C)	39.10	27.20	50.20	41.30
	Hot Summer (40 °C)	15.90	11.70	-10.30	4.40
Estimated Possible Capture Efficiency (%)	Normal Condition (25 °C)	**90.00**	**90.00**	**90.00**	**90.00**
	Hot Summer (40 °C)	**90.00**	**90.00**	**88.20**	**90.00**

The net heat demand at the hot side of the *Reboiler Unit* increases under hot summer conditions, when cooling water is available at 40 °C, because the mass flow rate of Exhaust Gas slightly increases due to lower volume of condensing water in *Exhaust Gas Cooler Unit* at elevated temperatures (due to slightly higher vapour pressure of water at at 50 °C compared to at 40 °C). Also, the temperatures of the Lean Liquid Mix (LLM) and Exhaust Gas streams entering the *Absorber Unit* are also higher (50 °C now instead of 40 °C under normal conditions). This is less favourable for the absorption CO_2 process, as it slightly shifts the CO_2–CESAR 1 Absorption Equilibrium curve to the left, causing the rich Liquid Mix CESAR–1 – CO_2 loading to decrease at the same CO_2 partial pressures in the Absorption Tower. Therefore, to maintain the same Capture Efficiency, the Liquid-to-Gas flow ratio (L/G Ratio) must be increased, which leads to a higher flow rate of Lean Liquid Mix entering the *Stripper Units* and, in turn, increases the Sensible and Overall heat demand of the *Reboiler Unit*. This can be estimated from the data presented in Table 27 below.

Table 27: Design and Operating parameters for Single and Two Genset(s) CCC Unit under Hot Temperature Conditions

Parameter	**Single Genset**	**Two Gensets**
Flue Gas (FG) Flow rate (kg/hr)	1992.39 (MPO) 1557.96 (CPO)	3984.78 (MPO) 3115.92 (CPO)
Flue Gas (FG) Inlet Temperature (°C)	50	50
Lean Liquid Mix (LLM) loading (mol CO_2/mol AMP+PZ)	0.10	0.10
Lean Liquid Mix (LLM) loading (% CO_2/AMP+PZ)	10.00	10.00
Lean Liquid Mix (LLM) Flow rate (kg/hr)	3040 (MPO) 2050 (CPO)	6700 (MPO) 4420 (CPO)
Lean Liquid Mix (LLM) Inlet Temperature (°C)	50	50
Absorber L/G Ratio (kg LLM/kg FG)	1.53 (MPO) 1.32 (CPO)	1.68 (MPO) 1.42 (CPO)
Absorber L/G Ratio (mol LLM/mol FG)	1.65 (MPO) 1.42 (CPO)	1.82 (MPO) 1.52 (CPO)
Amine Wash Tower Fresh Wash Water Flow rate (kg/hr)	40 (MPO) 28 (CPO)	86 (MPO) 60 (CPO)
Fresh Wash Water Inlet Temperature (°C)	50	50

In addition to that, the net available Thermal Energy also decreases under hot summer conditions because the Sink temperature of the Type II Absorption Heat Pump increases, causing the COP to decrease consequently. As a result, a even smaller fraction of the available Waste Heat can now be upgraded to 130 °C. Hence, as shown in Table 26, the net difference between Available and Required Heat Energy decreases, and there is also a scenario (II A) in which the Capture Efficiency is below 90% (88.20%) if no additional thermal source is provided. But for the results presented in this study for Hot Climatic Conditions Scenario, since the difference between available and required Heat demand is very small, it is assumed that the Genset system provides additional Thermal Energy by burning a small amount of extra diesel in an Oxy-Cat or Diesel Boiler unit and the results are calculated based on 90% Capture Efficiency.

5.1.2.2 CO_2 Compression Study Results

The CO_2 compression results from the Compressor Train are presented here. As evident from the results for Two Gensets Maximum Power Operation presented in Table 28, the CO_2 Compressor Train effectively compresses the CO_2 stream leaving the *Condenser Unit* to the desired pressure and also reduces the water content in the CO_2 gas stream before the Refrigeration cycle. This helps to minimize the risk of hydrate formation and water-ice crystallization in the CO_2 Chiller unit.

Table 28: CO_2 Compression Results from the Compressor Train for Operation under Normal Climatic Conditions (Two Gensets System – MPO Mode)

Parameter	**Two Gensets System Maximum Power Operation (MPO) Mode**
CO_2 Mix Stream Flow rate – Inlet (kg/hr)	528.73
CO_2 Flow rate – Inlet (kg/hr)	519.74
CO_2 Content – Inlet (wt%)	98.30
CO_2 Stream Temperature – Inlet (°C)	40
CO_2 Stream Pressure – Inlet (bar(a))	1.84
Compressor Isentropic Efficiency (%)	70
Compressor Electricity Demand (kW-elec)	34.85
Intercooler Cooling Duty Demand (kW-th)	42.96
CO_2 Mix Stream Flow rate – Outlet (kg/hr)	521.47
CO_2 Flow rate – Outlet (kg/hr)	519.70
CO_2 Content – Outlet (wt%)	99.70
CO_2 Stream Temperature – Outlet (°C)	40
CO_2 Stream Pressure – Outlet (bar(a))	25

5.1.2.3 CO_2 Liquefaction Study Results

The CO_2 Liquefaction results from the Chiller are presented here. As evident from the results in Table 29, the CO_2 Refrigeration Cycle efficiently Liquefies the high pressure CO_2 stream leaving the Compressor Train to the desired temperature and pressure conditions to make the CO_2 transport ready for Road Transport via Tank Trailer Trucks.

Table 29: CO_2 Liquefaction Results from the Refrigeration Cycle for Operation under Normal Climatic Conditions (Two Gensets System – MPO Mode)

Parameter	**Two Gensets System Maximum Power Operation (MPO) Mode**
CO_2 Mix Stream Flow rate – Outlet (kg/hr)	521.47
CO_2 Flow rate – Inlet (kg/hr)	519.70
CO_2 Content – Outlet (wt%)	99.70
CO_2 Stream Temperature – Inlet (°C)	40
CO_2 Stream Pressure – Inlet (bar(a))	25
Refrigerator Compressor Isentropic Efficiency (%)	70
Refrigerator Electricity Demand (kW-elec)	29.91
Refrigerator Cooling Duty Demand (kW-th)	80.80
Refrigerator Cooling Water Flow rate (kg/hr)	6948.59
Refrigerator COP_{Cooling}	1.701
CO_2 Stream Temperature – Outlet (°C)	-25
CO_2 Stream Pressure – Outlet (bar(a))	20

5.1.3 Sensitivity Study

The scenario selected for the sensitivity analysis corresponds to Two Gensets Maximum Power Operation mode with the highest Exhaust Gas flow rate, since this represents the upper design limit that the CCC unit must be able to handle effectively.

The first set of sensitivity studies was carried out to investigate the dependence of Specific Reboiler Duty on the *Absorber Unit* height, total *Stripper Unit* height, and Lean Liquid Mix CO_2 loading. These studies were used to optimize and fix the dimensions of the most important CCC components. In each case, a parametric sweep was performed for one variable while the remaining parameters were kept constant. This was followed by an analysis of the effect on Reboiler Heat duty due to changes in the *Recuperator Unit* Heat Transfer Coefficient, which can provide insight into the impact of fouling and long-term thermal resistance build-up in the Plate Type Heat Exchanger unit.

In addition, the *Recuperator Unit* Heat Transfer Area was optimized through a separate sensitivity study relating heat-exchanger area to Specific Reboiler Duty (SRD), with the aim of reducing the sensible heat demand and ultimately the overall heat demand in the Reboiler unit. A further sensitivity study was performed to optimize the number of Compressors and Intercoolers required in the Compression Train for the captured CO_2 gas based on the specific CO_2 Compression Energy demand and the specific CO_2 stream Intercooling Duty. The final selected values were then incorporated into the Aspen Model design, as discussed in the relevant sub-sections of the *Methodology* section (*Section 3*).

5.1.3.1 Absorber Column Height Optimization

The Absorber height was varied to examine its effect on the Specific Reboiler Duty, while the CO_2 Capture Rate, Lean Liquid Mix loading and total *Stripper Unit* height were kept constant at 0.10 mol CO_2/mol AMP+PZ and 4.0 m respectively. This indicates that the flowrate of Liquid Mix and the Rich Liquid Mix CO_2 loading varies with different scenarios of this sensitivity analysis.

As shown in Figure 21, the Specific Reboiler Duty decreases as the Absorber column height increases. This behaviour is expected, since a taller packed Absorber column provides a larger effective Mass Transfer area, more Theoretical Equilibrium Stages, and a longer residence time for Gas–Liquid contact, thereby improving CO_2 transfer from the Flue Gas phase to the Liquid Mix phase. For the same CO_2 capture target, the Liquid-to-Gas (L/G) ratio is therefore reduced, which in turn reduces the Lean Liquid Mix flow rate for the same Exhaust Gas flowrate. This lowers the sensible heat demand of the Liquid Mix stream at the Reboiler unit. At the same time, the Rich Liquid Mix CO_2 loading also increases.

From Figure 21, it can be seen that increasing the Absorber tower height beyond 1.95 m has a little effect on the Reboiler heat demand, with a decrease of less than 1.5%. This indicates that the Operating Line is approaching the Equilibrium Line at the bottom of the tower (or Lean Liquid Mix flow rate is approaching the theoretical minimum Solvent flow rate) and that the mass-transfer driving force is becoming smaller as the system approaches the Pinch Point. Moreover, a taller Absorber increases Rich Liquid Mix CO_2 loading further, which could lead to salt precipitation, corrosion, fouling, and deposition issues. Based on these considerations, the *Absorber Unit* height was fixed at 1.95 m. The rate of CO_2 captured is 519 $\mathrm{kg\,hr^{-1}}$ which is fixed for this study.

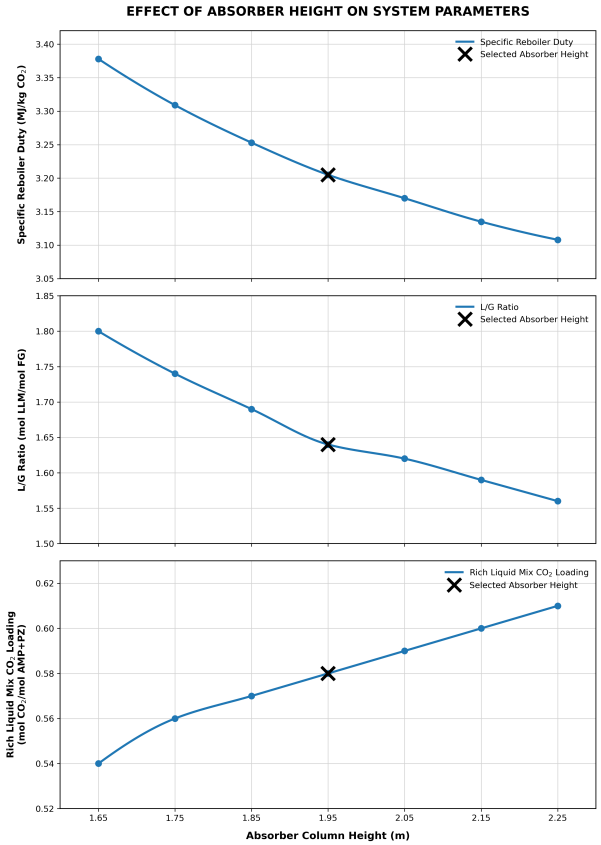


Figure 21: Absorber Column Height Sensitivity Analysis

5.1.3.2 Stripper Column Height Optimization

In this section, the sensitivity study for optimizing the total *Stripper Unit* height was executed. The total Stripper height was varied to determine its effect on the Specific Reboiler Duty, while the Capture Rate, Lean Liquid Mix loading and *Absorber Unit* height were kept constant at 0.10 mol CO_2/mol AMP+PZ and 1.95 m respectively. This means that the the Liquid-to-Gas (L/G) ratio and the Rich Liquid Mix CO_2 loading are also constant for all the simulated data points. The rate of CO_2 captured is 519 $\mathrm{kg\,hr^{-1}}$ which is fixed for this study.

As shown in Figure 22, the Specific Reboiler Duty decreases as the total Stripper column (*Stripper Unit 1* and *Stripper Unit 2*) height increases. This behaviour is consistent with the McCabe–Thiele Analogy which states that a lower Reboiler Duty means a lower available latent heat of vaporization which corresponds to a lower steam generation rate. As a result, the Operating Line approaches more closer to the Equilibrium Line from the bottom (operating line slope becomes steeper), which implies a reduced mass-transfer driving force and, subsequently, a larger number of Theoretical Equilibrium stages or a taller Packing Height. In other words, again a taller packed Desorber or Stripper column provides a larger effective Mass Transfer area and a longer residence time for Gas–Liquid contact, thereby improving CO_2 transfer from the Solvent phase to the Gas phase. For the same CO_2 desorption target, the Reboiler Heat Duty is therefore reduced, which in turn reduces the Specific Reboiler Heat Duty.

From Figure 22, it can be seen that increasing the Stripper tower height does not have a very prominent effect on the Reboiler Duty. When the Stripper height goes beyond 4.0 m a very minor change on the Reboiler heat demand, with a decrease of less than 0.5% was observed. This indicates that the Operating Line is very close to the Equilibrium Line and that the mass-transfer driving force is becoming smaller as the system approaches the Pinch Point which makes further increase of tower height unjustified. In addition, since the *Stripper* towers operate at high temperature, insulation becomes an important factor for maintaining operation close to the assumed adiabatic conditions. This again supports the idea of shorter individual towers so that they can be placed inside the 45-foot CCC unit container, which provides an additional layer of thermal protection. Based on these considerations, the total *Stripper Unit* height was fixed at 4.0 m.

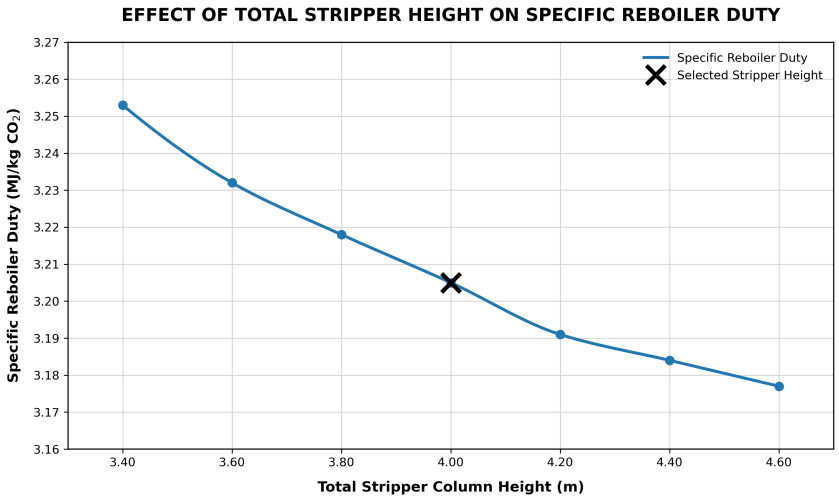


Figure 22: Total Stripper Column (Stripper 1 & Stripper 2) Height Sensitivity Analysis

5.1.3.3 Lean Liquid Mix CO_2 Loading Optimization

The Lean Liquid Mix loading was varied to determine its effect on the Specific Reboiler Duty, while the Capture Rate, *Absorber Unit* height and total *Stripper Unit* height were kept constant at 1.95 m and 4.0 m respectively. This means that the the Liquid-to-Gas (L/G) ratio is also constant for all the simulated data points. The rate of CO_2 captured is 519 kg/hr which is fixed for this study. The rate of CO_2 captured is 519 $\mathrm{kg\,hr^{-1}}$ which is fixed for this study.

As shown in Figure 23, the Specific Reboiler Duty decreases as the Lean Liquid Mix CO_2 loading increases and then increases again giving an asymmetric parabolic relationship between the two. This behaviour is obvious and also expected. At lower Lean Liquid Mix loadings, the amine solvent has a higher CO_2 carrying capacity, so the required Liquid-to-Gas or L/G ratio is lower. However, achieving very low CO_2 partial pressures (low CO_2 vapour phase concentrations) to reach extremely low Lean Liquid Mix loading values in the Stripper tower requires a larger amount of stripping vapour or steam, which increases the latent heat demand associated with water vaporization and, consequently, the total Reboiler Heat Duty. At higher Lean Liquid Mix loadings, the solvent has a lower CO_2 carrying capacity, so the L/G ratio increases and a larger solvent flow must be heated in the Reboiler unit. This raises the sensible heat demand and also increases the overall Reboiler Heat Duty. Since the latent heat of water vaporization is significantly higher than the sensible heat demand of the Liquid Mix, the resulting trend appears as a skewed parabola with a steeper left-hand side than right-hand side.

Hence, in all the scenarios the Liquid Mix captures exactly the same amount CO_2 of from the Flue Gas. From Figure 23, it can be seen that a global minima exists given the skewed parabolic nature of the Specific Reboiler Duty and the Lean Liquid Mix CO_2 loading relationship. Based on these considerations, the the Lean Liquid Mix CO_2 loading was fixed at 0.10 mol CO_2/mol AMP+PZ.

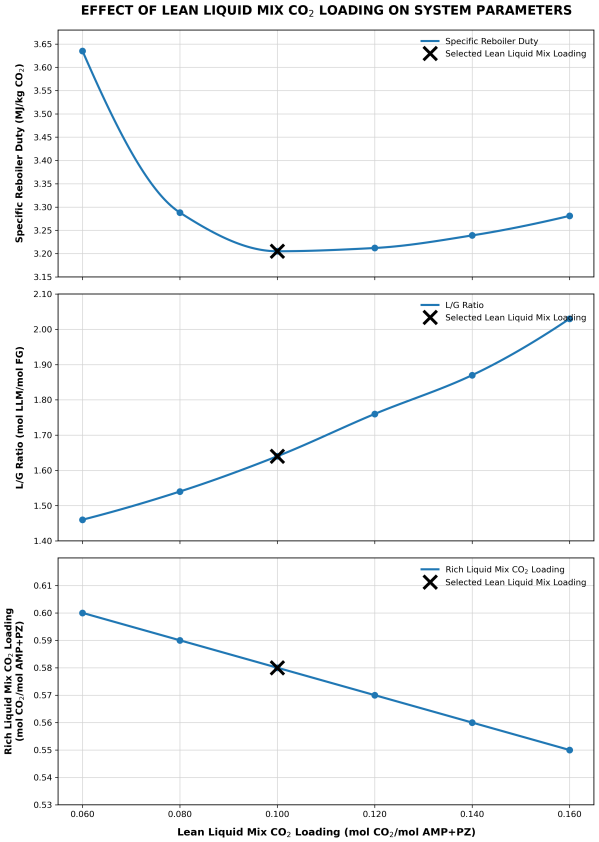


Figure 23: Lean Liquid Mix CO_2 loading Sensitivity Analysis

5.1.3.4 Sensitivity Study for Recuperator Unit

In this section, the sensitivity study for optimizing the *Recuperator Unit* was conducted. Both the Overall Heat Transfer Coefficient and Overall Heat Transfer Area were varied for the Recuperator considering one variable at a time to determine its effect on the Specific Reboiler Duty. First, the Overall Heat Transfer Area was varied to determine the optimum Recuperator area followed by keeping the Heat Transfer Area fixed and varying the Overall Heat Transfer Coefficient to simulate fouling conditions. The rate of CO_2 captured is 519 $\mathrm{kg\,hr^{-1}}$ which is fixed for this study.

As shown in Figure 24, the Specific Reboiler Duty decreases as the Overall Heat Transfer Area increases. This behaviour is observed because of higher sensible heat transfer between the Hot Lean Liquid Mix and the Cold Rich Liquid Mix streams reducing the sensible heat demand of the Reboiler. As a result, the Overall Reboiler Heat Duty decreases. But as we further increase the Heat Transfer Area and approach to an infinite Heat Exchanger, the Specific Reboiler Duty reaches a downward plateau indicating that though an Infinte Recuperator unit reduces the sensible heat demand in the Stripper section by a large extent, the principal share of Reboiler heat demand comes from the heat of CO_2 desorption reaction and the latent heat needed for water vapourization which cannot be satisfied by the Lean-Rich heat exchanger or Recuperator unit. Hence, based on these considerations, the Overall Heat Transfer Area for the *Recuperator Unit* was fixed at 65.0 $\mathrm{m^2}$.

Next, we study the effect of Overall Heat Transfer Coefficient of Recuperator on the Reboiler Heat Demand to replicate fouling like conditions. From Figure 25, it is evident that decreasing Overall Heat Transfer Coefficient increases the Specific Reboiler Duty due to reduced exchange of sensible heat but donot have a very prominent effect on the Specific Reboiler Duty. Based on the clean Recuperator Heat Transfer Coefficient values and design margin considerations as already explained in *Methodology* section (*Section 3*), the Overall Heat Transfer Coefficient of Recuperator unit was fixed at 850 $\mathrm{W\,m^{-2}.K^{-1}}$.

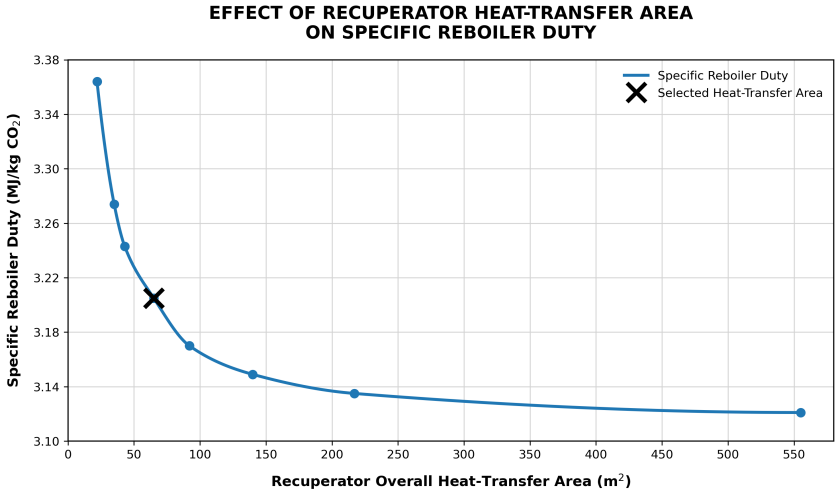


Figure 24: Recuperator Unit Overall Heat Transfer Area Sensitivity Analysis

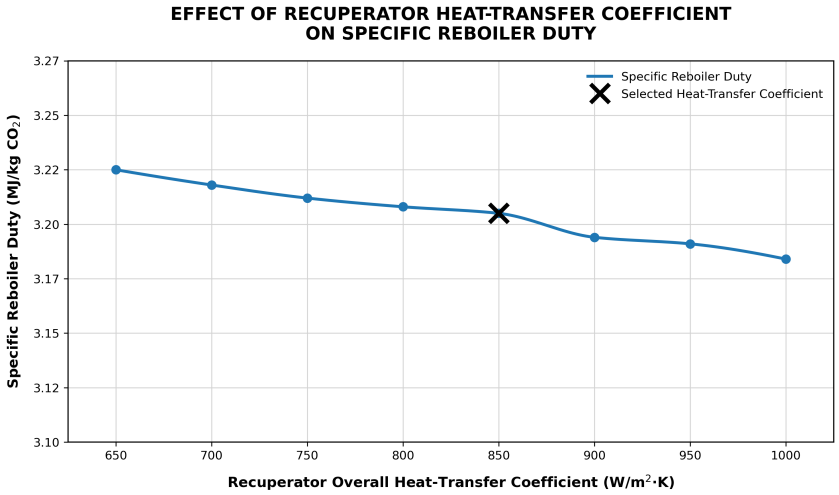


Figure 25: Recuperator Unit Overall Heat Transfer Coefficient Sensitivity Analysis

5.1.3.5 Sensitivity Study for Compressor Train

In this section, the sensitivity study for optimizing the *Multi-stage Compressor Unit* was carried out. From Figure 26 and 27, it is clear that increasing the number of Compressor and Inter-cooler units in a Compressor Train reduces both the Compressor Power and Intercooler Heat Duty. But going beyond 3 Compressors and Intercoolers is unjustified because the decrease in the Energy Demand drops below 3% for both the components which is not sufficient to overcome the additional Capital Cost requirement and the occasional Maintenance expenditures incurred for the extra units. So, based on the design considerations and cost constraints, the *Multi-stage Compressor Unit* was fixed at 3 sets of Compressors and Intercoolers. The rate of CO_2 captured is 519 $\mathrm{kg\,hr^{-1}}$ which is fixed for this study.

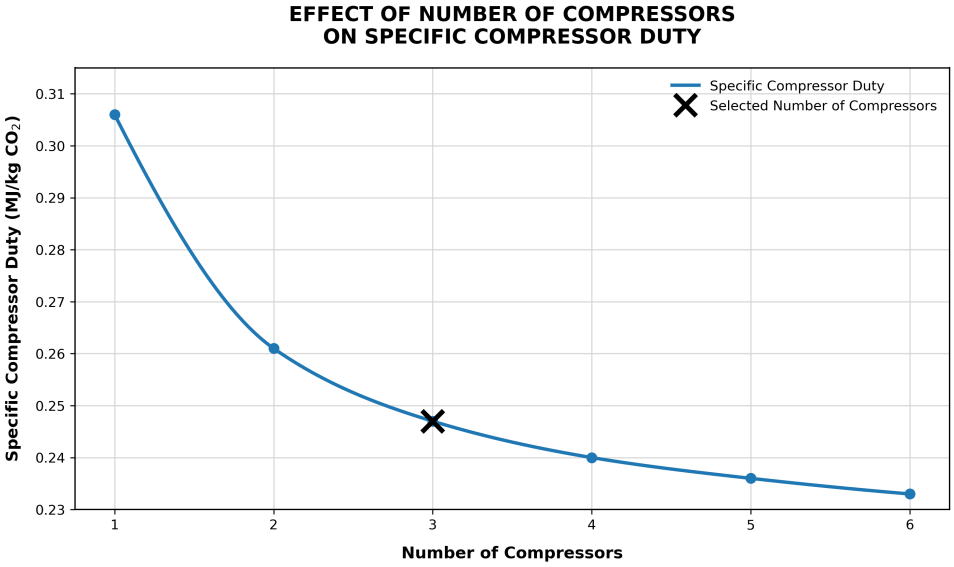


Figure 26: Compressor Unit Specific Energy Demand Sensitivity Analysis as a Function of Compressor Stages

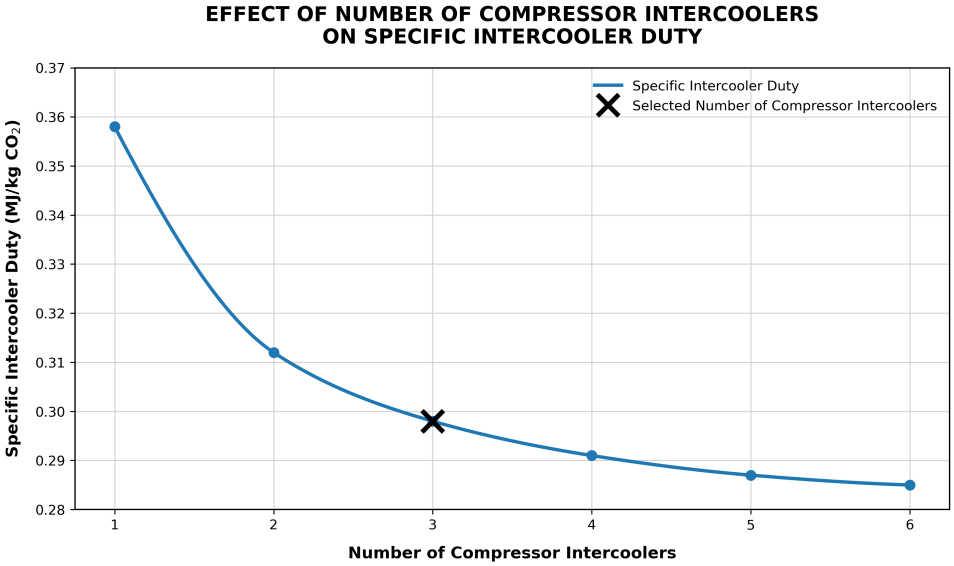


Figure 27: Intercooler Unit Specific Heat Duty Sensitivity Analysis as a Function of Intercooler Stages

5.2 Economic Analysis Results

The results from the Economic Analysis study of the CCC plant are presented in this section. The discussion begins with the CAPEX assessment, in which the capital costs of the major process equipment and supporting units are estimated and added to obtain the total investment required for constructing the proposed CCC System. This is followed by the OPEX analysis, where the recurring operating costs associated with energy consumption, solvent usage, process water, and other relevant operating requirements are evaluated and presented. Together, these results provide a complete overview of the economic performance of the CCC plant and form the basis for assessing its overall techno-economic feasibility based on the specific CO_2 Capture Cost (€$_{2024}$/tonne CO_2).

5.2.1 Capital Expenditure (CAPEX) Analysis Results

Pie Chart 28 shows the capital cost breakup for major components of a Two Genset CCC system. A summary of the approximate bill of quantities for different Process Flowsheet Equipments, their Capacity parameter values, their units, and the corresponding Purchased Equipment Cost (PEC) in €$_{2024}$ million (C_{2024}^{MEuro}) for the Single Genset and Two Gensets System CCC units can be found in Table 50 and Table 51 of *Appendix B*. The full Purchased Equipment Cost comes out to be around 0.943 million €$_{2024}$ for Single Genset and 1.344 million €$_{2024}$ for Two Genset systems. However, the cost for Pumps and Pumping cost for the cooling water, chemical tanks and water containers in the CCC unit is not calculated due to lack of data and knowledge on piping length and geometry estimates at this primary stage of design. Also as mentioned in the *CO_2 Purification* subsection (*Section 2.5.1*) under the literature review, the design of CO_2 dehumidifier unit and its CAPEX and OPEX are not analyzed in the current study.

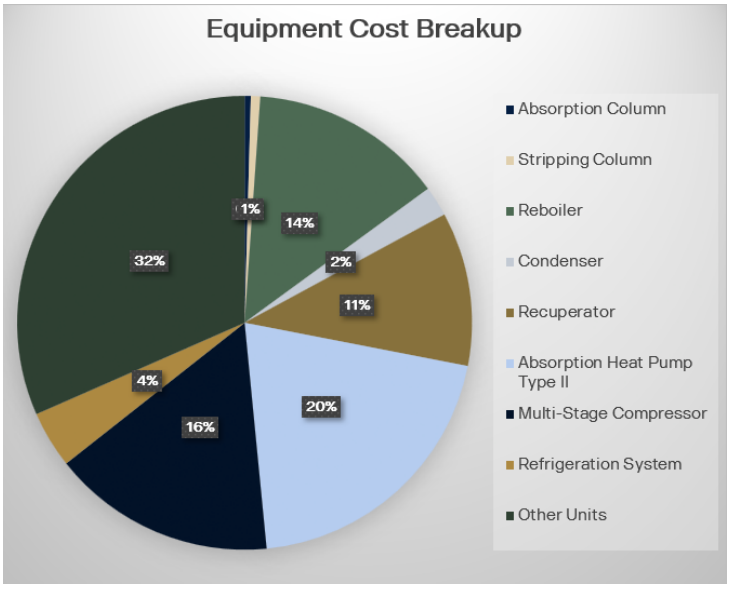


Figure 28: Capital Cost breakup for a Two Gensets Carbon Capture Container (CCC) System

The initial stock quantities for chemical and process water, as well as cooling water, were calculated based on the required material flow rate at Maximum Power Operation mode and a storage-tank residence time of 10 minutes, in accordance with literature sources (Cantrell et al., 2017). In addition, a safety or overdesign factor of 8% was applied to the heat exchangers, while that of 15% was applied to the remaining process units, following common engineering practice (Peters et al., 2003).

Finally, the Purchased Equipment Cost (PEC) was multiplied by the Total Capital Investment Factor (f_{TCI}) of 1.75 to obtain the Final Equipment Cost (FEC) or Bare Module Cost (BMC) including the Contingency, Fees and Working Capital Investment. The final values for Total Capital Investment (TCI) were obtained as 1.6511 and 2.3554 (in €$_{2024}$ million, C_{2024}^{MEuro}) for Single Genset and Two Gensets CCC units respectively using the following equation.

$$TCI = (f_{\text{TCI}} \times PEC) + Cost_{\text{AMP}} + Cost_{\text{PZ}} + Cost_{\text{PW}} + Cost_{\text{CW}} \tag{20}$$

5.2.2 Operational Expenditure (OPEX) Analysis Results

Four scenarios involving the Operating Cost (OPEX) analysis were evaluated in this study, including Single and Two Genset Continuous Power Operating modes, together with special OPEX cases without a Absorption Heat Transformer for the Single and Two Genset(s) CCC unit. These cases are included to analyze the economic benefit of Engine modifications that enable higher-temperature Coolant outlet from the Genset and hotter Tailpipe Exhaust Flue Gases, thereby partly reducing the need for a separate Heat upgrading Circuit and Absorption Heat Transformer unit.

Since a detailed CESAR–1 solvent degradation study was not carried out, the annual make-up chemical and material requirement was assumed to be 2% of the total storage-tank inventory for each respective material as a conservative measure. An example case for the Two Genset CCC unit is presented in Table 30, while the OPEX for remaining operational cases are provided in *Appendix B* under Tables 52 and 53. The levelized CO_2 abatement cost is calculated to be **1200.05** and **864.00 €$_{2024}$/tonne CO$_2$** for the Single and Two Genset(s) CCC units, respectively.

Table 30: Main OPEX parameters for Two Gensets Carbon Capture Container Design

Parameter	**Total Cost (€$_{2024}$)**
Annual Raw Material Cost	71
Annual Electricity Cost	19737
Annual Maintenance and Repair Cost	188435
Total Annual Operating Supplies Cost	211989
Annual Taxes and Insurance	75374
Total Annual OPEX	**304816**

When the Absorption Heat Transformer is removed, the CAPEX associated with purchasing the Absorption Heat Transformer and the OPEX associated with operating the Absorption Heat Transformer Solution Pump are not considered. Under these conditions, the levelized CO_2 abatement cost becomes **1027.53** and **691.61 €$_{2024}$/tonne CO$_2$** for the Single and Two Genset(s) CCC units, respectively.

6 Environmental Assesment and Chemical Risk Analysis

This chapter addresses the Environmental Aspects and associated Chemical Risks during operating the proposed CCC unit for Scania Gensets. Environmental Analysis and Regulatory Assessment are important and integral parts of any Engineering Process Design before proceeding with Pilot or Industrial Unit construction. The discussion covers key environmental aspects such as Air Emissions, Water Usage, and Energy Consumption. Special attention is also given to Environmental Regulations and considerations related to Non-Methane Volatile Organic Compounds (NMVOCs) emissions, particularly in relation to Amine slip phenomena. However, a detailed regulatory assessment is outside the scope of this present thesis study.

A general Chemical Risk and Safety Analysis is also presented at the end of the chapter. However, a more extensive study of these aspects has not been carried out due to scope limitations and time constraints.

6.1 Air Emissions and Water Usage

Typical Genset Exhausts contain certain levels of SO_x, NO_x, CO, particulates, and volatile organic compounds (VOCs) compared with normal atmospheric limits. But with stricter environmental regulations and modern Engine Tailpipe emissions exhaust after-treatment techniques present in EU Stage V compliant engines like DC13-505A, the levels of SO_x, NO_x, CO, particulates, and volatile organic compounds (VOCs) are ensured to be within acceptable limits. Moreover, the Exhaust Gas data provided by Scania do not contain traces of SO_x, NO_x, CO or VOCs. Hence, the study and analysis of Nitrosamines and Nitramines are not considered in the current investigation (Languille et al., 2021).

The Amine Slip behaviour of the CESAR–1 solvent due to higher volatility of AMP is a usual problem encountered in Pilot Plants. This problem becomes dominant as the temperature goes up higher. Available Regulations suggest AMP and PZ emissions from CESAR–1 based Carbon Capture plants to kept below 1 mg Nm^{-3} of Exhaust Gas in normal operational conditions which is roughly around 0.3 PPMV concentration of AMP and PZ amines (Herraiz et al., 2024; CCS Norway, 2023). For EU Stage V compliant stationary Gensets though, regulations suggest the limit for total Non-Methane Volatile Organic Compounds emission should be below 0.19 $g\,kWh^{-1}$ (International Council on Clean Transportation (ICCT), 2016). This corresponds to 83.03 $g\,h^{-1}$ and 59.28 $g\,h^{-1}$ of total quantity of Amines in the Final Exhaust Gas Out for Single Genset MPO and CPO modes respectively and to 166.06 $g\,h^{-1}$ and 118.56 $g\,h^{-1}$ of total quantity of Amines in the Final Exhaust Gas Out for Two Gensets MPO and CPO modes respectively. The final outlet amine concentrations for both Normal and Hot Climatic Conditions are therefore presented in Table 31 for the Single Genset System and Table 32 for the Two Genset System below.

Table 31: Simulated Amine Slip Results for Single Genset System Operation

Parameter	**Maximum Power Operation (MPO) Mode**		**Continuous Power Operation (CPO) Mode**	
	Normal Condition (25 °C)	**Hot Summer (40 °C)**	**Normal Condition (25 °C)**	**Hot Summer (40 °C)**
AMP in Final Exhaust Gas Out (mg/hr)	48.7	272.0	47.9	256.0
PZ in Final Exhaust Gas Out (mg/hr)	0.002	0.025	0.001	0.014
AMP Conc. in Final Exhaust Gas Out (PPBV)	9.1	48.1	11.3	57.0
PZ Conc. in Final Exhaust Gas Gas Out (PPBV)	0.0004	0.005	0.0003	0.003

Table 32: Simulated Amine Slip Results for Two Gensets System Operation

Parameter	**Maximum Power Operation (MPO) Mode**		**Continuous Power Operation (CPO) Mode**	
	Normal Condition (25 °C)	**Hot Summer (40 °C)**	**Normal Condition (25 °C)**	**Hot Summer (40 °C)**
AMP in Final Exhaust Gas Out (mg/hr)	113.8	603.0	97.9	594.0
PZ in Final Exhaust Gas Out (mg/hr)	0.02	0.19	0.01	0.12
AMP Conc. in Final Exhaust Gas Out (PPBV)	10.7	53.4	11.5	66.1
PZ Conc. in Final Exhaust Gas Out (PPBV)	0.002	0.017	0.001	0.014

The values from the Tables 31 and 32 suggest that the entrained amine concentration in the Final Exhaust Gas Out is well within the regulated limits. But since the numbers are too small and Pilot

Plant operation conditions can highly differ from the simulated results due to difference in modelled and real life Vapor Liquid Equilibrium behaviour of the amine blends, further investigation and Pilot-scale model validation are still recommended to check and reassure the accuracy of the predicted Amine-slip behaviour.

From the Water Usage and Water Treatment perspective, the most common toxic substances found in aqueous effluent streams from an amine based CCUS plant are Nitrosamines, Nitramines, Amine residues and fragments, Amides, Sulfates, Aldehydes, Ammonium compounds, Paints and Grease to name a few (Languille et al., 2021). But since this is a Stand-alone Containerized Capture system, there is no path for effluents to escape out of the system boundary which must hence be treated and removed occasionally as the concentration of toxic elements build up in the Process Water Storage Tanks. Bigger CCUS plants are equipped with modern equipments and advanced wastewater treatment facilities are capable of handling these challenges efficiently and effectively.

6.2 Energy Usage

The main energy demand of the proposed CCC unit is dominated by the Reboiler Heat Duty, which constitutes a substantial share of the overall Thermal Energy requirement of the Capture plant. As discussed in this study, the simulation results and supporting calculations indicate that Waste Heat recovery, subsequent Heat upgrading, and supply to the hot side of the *Reboiler Unit* are feasible, and that the upgraded waste heat available is sufficient to meet the thermal energy demand of the system. In addition to the thermal load, several units in the CCC system also require electrical energy for operation. The largest electricity consumer is the CO_2 Compressor Train, followed by the CO_2 Refrigerator or Chiller, while the Amine and Absorption Heat Transformer Solvent Pumps and Water circulation Pumps contribute only a relatively small fraction of the total electrical demand. Overall, the electrical power requirement is estimated to account for approximately 10–15% of the total energy demand of the system and may be supplied through additional diesel combustion in the Genset unit when required during the process.

6.3 Safety and Chemical Risk Aspects

Capturing the tailpipe carbon dioxide from Industrial Gensets seems like a sustainable and environment friendly solution for carbon neutrality. However, this complicated process comes with significant risks associated with chemical plants which work with hazardous and corrosive materials. The manufacturing involves handling large quantities of corrosive materials under moderately high temperature and pressure conditions, creating potential chemical risks that must be carefully managed to ensure worker safety and environmental protection.

Safety and chemical-risk considerations are important in CESAR–1-based CO_2 capture because the solvent contains volatile AMP and reactive PZ, both of which may contribute to amine slip, aerosol formation, and solvent losses if emission-control measures are insufficient. Pilot-plant studies show that CESAR–1 performs well relative to MEA, but degradation products can still accumulate during long-term operation, especially under oxidative conditions and in the presence of NO_2, which has been linked to higher degradation rates and the formation of Nitrosamines and Nitramines. The main operational risks therefore include solvent degradation, corrosion, contamination of the CO_2 product, and emissions of amines and degradation by-products to air and wastewater. Equipment operating under high pressure and temperature conditions is far more likely to fail. A blocked control valve on the Stripper Column outlet can cause a rapid pressure or liquid build-up, which could cause the opening of the safety valves to open, releasing a hot and hazardous mixture of Amines and other reactants, leading to a chemical hazard. Furthermore, the rotating equipments especially the Compressor, Chiller and the Liquid Pumps and the corresponding valves and pipelines should be properly handled and must be frequently maintained. This keeps the prone-to-failure equipments in good operational conditions in the long term, minimizing the chances of possible leaks and failures. The risks can also be reduced through careful temperature control, wash sections, solvent management, and polishing technologies such as dry beds and acid wash systems.

Safety procedures are just as important as technical controls in any chemical plant. A comprehensive risk assessment must identify all potential safety issues to create a disaster management plan. This includes strict safety procedures, proper labelling of hazardous materials, and continuous training for the employees and maintenance personnel.

7 Conclusion

This thesis examined whether a compact, moderate–temperature and ambient–pressure Carbon Capture Container (CCC) could be integrated with one or two Scania DC13-505A Generator sets without compromising Generator operation. A small portion of electricity produced by the Gensets (around 7.50%) is consumed by the CCC unit for operation. Within the domain and assumptions of the study, the concept is technically feasible. A common process configuration was identified for both CPO and MPO operation, and the model achieved the 90% CO_2-capture target while maintaining low exhaust pressure drop and producing a CO_2 stream suitable for liquefaction and road transport. Economically, however, the design is not yet competitive at the assessed scale and annual utilisation.

CESAR–1 was selected as the working solvent because it combines the rapid absorption kinetics of Piperazine with the high cyclic capacity and comparatively low regeneration thermal energy demand of AMP. The robustness and high performance efficiency of CESAR–1 is evident from the relatively lower Liquid-to-Gas (L/G) ratios when compared to a classical MEA–based CO_2 absorption system. Its predicted Specific Reboiler Duty (SRD) was compatible with the recoverable Genset waste heat. The rate-based Aspen Plus framework reproduced the principal pilot-plant indicators with sufficient accuracy for conceptual design, although local temperature profiles and very low amine-emission predictions still require pilot-scale verification.

The final common design uses a 0.77 m internal-diameter Absorber with 1.95 m of structured packing and a total Stripper packing height of 4.0 m divided between two compact sections. A single-stage *Absorber* tower combined with two half-*Stripper* units was found to perform efficiently in the CO_2 Capture system flowsheet design. Sensitivity studies identified a lean loading of 0.10 mol CO_2 per mol AMP+PZ and a 31 m^2 and 65 m^2 lean–rich recuperator for the Single and Two-Genset designs respectively. Larger columns or recuperator areas produced only modest further energy savings but were found to be cost intensive. Three-stage Compressor train with Intercoolers was selected for the same reason, since additional stages provided less than approximately 3% incremental energy saving and contributed to additional capital investments. The idea of using a portion of the total Reflux water stream from the *Condenser unit* of the *Stripper 1 Unit* as the Wash-water source for the *Amine Wash Tower* inlet, and then returning it to the *Stripper 1 Amine Wash Section*, showed improved system efficiency and promising water management and operational performance.

At normal cooling conditions, specific reboiler duty ranged from 2.97 to 3.20 $MJ_{th}\ kg_{CO_2}^{-1}$. The integrated Exhaust Gas, Coolant and Charge-air heat-recovery network, supported by a Type II Absorption Heat Pump, supplied 3.51–3.55 $MJ_{th}\ kg_{CO_2}^{-1}$ and was sufficient for all normal cases. Under hot-summer conditions, available upgraded heat decreased to 3.42–3.45 $MJ_{th}\ kg_{CO_2}^{-1}$, while Reboiler energy demand increased to 3.18–3.52 $MJ_{th}\ kg_{CO_2}^{-1}$. The Two-genset MPO case would therefore achieve approximately 88.2% capture without supplementary heat; a small auxiliary input, decreasing Engine efficiency to produce hotter Exhaust Gas or improved Engine-side heat recovery would close this gap. Simulated Exhaust pressure drop remained between 2.6 and 11.4 mbar(a), and the wash-and-cooling system reduced predicted AMP emissions to 9.1–11.5 ppbv under normal conditions, with negligible PZ slip. However, the amine-slip results still need to be verified in an actual Pilot-scale operating setup.

The fixed capital investment was estimated at € 1.65 million for one genset and € 2.36 million for Two Gensets. At 2000 operating hours per year, the specific CO_2 Capture costs were € 1200.05 $tonne_{CO_2}^{-1}$ and € 864.00 $tonne_{CO_2}^{-1}$, respectively, before transport and storage for the year 2024. The Two-Genset case benefits from economies of scale, but both costs remain high. The most obvious next step is one Single Genset CCC unit construction followed by dynamic testing, measurement of real heat losses and backpressure, solvent-degradation and corrosion studies, validated emission monitoring, and review and refinement of the cost model. Higher utilisation, serial modular manufacture, engine-side heat-recovery improvements and future integration with sustainably sourced biofuels could substantially strengthen both the commercial and climate case.

7.1 Future Scope and Prospects

Since CO_2 capture is a broad and rapidly developing field, future prospects are an important part of any CCUS study, as in other rapidly expanding research areas. There are still several possible directions for future work that could build on the present design and study findings.

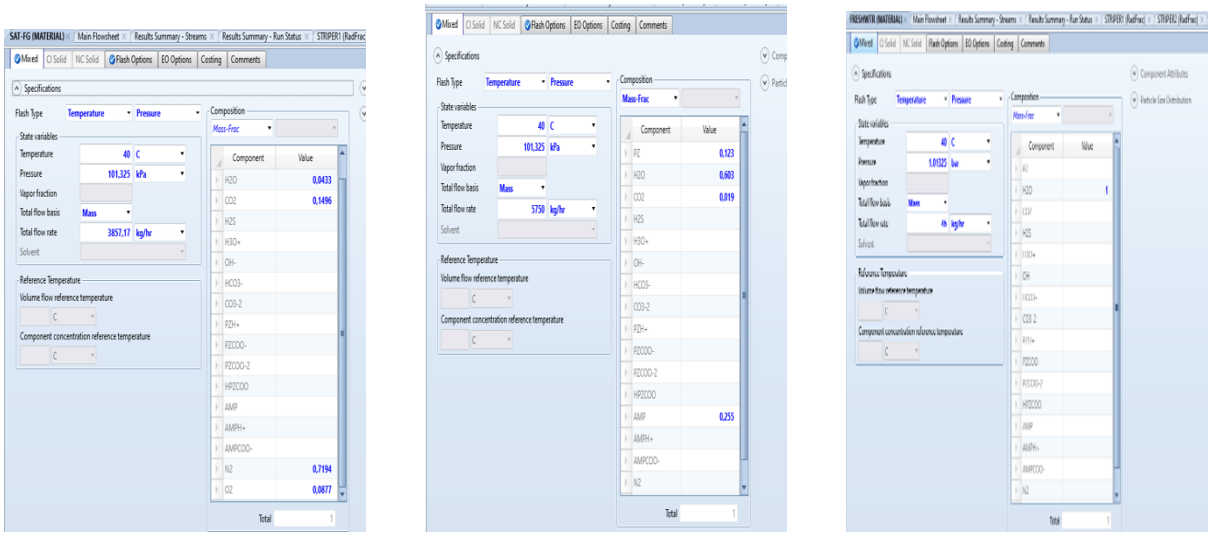
One of the most important future developments could happen on the Genset Engine side of the setup. Modifying the Genset Engine Charge-Air intake system to simplify Heat Recovery and improve Heat Extraction from the Compressed Charge Air Intercooler could provide greater energy security and make the Waste-Heat supply to the Carbon Capture circuit more reliable. In addition, modifying the Genset Engine internals to reduce the engine lambda value (λ) (air-to-fuel mass ratio) may be important in the long term for smoother operation of the CCC unit. A lower lambda would reduce the stoichiometric excess O_2 content in the intake air, which would in turn lower the O_2 concentration (wt%) in the outlet flue gas. This could help reduce solvent degradation and extend the amine cycle life in the CCC unit. At the same time, a lower lambda (λ) would reduce the N_2 content in the exhaust gas and increase the CO_2 concentration (wt%), making CO_2 capture easier and lowering the Specific Reboiler Heat Duty requirement. In addition to this, modifications could also be made on the Genset Engine side to investigate the possibility of increasing the outlet Exhaust Gas temperature and allowing the Engine Coolant to exit at 130–135 °C. In that case, the Absorption Heat Transformer could potentially be removed from the design, since waste-heat upgrading would no longer be necessary. This would reduce both the CAPEX and OPEX of the overall system.

From the CCC unit side, the next step would be to scale the design up from a Aspen Simulation Flowsheet to a detailed Process and Instrumentation Diagram (P&ID), followed by a Pilot-Plant or one Single Genset CCC prototype unit construction and design implementation. This would give a better understanding of system dynamics, improve confidence in the equipment design, and allow the operators to carry out more accurate measurement of Amine-slip behaviour under actual operating conditions. A full-scale setup would also make it possible to study energy balance of the plant, water management and solvent degradation in more detail under different loads and operating conditions. In addition, the AMP and PZ composition in the CESAR–1 solvent could be varied in future sensitivity studies to verify whether or not the mass transfer rate, Reboiler thermal energy demand and solvent degradation rate can be is slightly or heavily influenced by changing solvent composition.

Finally, the transport logistics, storage and possible utilization pathways for the Captured CO_2 could be explored in more detail. A future master's thesis could also extend this work by including a more complete techno-economic assessment of the full CO_2 capture, transport, storage, and utilization chain.

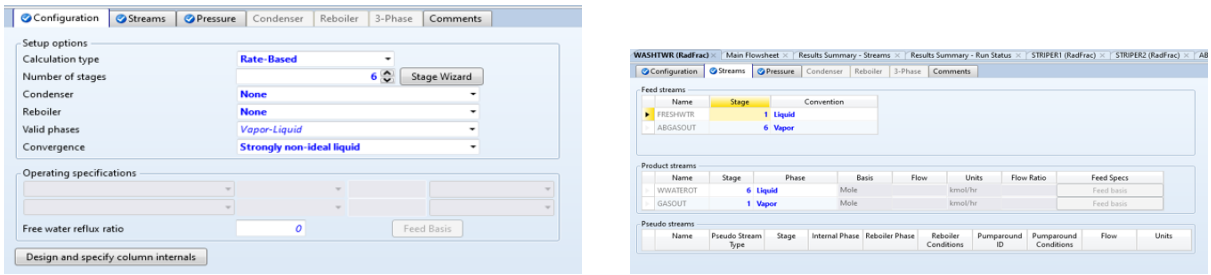
A Aspen Plus Simulation Graphic User Interface

The Aspen Plus Simulation Environment and Graphic User Interface (GUI) for Two Genset Maximum Power Operation mode are summarized here in this Appendix A section.

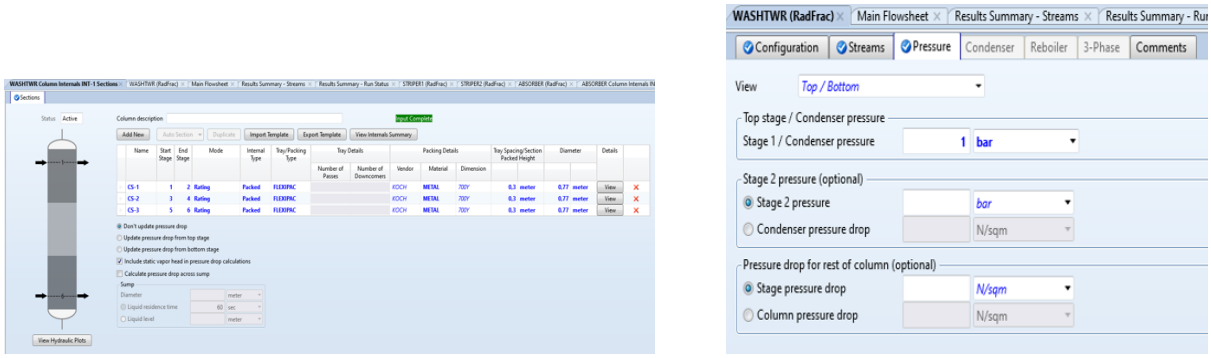


(a) (b) (c)

Figure 29: Aspen Plus GUI for Input Material Streams



(a) (b)



(c) (d)

Figure 30: Aspen Plus GUI for Absorber Unit Input

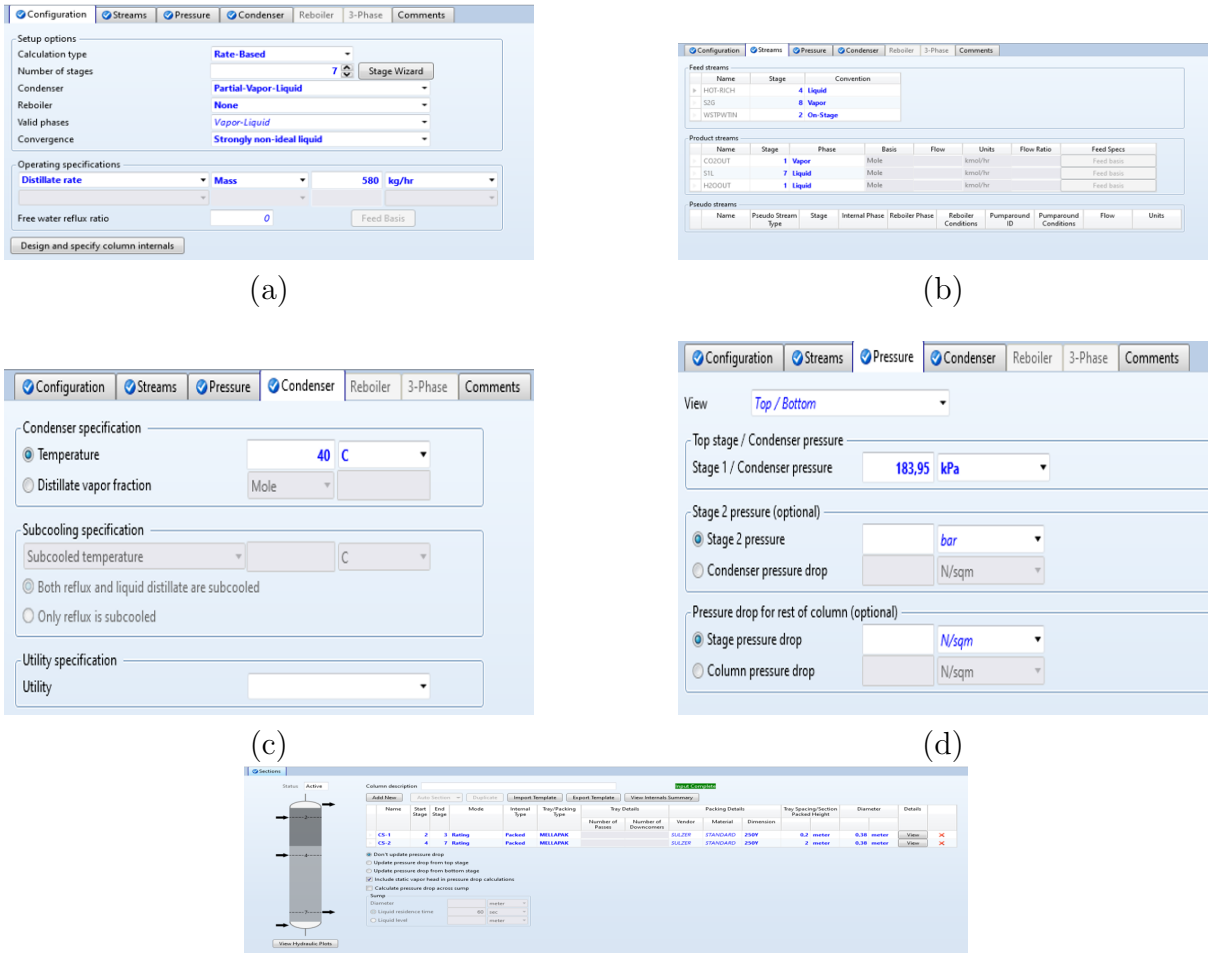
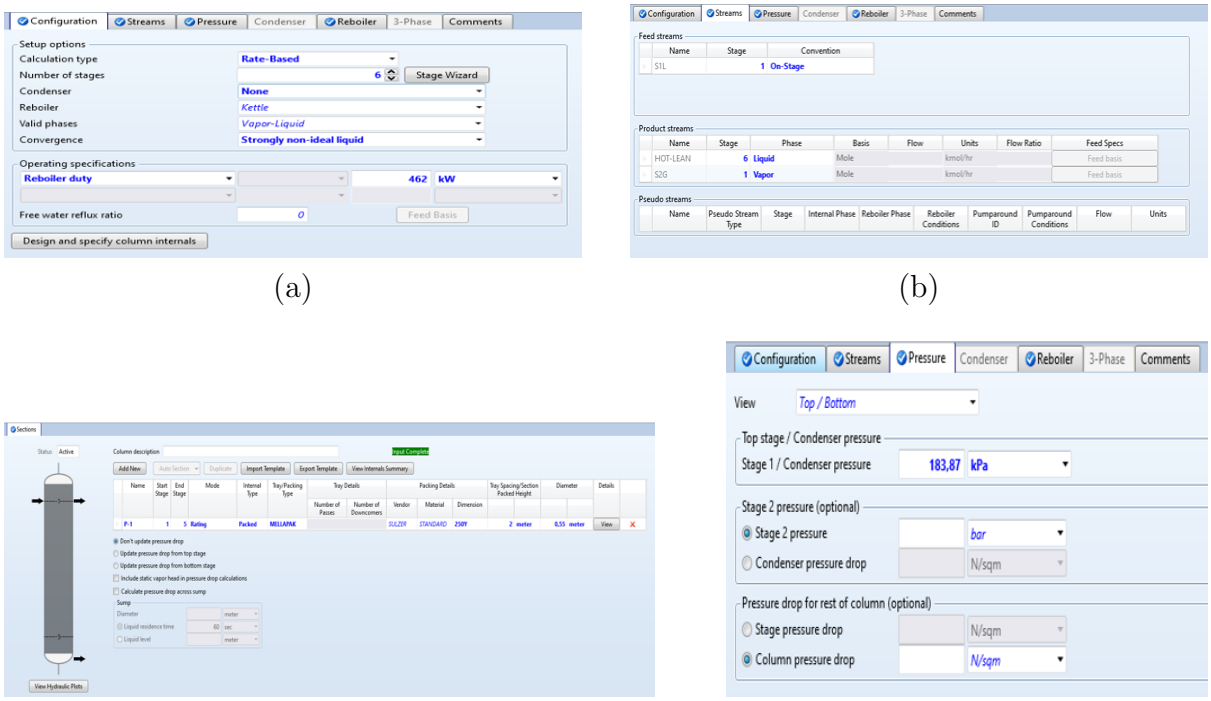


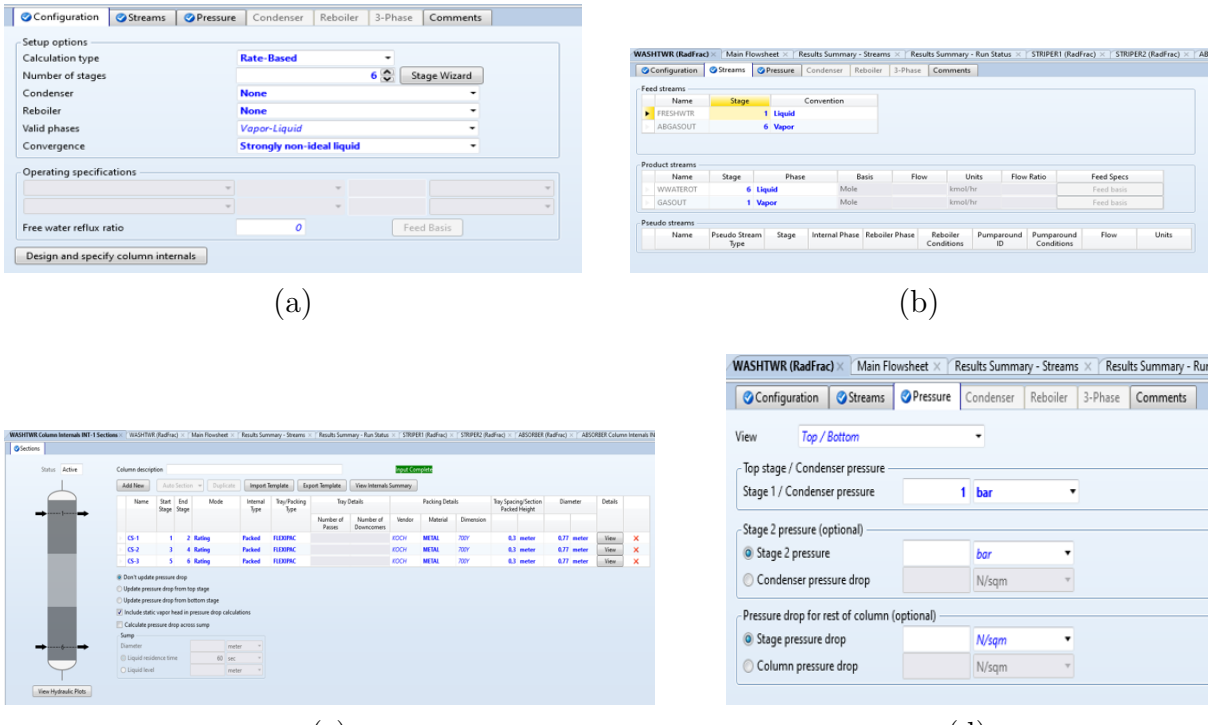
Figure 31: Aspen Plus GUI for Stripper 1 Unit Input



(a) (b)

(c) (d)

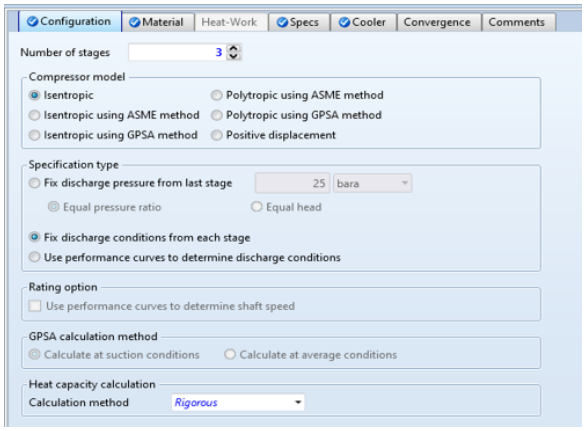
Figure 32: Aspen Plus GUI for Stripper 2 Unit Input



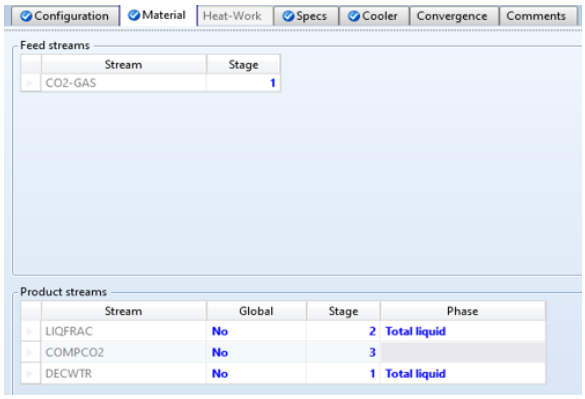
(a) (b)

(c) (d)

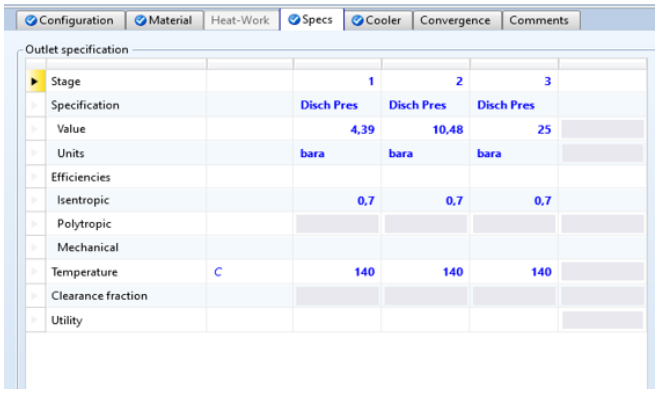
Figure 33: Aspen Plus GUI for Amine Wash Tower Unit Input



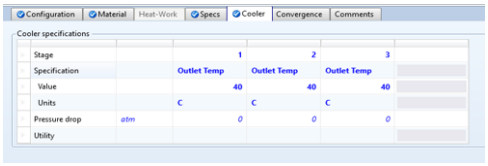
(a)



(b)



(c)



(d)

Figure 34: Aspen Plus GUI for Compressor Train Unit Input

B Aspen Plus Simulation Parameter Results

The Aspen Plus Simulation Parameter Results for the important Parameters of Single Genset and Two Gensets Continuous Power Operation and Maximum Power Operation modes are summarized here in this Appendix B section.

Table 33: Comparison of Carbon Capture Circuit Results and Performance for Single Genset System Operation

Parameter	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
Captured CO_2 – Absorber (kg/hr)	259.72	178.74
Desorbed CO_2 – Stripper (kg/hr)	258.92	178.10
CO_2 Capture Efficiency – Absorber (%)	∼90.00	∼90.00
CO_2 Capture Efficiency – Overall (%)	∼90.00	∼90.00
Reboiler Thermal Energy Demand (kW-th)	217	147
Specific Reboiler Heat Duty (MJ-th/kg CO_2)	3.02	2.97
AMP Concentration in Absorber Gas out (ppmv)	914.30	831.30
PZ Concentration in Absorber Gas out (ppmv)	27.90	25.50
Rich Liquid Mix loading (mol CO_2/mol AMP+PZ)	0.623	0.635
Rich Liquid Mix loading (% CO_2/AMP+PZ)	62.30	63.50
Specific Stripper Steam Demand (kg Steam/kg CO_2 Captured)	1.154	1.134

Table 34: Comparison of Carbon Capture Circuit Results and Performance for Two Genset System Operation

Parameter	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
Captured CO_2 – Absorber (kg/hr)	518.67	356.99
Desorbed CO_2 – Stripper (kg/hr)	519.14	357.30
CO_2 Capture Efficiency – Absorber (%)	∼90.00	∼90.00
CO_2 Capture Efficiency – Overall (%)	∼90.00	∼90.00
Reboiler Thermal Energy Demand (kW-th)	462	307
Specific Reboiler Heat Duty (MJ-th/kg CO_2)	3.20	3.09
AMP Concentration in Absorber Gas out (ppmv)	846.00	783.50
PZ Concentration in Absorber Gas out (ppmv)	26.00	24.40
Rich Liquid Mix loading (mol CO_2/mol AMP+PZ)	0.581	0.604
Rich Liquid Mix loading (% CO_2/AMP+PZ)	58.10	60.40
Specific Stripper Steam Demand (kg Steam/kg CO_2 Captured)	1.239	1.191

Table 35: Aspen Plus Simulation Backpressure Results for Single Genset System Operation

Parameter	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
Exhaust Gas Pressure Drop in Absorber (mbar(a))	0.957	0.957
Exhaust Gas Pressure Drop in Amine Wash Tower (mbar(a))	2.396	1.635
Exhaust Gas Total Pressure Drop (mbar(a))	3.353	2.592
CO_2 Stream Pressure Drop in the Stripper 1 (mbar(a))	0.703	0.574
CO_2 Stream Pressure Drop in the Stripper 2 (mbar(a))	0.126	0.054
CO_2 Stream Total Pressure Drop (mbar(a))	0.829	0.628

Table 36: Aspen Plus Simulation Backpressure Results for Two Genset System Operation

Parameter	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
Exhaust Gas Pressure Drop in Absorber (mbar(a))	3.677	3.560
Exhaust Gas Pressure Drop in Amine Wash Tower (mbar(a))	7.708	5.029
Exhaust Gas Total Pressure Drop (mbar(a))	11.385	8.589
CO_2 Stream Pressure Drop in the Stripper 1 (mbar(a))	2.509	0.984
CO_2 Stream Pressure Drop in the Stripper 2 (mbar(a))	0.479	0.201
CO_2 Stream Total Pressure Drop (mbar(a))	2.988	1.185

Table 37: Aspen Plus Simulation Amine Slip Results for Single Genset System Operation

Parameter	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
AMP Conc. in Amine Wash Tower Gas Out (PPBV)	19700	18700
PZ Conc. in Amine Wash Tower Gas Out (PPBV)	6.30	3.30
AMP Conc. in Cooler 2 Gas Out (PPBV)	9.10	11.30
PZ Conc. in Cooler 2 Gas Out (PPBV)	0.0004	0.0003

Table 38: Aspen Plus Simulation Amine Slip Results for Two Genset System Operation

Parameter	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
AMP Conc. in Amine Wash Tower Gas Out (PPBV)	20300	18200
PZ Conc. in Amine Wash Tower Gas Out (PPBV)	21.90	13.60
AMP Conc. in Cooler 2 Gas Out (PPBV)	10.70	11.50
PZ Conc. in Cooler 2 Gas Out (PPBV)	0.002	0.001

Table 39: Comparison of Carbon Capture Circuit Results and Performance for Operation under Hot Climatic Conditions

Parameter	**Single Genset System**		**Two Gensets System**	
	Maximum Power Operation (MPO) Mode	**Continuous Power Operation (CPO) Mode**	**Maximum Power Operation (MPO) Mode**	**Continuous Power Operation (CPO) Mode**
Captured CO_2 – Absorber (kg/hr)	258.89	178.16	517.47	356.19
Desorbed CO_2 – Stripper (kg/hr)	258.93	178.48	518.76	357.09
CO_2 Capture Efficiency – Absorber (%)	∼90.00	∼90.00	∼90.00	∼90.00
CO_2 Capture Efficiency – Overall (%)	∼90.00	∼90.00	∼90.00	∼90.00
Reboiler Thermal Energy Demand (kW-th)	233	158	508	335
Specific Reboiler Heat Duty (MJ-th/kg CO_2)	3.23	3.18	3.52	3.38
Rich Liquid Mix loading (mol CO_2/mol AMP+PZ)	0.556	0.566	0.513	0.532
Rich Liquid Mix loading (% CO_2/AMP+PZ)	55.60	56.60	51.30	53.20
Specific Stripper Steam Demand (kg Steam/kg CO_2 Captured)	1.245	1.218	1.364	1.305

Table 40: CO_2 Compression Results from the Compressor Train for Operation under Normal Climatic Conditions

Parameter	Single Genset System		Two Gensets System	
	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode
CO_2 Mix Stream Flow rate – Inlet (kg/hr)	263.51	181.22	528.73	363.60
CO_2 Flow rate – Inlet (kg/hr)	259.01	178.13	519.74	357.41
CO_2 Content – Inlet (wt%)	98.30	98.30	98.30	98.30
CO_2 Stream Temperature – Inlet (°C)	40	40	40	40
CO_2 Stream Pressure – Inlet (bar(a))	1.84	1.84	1.84	1.84
Compressor Isentropic Efficiency (%)	70	70	70	70
Compressor Electricity Demand (kW-elec)	17.37	11.94	34.85	23.97
Intercooler Cooling Duty Demand (kW-th)	21.42	14.73	42.96	29.55
CO_2 Mix Stream Flow rate – Outlet (kg/hr)	259.88	178.72	521.47	358.60
CO_2 Flow rate – Outlet (kg/hr)	258.99	178.11	519.70	357.38
CO_2 Content – Outlet (wt%)	99.70	99.70	99.70	99.70
CO_2 Stream Temperature – Outlet (°C)	40	40	40	40
CO_2 Stream Pressure – Outlet (bar(a))	25	25	25	25

Table 41: CO_2 Compression Results from the Compressor Train for Operation under Hot Climatic Conditions

Parameter	Single Genset System		Two Gensets System	
	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode
CO_2 Mix Stream Flow rate – Inlet (kg/hr)	266.57	183.76	534.06	367.63
CO_2 Flow rate – Inlet (kg/hr)	258.93	178.48	518.77	357.09
CO_2 Content – Inlet (wt%)	97.10	97.10	97.10	97.10
CO_2 Stream Temperature – Inlet (°C)	50	50	50	50
CO_2 Stream Pressure – Inlet (bar(a))	1.84	1.84	1.84	1.84
Compressor Isentropic Efficiency (%)	70	70	70	70
Compressor Electricity Demand (kW-elec)	17.92	12.35	35.90	24.72
Intercooler Cooling Duty Demand (kW-th)	23.72	16.35	47.49	32.70
CO_2 Mix Stream Flow rate – Outlet (kg/hr)	259.62	178.96	521.66	359.08
CO_2 Flow rate – Outlet (kg/hr)	258.89	178.44	518.71	357.05
CO_2 Content – Outlet (wt%)	99.70	99.70	99.40	99.40
CO_2 Stream Temperature – Outlet (°C)	50	50	50	50
CO_2 Stream Pressure – Outlet (bar(a))	25	25	25	25

Table 42: CO_2 Liquefaction Results from the Refrigeration Cycle for Operation under Normal Climatic Conditions

Parameter	Single Genset System		Two Gensets System	
	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode
CO_2 Mix Stream Flow rate – Outlet (kg/hr)	259.88	178.72	521.47	358.60
CO_2 Flow rate – Inlet (kg/hr)	258.99	178.11	519.70	357.38
CO_2 Content – Outlet (wt%)	99.70	99.70	99.70	99.70
CO_2 Stream Temperature – Inlet (°C)	40	40	40	40
CO_2 Stream Pressure – Inlet (bar(a))	25	25	25	25
Refrigerator Compressor Isentropic Efficiency (%)	70	70	70	70
Refrigerator Electricity Demand (kW-elec)	14.96	10.29	29.91	20.58
Refrigerator Cooling Duty Demand (kW-th)	40.40	27.80	80.80	55.59
Refrigerator Cooling Water Flow rate (kg/hr)	2316.30	1593.63	6948.59	3187.26
Refrigerator COP_{Cooling}	1.701	1.701	1.701	1.701
CO_2 Stream Temperature – Outlet (°C)	-25	-25	-25	-25
CO_2 Stream Pressure – Outlet (bar(a))	20	20	20	20

Table 43: CO_2 Liquefaction Results from the Refrigeration Cycle for Operation under Hot Climatic Conditions

Parameter	Single Genset System		Two Gensets System	
	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode	Maximum Power Operation (MPO) Mode	Continuous Power Operation (CPO) Mode
CO_2 Mix Stream Flow rate – Outlet (kg/hr)	259.62	178.96	521.66	359.08
CO_2 Flow rate – Inlet (kg/hr)	258.89	178.44	518.71	357.05
CO_2 Content – Inlet (wt%)	99.70	99.70	99.40	99.40
CO_2 Stream Temperature – Inlet (°C)	50	50	50	50
CO_2 Stream Pressure – Inlet (bar(a))	25	25	25	25
Refrigerator Compressor Isentropic Efficiency (%)	70	70	70	70
Refrigerator Electricity Demand (kW-elec)	15.40	10.59	30.79	21.19
Refrigerator Cooling Duty Demand (kW-th)	41.59	28.62	83.19	57.23
Refrigerator Cooling Water Flow rate (kg/hr)	3577.07	2461.05	7154.13	4922.09
Refrigerator COP_{Cooling}	1.701	1.701	1.701	1.701
CO_2 Stream Temperature – Outlet (°C)	-25	-25	-25	-25
CO_2 Stream Pressure – Outlet (bar(a))	20	20	20	20

Table 44: Absorber Column Height Sensitivity Analysis

Absorber Height (m)	L/G Ratio (kg LLM/kg FG)	L/G Ratio (mol LLM/mol FG)	Rich Liquid Mix CO_2 loading (mol CO_2/mol AMP+PZ)	Reboiler duty (kW_{th})	Specific Reboiler Duty (MJ_{th}/kg CO_2)	% SRD Change
1.65	1.63	1.80	0.54	487	3.378	–
1.75	1.57	1.74	0.56	477	3.309	2.096
1.85	1.53	1.69	0.57	469	3.253	1.706
1.95	1.49	1.64	0.58	462	3.205	1.515
2.05	1.46	1.62	0.59	457	3.170	1.094
2.15	1.44	1.59	0.60	452	3.135	1.106
2.25	1.42	1.56	0.61	448	3.108	0.893

Table 45: Total Stripper Column (Stripper 1 & Stripper 2) Height Sensitivity Analysis

Stripper Column Total Height (m)	Reboiler Duty (kW_{th})	Specific Reboiler Duty (MJ_{th}/kg CO_2)	% SRD Change
3.40	469	3.253	–
3.60	466	3.232	0.644
3.80	464	3.218	0.431
4.00	462	3.205	0.433
4.20	460	3.191	0.435
4.40	459	3.184	0.218
4.60	458	3.177	0.218

Table 46: Lean Liquid Mix CO_2 loading Sensitivity Analysis

Lean Liquid Mix CO_2 loading (mol CO_2/mol AMP+PZ)	L/G Ratio (kg LLM/kg FG)	L/G Ratio (mol LLM/mol FG)	Rich Liquid Mix CO_2 loading (mol CO_2/mol AMP+PZ)	Reboiler Duty (kW_{th})	Specific Reboiler Duty (MJ_{th}/kg CO_2)	% SRD Change
0.060	1.32	1.46	0.60	524	3.635	–
0.080	1.39	1.54	0.59	474	3.288	10.549
0.100	1.49	1.64	0.58	462	3.205	2.597
0.120	1.60	1.76	0.57	463	3.212	0.216
0.140	1.71	1.87	0.56	467	3.239	0.857
0.160	1.86	2.03	0.55	473	3.281	1.268

Table 47: Recuperator Unit Overall Heat Transfer Area Sensitivity Analysis

Recuperator Overall Heat Transfer Coefficient ($W/m^2 \cdot K$)	Heat Transfer Area × Heat Transfer Coefficient (W/K)	Reboiler Duty (kW_{th})	Specific Reboiler Duty (MJ_{th}/kg CO_2)	% SRD Change
22.0	18700	485.0	3.364	–
35.0	29750	472.0	3.274	2.754
43.0	36550	467.5	3.243	0.963
65.0	55250	462.0	3.205	1.190
92.0	78200	457.0	3.170	1.094
140.0	119000	454.0	3.149	0.661
217.0	184450	452.0	3.135	0.442
555.0	471750	450.0	3.121	0.444

Table 48: Recuperator Unit Overall Heat Transfer Coefficient Sensitivity Analysis

Recuperator Overall Heat Transfer Coefficient ($W/m^2 \cdot K$)	Heat Transfer Area × Heat Transfer Coefficient (W/K)	Reboiler Duty (kW_{th})	Specific Reboiler Duty (MJ_{th}/kg CO_2)	% SRD Change
1000	65000	459.0	3.184	–
950	61750	460.0	3.191	0.217
900	58500	460.5	3.194	0.109
850	55250	462.0	3.205	0.325
800	52000	462.5	3.208	0.108
750	48750	463.0	3.212	0.108
700	45500	464.0	3.218	0.216
650	42250	465.0	3.225	0.215

Table 49: Specific Compressor Power and Specific Intercooler Heat Duty sensitivity analysis

No. of Compressors /Intercoolers	Gas Flow rate (kg/hr)	Compressor Duty (kW_{elec})	Specific Compressor Duty (MJ_{elec}/kg CO_2)	% Change	Intercooler Heat Duty (kW_{th})	Specific Intercooler Heat Duty (MJ_{th}/kg CO_2)	% Change
1	528.725	43.43	0.301	–	-51.55	0.358	–
2	528.725	36.86	0.256	17.814	-44.97	0.312	14.623
3	528.725	34.85	0.242	5.768	-42.96	0.298	4.683
4	528.725	33.87	0.235	2.878	-41.98	0.291	2.323
5	528.725	33.30	0.231	1.728	-41.41	0.287	1.391
6	528.725	32.92	0.228	1.154	-41.03	0.285	0.927

Table 50: CAPEX Breakdown for the Single Genset CCC unit

Equipment Name	Capacity (A)	Capacity Unit	Price (M€$_{2024}$)	Reference Data Year	Reference
Absorber Column	1.045	m^3	0.0060	2001	(Turton et al., 2012)
Absorber Column Packing	0.908	m^3	0.0004	2024	(Haiyan Zhongxin Chemical Equipment Co., Ltd., 2026)
Stripper Column 1	0.287	m^3	0.0036	2001	(Turton et al., 2012)
Stripper Column 2	0.547	m^3	0.0046	2001	(Turton et al., 2012)
Stripper Column Packing	0.725	m^3	0.0003	2024	(Haiyan Zhongxin Chemical Equipment Co., Ltd., 2026)
Condenser	4.706	m^2	0.0209	2018	(Kim & Léonard, 2025)
Reboiler	47.782	m^2	0.0921	2001	(Turton et al., 2012)
Recuperator	32.811	m^2	0.1137	2001	(Turton et al., 2012)
Amine Wash Tower	0.482	m^3	0.0044	2001	(Turton et al., 2012)
Amine Wash Tower Packing	0.419	m^3	0.0002	2024	(Haiyan Zhongxin Chemical Equipment Co., Ltd., 2026)
Lean Liquid Mix Cooler (Cooler 1)	7.168	m^2	0.0324	2018	(Aromada, Eldrup, Normann, et al., 2020)
CO_2–Lean Gas Out Cooler (Cooler 2)	190.706	m^2	0.0568	2001	(Turton et al., 2012)
Exhaust Gas Cooler	27.011	m^2	0.0314	2001	(Turton et al., 2012)
Absorption Heat Pump Type II	75.790	kW-th	0.1384	2019	(Cudok et al., 2021)
Exhaust Gas to Process Water Heat Exchanger	49.327	m^2	0.0355	2001	(Turton et al., 2012)
Charge Air to Coolant Heat Exchanger	26.949	m^2	0.0314	2001	(Turton et al., 2012)
Exhaust Gas to Coolant Heat Exchanger	16.661	m^2	0.0294	2001	(Turton et al., 2012)
Cooling Water to Air Heat Exchanger	51.212	m^2	0.0707	2001	(Turton et al., 2012)
Multi-stage Compressor	19.976	kW-elec	0.1823	2020	(Shamoushaki et al., 2021)
Compressor Train Intercoolers	2.763	m^2	0.0152	2018	(Aromada, Eldrup, Normann, et al., 2020)
Refrigeration System	27.995	kW-th	0.0362	2014	(Moray Council, 2015)
Rich Liquid Mix Pump (Pump 1)	0.2262	kW-elec	0.0124	2010	(Towler & Sinnott, 2022)
Lean Liquid Mix Pump (Pump 2)	0.0004	kW-elec	0.0124	2010	(Towler & Sinnott, 2022)
Amine Wash Water Pump (Pump 3)	0.0024	kW-elec	0.0121	2010	(Towler & Sinnott, 2022)
Initial AMP	112.840	kg	0.0012	2015	(Manzolini et al., 2015)
Initial PZ	54.43	kg	0.0004	2015	(Manzolini et al., 2015)
Initial Process Water (PW)	4.70	m^3	0.00004	2015	(Manzolini et al., 2015)
Initial Cooling Water (CW)	3.06	m^3	0.000001	2020	(Nazir et al., 2020)
Total Cost	–	–	**0.9442**	–	–

Table 51: CAPEX Breakdown for the Two Gensets CCC unit

Equipment Name	Capacity (A)	Capacity Unit	Price (M€$_{2024}$)	Reference Data Year	Reference
Absorber Column	1.045	m^3	0.0060	2001	(Turton et al., 2012)
Absorber Column Packing	0.908	m^3	0.0004	2024	(Haiyan Zhongxin Chemical Equipment Co., Ltd., 2026)
Stripper Column 1	0.287	m^3	0.0036	2001	(Turton et al., 2012)
Stripper Column 2	0.547	m^3	0.0046	2001	(Turton et al., 2012)
Stripper Column Packing	0.725	m^3	0.0003	2024	(Haiyan Zhongxin Chemical Equipment Co., Ltd., 2026)
Condenser	10.950	m^2	0.0286	2001	(Turton et al., 2012)
Reboiler	102.471	m^2	0.1883	2018	(Aromada, Eldrup, Normann, et al., 2020)
Recuperator	68.944	m^2	0.1498	2001	(Turton et al., 2012)
Amine Wash Tower	0.482	m^3	0.0044	2001	(Turton et al., 2012)
Amine Wash Tower Packing	0.419	m^3	0.0002	2024	(Haiyan Zhongxin Chemical Equipment Co., Ltd., 2026)
Lean Liquid Mix Cooler (Cooler 1)	15.622	m^2	0.0513	2018	(Aromada, Eldrup, Normann, et al., 2020)
CO_2–Lean Gas Out Cooler (Cooler 2)	329.711	m^2	0.0769	2001	(Turton et al., 2012)
Exhaust Gas Cooler	54.022	m^2	0.0364	2001	(Turton et al., 2012)
Absorption Heat Pump Type II	151.470	kW-th	0.2766	2019	(Cudok et al., 2021)
Exhaust Gas to Process Water Heat Exchanger	98.654	m^2	0.0437	2001	(Turton et al., 2012)
Charge Air to Coolant Heat Exchanger	53.898	m^2	0.0364	2001	(Turton et al., 2012)
Exhaust Gas to Coolant Heat Exchanger	30.854	m^2	0.0326	2001	(Turton et al., 2012)
Cooling Water to Air Heat Exchanger	106.678	m^2	0.0962	2001	(Turton et al., 2012)
Multi-stage Compressor	40.076	kW-elec	0.1935	2020	(Shamoushaki et al., 2021)
Compressor Train Intercoolers	5.132	m^2	0.0229	2018	(Aromada, Eldrup, Normann, et al., 2020)
Refrigeration System	55.990	kW-th	0.0549	2014	(Moray Council, 2015)
Rich Liquid Mix Pump (Pump 1)	0.4490	kW-elec	0.0126	2010	(Towler & Sinnott, 2022)
Lean Liquid Mix Pump (Pump 2)	0.0007	kW-elec	0.0126	2010	(Towler & Sinnott, 2022)
Amine Wash Water Pump (Pump 3)	0.0047	kW-elec	0.0121	2010	(Towler & Sinnott, 2022)
Initial AMP	244.38	kg	0.0026	2015	(Manzolini et al., 2015)
Initial PZ	117.88	kg	0.0009	2015	(Manzolini et al., 2015)
Initial Process Water (PW)	10.01	m^3	0.00008	2015	(Manzolini et al., 2015)
Initial Cooling Water (CW)	6.37	m^3	0.000002	2020	(Nazir et al., 2020)
Total Cost	–	–	**1.3475**	–	–

Table 52: Full OPEX Analysis for Single Genset CCC Unit

Parameter	**Value**	**Unit**	**Yearly Operating Cost (€$_{2024}$/year)**
Makeup Materials			
Makeup AMP	2.26	kg	23.51
Makeup PZ	1.09	kg	8.50
Makeup Process Water	0.09	m^3	0.73
Makeup Cooling Water	0.06	m^3	0.02
Electricity Consumption			
Heat Pump Electricity	0.4260	kW-elec	184.03
CO_2 Compressor Train Electricity	11.9400	kW-elec	5158.08
Refrigeration Cycle Compressor Electricity	10.2900	kW-elec	4445.28
Pump 1 Duty	0.1315	kW-elec	56.81
Pump 2 Duty	0.0002	kW-elec	0.09
Pump 3 Duty	0.0014	kW-elec	0.60
Capital and Annual Operating Costs			
Total Maintenance and Repair	–	–	132086
Total Operating Supplies	–	–	14859.63
Total Local Taxes and Insurance	–	–	52834.23
Annual Operational Expenditure (OPEX)	–	–	209657
Total Capital Investment (TCI)	1651070	€$_{2024}$	–
Annualized Capital Investment	219097	€$_{2024}$	–
Total CO_2 Captured	357.28	Metric Tonne CO_2/year	–
Specific Cost of CO_2 Captured	**1200.05**	**€$_{2024}$/tonne CO_2**	–

Table 53: Full OPEX Analysis for Two Gensets CCC Unit

Parameter	**Value**	**Unit**	**Yearly Operating Cost (€$_{2024}$/year)**
Makeup Materials			
Makeup AMP	4.89	kg	50.91
Makeup PZ	2.36	kg	18.42
Makeup Process Water	0.20	m^3	1.56
Makeup Cooling Water	0.13	m^3	0.04
Electricity Consumption			
Heat Pump Electricity	0.8520	kW-elec	368.06
CO_2 Compressor Train Electricity	23.9700	kW-elec	10355.04
Refrigeration Cycle Compressor Electricity	20.5800	kW-elec	8890.56
Pump 1 Duty	0.2812	kW-elec	121.48
Pump 2 Duty	0.0005	kW-elec	0.22
Pump 3 Duty	0.0029	kW-elec	1.25
Capital and Annual Operating Costs			
Total Maintenance and Repair	–	–	188436
Total Operating Supplies	–	–	21198.98
Total Local Taxes and Insurance	–	–	75374.15
Annual Operational Expenditure (OPEX)	–	–	304816
Total Capital Investment (TCI)	2355442	€$_{2024}$	–
Annualized Capital Investment	312567	€$_{2024}$	–
Total CO_2 Captured	714.56	Metric Tonne CO_2/year	–
Specific Cost of CO_2 Captured	**864.00**	**€$_{2024}$/tonne CO_2**	–

C Aspen Plus Simulation Stream Results

The Aspen Plus Simulation Stream Results for some of the important streams of Two Gensets Maximum Power Operation mode are summarized here in this Appendix C section.

Table 54: Aspen Plus Stream Report: Important Streams and Mixed Sub-streams

STREAMS	UNIT	ABGASOUT	CLD-LEAN	CLDGAS	CO2OUT	HOT-LEAN	HOT-RICH	LEAN-LM	LEANLM-C	RICH-LM	SAT-FG
From		ABSORBER	RECUP-HX	COOLER2	STRIPER1	STRIPER2	RECUP-HX		COOLER1	ABSORBER	
To		WASHTWR	MIX1			PUMP-2	STRIPER1	ABSORBER		PUMP-1	ABSORBER
Stream Class		CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Phase		Vapor Phase	Liquid Phase		Vapor Phase	Liquid Phase	Liquid Phase	Liquid Phase	Liquid Phase	Liquid Phase	Vapor Phase
Temperature	C	60.805	56.991	40.0	40.0	119.065	111.341	40.0	40.0	51.626	40.0
Pressure	bar(a)	1	2	1	1.8395	1.8387	1.8387	1.01325	1.01325	1	1.01325
Molar Vapor Fraction		1	0	0.8794	1	0	0	0	0	0	1
Molar Liquid Fraction		0	1	0.1206	0	1	1	1	1	1	0
Molar Solid Fraction		0	0	0	0	0	0	0	0	0	0
Mass Vapor Fraction		1	0	0.9183	1	0	0	0	0	0	1
Mass Liquid Fraction		0	1	0.0817	0	1	1	1	1	1	0
Mass Solid Fraction		0	0	0	0	0	0	0	0	0	0
Molar Enthalpy	kJ/mol	-47.3600	-283.6041	-53.5309	-386.8313	-277.3083	-305.8256	-285.3616	-285.3146	-312.2331	-55.6235
Mass Enthalpy	MJ/kg	-1.7739	-10.4301	-2.0110	-9.0039	-10.1988	-10.3087	-10.7753	-10.5020	-10.5193	-1.9037
Molar Entropy	kJ/kmol-K	1.4903	-221.0564	-16.2921	-0.8460	-203.5324	-205.7846	-221.0185	-226.0452	-226.3748	5.5592
Mass Entropy	kJ/kg-K	0.0558	-8.1298	-0.6121	-0.0197	-7.4855	-6.9365	-8.3457	-8.3203	-7.6267	0.1903
Molar Density	kmol/cum	0.0361	36.9777	0.0437	0.0712	34.9604	35.1419	38.3813	37.4461	36.8920	0.0390
Mass Density	kg/cum	0.9627	1005.4591	1.1634	3.0607	950.5809	1042.5490	1016.4464	1017.3276	1095.0249	1.1383
Enthalpy Flow	MW	-1.7895	-15.7494	-2.0262	-1.3224	-15.4002	-17.1111	-17.2106	-15.8905	-17.4608	-2.0396
Average MW		26.6978	27.1909	26.6184	42.9627	27.1902	29.6668	26.4828	27.1678	29.6819	29.2193
Mole Flows	kmol/hr	136.0266	199.9190	136.2660	12.3066	199.9242	201.4223	217.1217	200.5005	201.3198	132.0076
PZ	kmol/hr	0.0035	5.4592	8.10E-08	2.42E-10	5.7048	0.7675	5.4162	5.4418	0.4445	0
H_2O	kmol/hr	24.9590	175.1306	25.3362	0.4941	174.8173	172.4397	192.3085	175.7403	173.9793	9.2708
CO_2	kmol/hr	1.3262	4.08E-05	1.3017	11.8097	0.0052	0.1047	8.04E-06	7.23E-06	0.0023	13.1115
H2S	kmol/hr	0	0	0	0	0	0	0	0	0	0
H_3O^+	kmol/hr	0	6.81E-10	4.68E-09	0	1.90E-08	1.09E-07	2.21E-10	1.99E-10	6.97E-09	0
OH^-	kmol/hr	0	0.0019	6.44E-07	0	0.0011	0.0003	0.0024	0.0022	0.0003	0
HCO_3^-	kmol/hr	0	0.1217	0.0027	0	0.4716	4.2855	0.0718	0.0622	2.2635	0
CO_3^{2-}	kmol/hr	0	0.0395	1.48E-05	0	0.0038	0.0479	0.0789	0.0647	0.5303	0
PZH^+	kmol/hr	0	0.6385	2.70E-06	0	0.8421	1.0042	0.5789	0.5535	0.5960	0
$PZCOO^-$	kmol/hr	0	1.9516	8.83E-09	0	1.3535	1.5858	2.1122	2.1200	1.9460	0
$PZ(COO^-)_2$	kmol/hr	0	0.1506	2.03E-10	0	0.2862	3.3045	0.0961	0.0895	4.1284	0
HPZCOO	kmol/hr	0	0.0105	2.00E-07	0	0.0239	1.5452	0.0073	0.0057	1.0923	0
AMP	kmol/hr	0.1151	14.5560	7.74E-05	8.62E-07	14.8205	4.5831	14.4519	14.4410	3.0502	0
$AMPH^+$	kmol/hr	0	1.8379	0.0027	0	1.5791	11.6617	1.9775	1.9594	13.1076	0
$AMPCOO^-$	kmol/hr	0	0.0208	2.58E-08	0	0.0151	0.0895	0.0199	0.0201	0.1765	0
N_2	kmol/hr	99.0517	0	99.0517	0.0023	7.03E-15	0.0023	0	2.97E-09	0.0023	99.0540
O_2	kmol/hr	10.5710	5.10E-05	10.5710	0.0005	5.10E-05	0.0005	0	5.10E-05	0.0005	10.5714
Mole Fractions											
PZ		2.60E-05	0.0273	5.94E-10	1.97E-11	0.0285	0.0038	0.0249	0.0271	0.0022	0
H_2O		0.1835	0.8760	0.1859	0.0401	0.8744	0.8561	0.8857	0.8765	0.8642	0.0702
CO_2		0.0097	2.04E-07	0.0096	0.9596	2.60E-05	0.0005	3.70E-08	3.60E-08	1.13E-05	0.0993
H2S		0	0	0	0	0	0	0	0	0	0
H_3O^+		0	3.40E-12	3.43E-11	0	9.50E-11	5.42E-10	1.02E-12	9.94E-13	3.46E-11	0
OH^-		0	9.59E-06	4.72E-09	0	5.31E-06	1.38E-06	1.10E-05	1.09E-05	1.63E-06	0
HCO_3^-		0	0.0006	1.96E-05	0	0.0024	0.0213	0.0003	0.0003	0.0112	0
CO_3^{2-}		0	0.0002	1.09E-07	0	1.90E-05	0.0002	0.0004	0.0003	0.0026	0

STREAMS		ABGASOUT	CLD-LEAN	CLDGAS	CO2OUT	HOT-LEAN	HOT-RICH	LEAN-LM	LEANLM-C	RICH-LM	SAT-FG
PZH^+		0	0.0032	1.98E-08	0	0.0042	0.0050	0.0027	0.0028	0.0030	0
$PZCOO^-$		0	0.0098	6.48E-11	0	0.0068	0.0079	0.0097	0.0106	0.0097	0
$PZ(COO-)_2$		0	0.0008	1.49E-12	0	0.0014	0.0164	0.0004	0.0004	0.0205	0
HPZCOO		0	5.25E-05	1.47E-09	0	0.0001	0.0077	3.36E-05	2.84E-05	0.0054	0
AMP		0.0008	0.0728	5.68E-07	7.00E-08	0.0741	0.0228	0.0666	0.0720	0.0152	0
$AMPH^+$		0	0.0092	1.98E-05	0	0.0079	0.0579	0.0091	0.0098	0.0651	0
$AMPCOO^-$		0	0.0001	1.89E-10	0	7.55E-05	0.0004	9.15E-05	0.0001	0.0009	0
N_2		0.7282	0	0.7269	0.0002	3.51E-17	1.14E-05	0	1.48E-11	1.14E-05	0.7504
O_2		0.0777	2.55E-07	0.0776	3.68E-05	2.55E-07	2.24E-06	0	2.54E-07	2.24E-06	0.0801
Mass Flows	kg/hr	3631.6153	5435.9835	3627.1815	528.7249	5435.9835	5975.5531	5750.0001	5447.1486	5975.5532	3857.1680
PZ	kg/hr	0.3042	470.2415	6.98E-06	2.09E-08	491.3913	66.1075	466.5325	468.7406	38.2845	0
H_2O	kg/hr	449.6441	3155.0263	456.4379	8.9012	3149.3822	3106.5492	3464.4915	3166.0100	3134.2858	167.0154
CO_2	kg/hr	58.3671	0.0018	57.2892	519.7442	0.2288	4.6089	0.0004	0.0003	0.1002	577.0323
H2S	kg/hr	0	0	0	0	0	0	0	0	0	0
H_3O^+	kg/hr	0	1.29E-08	8.89E-08	0	3.61E-07	2.08E-06	4.21E-09	3.79E-09	1.33E-07	0
OH^-	kg/hr	0	0.0326	1.09E-05	0	0.0180	0.0047	0.0407	0.0371	0.0056	0
$HCO_3{}^-$	kg/hr	0	7.4276	0.1627	0	28.7773	261.4887	4.3800	3.7948	138.1108	0
$CO_3{}^{2-}$	kg/hr	0	2.3729	0.0009	0	0.2278	2.8758	4.7376	3.8852	31.8209	0
PZH^+	kg/hr	0	55.6434	0.0002	0	73.3847	87.5106	50.4514	48.2385	51.9419	0
$PZCOO^-$	kg/hr	0	252.0255	1.14E-06	0	174.7905	204.7953	272.7711	273.7693	251.3069	0
$PZ(COO-)_2$	kg/hr	0	25.9329	3.49E-08	0	49.2644	568.8391	16.5500	15.4120	710.6780	0
HPZCOO	kg/hr	0	1.3669	2.60E-05	0	3.1131	201.1070	0.9482	0.7413	142.1550	0
AMP	kg/hr	10.2582	1297.4840	0.0069	7.68E-05	1321.0595	408.5252	1288.2086	1287.2382	271.8827	0
$AMPH^+$	kg/hr	0	165.6766	0.2428	0	142.3507	1051.2393	178.2622	176.6289	1181.5850	0
$AMPCOO^-$	kg/hr	0	2.7498	3.41E-06	0	1.9935	11.8230	2.6258	2.6509	23.3170	0
N_2	kg/hr	2774.7823	0	2774.7817	0.0649	1.97E-13	0.0643	0	8.31E-08	0.0643	2774.8467
O_2	kg/hr	338.2592	0.0016	338.2591	0.0145	0.0016	0.0144	0	0.0016	0.0144	338.2736
Mass Fractions											
PZ		8.38E-05	0.0865	1.92E-09	3.95E-11	0.0904	0.0111	0.0811	0.0861	0.0064	0
H_2O		0.1238	0.5804	0.1258	0.0168	0.5794	0.5199	0.6025	0.5812	0.5245	0.0433
CO_2		0.0161	3.30E-07	0.0158	0.9830	4.21E-05	0.0008	6.16E-08	5.84E-08	1.68E-05	0.1496
H2S		0	0	0	0	0	0	0	0	0	0
H_3O^+		0	2.38E-12	2.45E-11	0	6.64E-11	3.47E-10	7.32E-13	6.96E-13	2.22E-11	0
OH^-		0	6.00E-06	3.02E-09	0	3.32E-06	7.92E-07	7.08E-06	6.81E-06	9.36E-07	0
$HCO_3{}^-$		0	0.0014	4.48E-05	0	0.0053	0.0438	0.0008	0.0007	0.0231	0
$CO_3{}^{2-}$		0	0.0004	2.45E-07	0	4.19E-05	0.0005	0.0008	0.0007	0.0053	0
PZH^+		0	0.0102	6.48E-08	0	0.0135	0.0146	0.0088	0.0089	0.0087	0
$PZCOO^-$		0	0.0464	3.15E-10	0	0.0322	0.0343	0.0474	0.0503	0.0421	0
$PZ(COO-)_2$		0	0.0048	9.63E-12	0	0.0091	0.0952	0.0029	0.0028	0.1189	0
HPZCOO		0	0.0003	7.17E-09	0	0.0006	0.0337	0.0002	0.0001	0.0238	0
AMP		0.0028	0.2387	1.90E-06	1.45E-07	0.2430	0.0684	0.2240	0.2363	0.0455	0
$AMPH^+$		0	0.0305	6.69E-05	0	0.0262	0.1759	0.0310	0.0324	0.1977	0
$AMPCOO^-$		0	0.0005	9.39E-10	0	0.0004	0.0020	0.0005	0.0005	0.0039	0
N_2		0.7641	0	0.7650	0.0001	3.62E-17	1.08E-05	0	1.53E-11	1.08E-05	0.7194
O_2		0.0931	3.00E-07	0.0933	2.74E-05	3.00E-07	2.42E-06	0	3.00E-07	2.42E-06	0.0877
Volume Flow	cum/hr	3772.3242	5.4065	3117.8153	172.7483	5.7186	5.7317	5.6570	5.3544	5.4570	3388.5587

Table 55: Aspen Plus Stream Report: Vapor Phase

Vapor Phase	UNIT	ABGASOUT	CLD-LEAN	CLDGAS	CO2OUT	HOT-LEAN	HOT-RICH	LEAN-LM	LEANLM-C	RICH-LM	SAT-FG
Molar Enthalpy	kJ/mol	-47.3600		-21.8103	-386.8313						-55.6235
Mass Enthalpy	MJ/kg	-1.7739		-0.7847	-9.0039						-1.9037
Molar Entropy	kJ/kmol-K	1.4903		3.3732	-0.8460						5.5592
Mass Entropy	kJ/kg-K	0.0558		0.1214	-0.0197						0.1903
Molar Density	kmol/cum	0.0361		0.0384	0.0712						0.0390
Mass Density	kg/cum	0.9627		1.0684	3.0607						1.1383
Enthalpy Flow	MW	-1.7895		-0.7260	-1.3224						-2.0396
Average MW		26.6978		27.7958	42.9627						29.2193
Mole Flows	kmol/hr	136.0266		119.8292	12.3066						132.0076
PZ	kmol/hr	0.0035		2.05E-10	2.42E-10						0
H_2O	kmol/hr	24.9590		8.9050	0.4941						9.2708
CO_2	kmol/hr	1.3262		1.3017	11.8097						13.1115
H2S	kmol/hr	0		0	0						0
H_3O^+	kmol/hr	0		0	0						0
OH^-	kmol/hr	0		0	0						0
$HCO_3{}^-$	kmol/hr	0		0	0						0
$CO_3{}^{2-}$	kmol/hr	0		0	0						0
PZH^+	kmol/hr	0		0	0						0
$PZCOO^-$	kmol/hr	0		0	0						0
$PZ(COO-)_2$	kmol/hr	0		0	0						0
HPZCOO	kmol/hr	0		0	0						0
AMP	kmol/hr	0.1151		1.28E-06	8.62E-07						0
$AMPH^+$	kmol/hr	0		0	0						0
$AMPCOO^-$	kmol/hr	0		0	0						0
N_2	kmol/hr	99.0517		99.0515	0.0023						99.0540
O_2	kmol/hr	10.5710		10.5710	0.0005						10.5714
Mole Fractions											
PZ		2.60E-05		1.71E-12	1.97E-11						0
H_2O		0.1835		0.0743	0.0401						0.0702
CO_2		0.0097		0.0109	0.9596						0.0993
H2S		0		0	0						0
H_3O^+		0		0	0						0
OH^-		0		0	0						0
$HCO_3{}^-$		0		0	0						0
$CO_3{}^{2-}$		0		0	0						0
PZH^+		0		0	0						0
$PZCOO^-$		0		0	0						0
$PZ(COO-)_2$		0		0	0						0
HPZCOO		0		0	0						0
AMP		0.0008		1.07E-08	7.00E-08						0
$AMPH^+$		0		0	0						0
$AMPCOO^-$		0		0	0						0
N_2		0.7282		0.8266	0.0002						0.7504
O_2		0.0777		0.0882	3.68E-05						0.0801
Mass Flows	kg/hr	3631.6153		3330.7484	528.7249						3857.168
PZ	kg/hr	0.3042		1.77E-08	2.09E-08						0
H_2O	kg/hr	449.6441		160.4260	8.9012						167.0154

Vapor Phase	UNIT	ABGASOUT	CLD-LEAN	CLDGAS	CO2OUT	HOT-LEAN	HOT-RICH	LEAN-LM	LEANLM-C	RICH-LM	SAT-FG
CO_2	kg/hr	58.3671		57.2861	519.7442						577.0323
H2S	kg/hr	0		0	0						0
H_3O^+	kg/hr	0		0	0						0
OH^-	kg/hr	0		0	0						0
HCO_3^-	kg/hr	0		0	0						0
CO_3^{2-}	kg/hr	0		0	0						0
PZH^+	kg/hr	0		0	0						0
$PZCOO^-$	kg/hr	0		0	0						0
$PZ(COO^-)_2$	kg/hr	0		0	0						0
HPZCOO	kg/hr	0		0	0						0
AMP	kg/hr	10.2582		0.0001	7.68E-05						0
$AMPH^+$	kg/hr	0		0	0						0
$AMPCOO^-$	kg/hr	0		0	0						0
N_2	kg/hr	2774.7823		2774.7780	0.0649						2774.8467
O_2	kg/hr	338.2592		338.2582	0.0145						338.2736
Mass Fractions											
PZ		8.38E-05		5.30E-12	3.95E-11						0
H_2O		0.1238		0.0482	0.0168						0.0433
CO_2		0.0161		0.0172	0.9830						0.1496
H2S		0		0	0						0
H_3O^+		0		0	0						0
OH^-		0		0	0						0
HCO_3^-		0		0	0						0
CO_3^{2-}		0		0	0						0
PZH^+		0		0	0						0
$PZCOO^-$		0		0	0						0
$PZ(COO^-)_2$		0		0	0						0
HPZCOO		0		0	0						0
AMP		0.0028		3.42E-08	1.45E-07						0
$AMPH^+$		0		0	0						0
$AMPCOO^-$		0		0	0						0
N_2		0.7641		0.8331	0.0001						0.7194
O_2		0.0931		0.1016	2.74E-05						0.0877
Volume Flow	cum/hr	3772.3242		3117.5166	172.7483						3388.5587

Table 56: Aspen Plus Stream Report: Liquid Phase

Liquid Phase	UNIT	ABGASOUT	CLD-LEAN	CLDGAS	CO2OUT	HOT-LEAN	HOT-RICH	LEAN-LM	LEANLM-C	RICH-LM	SAT-FG
Molar Enthalpy	kJ/mol		-283.6041	-284.7827		-277.3083	-305.8256	-285.3616	-285.3146	-312.2331	
Mass Enthalpy	MJ/kg		-10.4301	-15.7908		-10.1988	-10.3087	-10.7753	-10.5020	-10.5193	
Molar Entropy	kJ/kmol-K		-221.0564	-159.6573		-203.5324	-205.7846	-221.0185	-226.0452	-226.3748	
Mass Entropy	kJ/kg-K		-8.1298	-8.8528		-7.4855	-6.9365	-8.3457	-8.3203	-7.6267	
Molar Density	kmol/cum		36.9777	55.0303		34.9604	35.1419	38.3813	37.4461	36.8920	
Mass Density	kg/cum		1005.4591	992.4542		950.5809	1042.5490	1016.4464	1017.3276	1095.0249	
Enthalpy Flow	MW		-15.7494	-1.3003		-15.4002	-17.1111	-17.2106	-15.8905	-17.4608	
Average MW			27.1909	18.0347		27.1902	29.6668	26.4828	27.1678	29.6819	
Mole Flows	kmol/hr		199.9190	16.4368		199.9242	201.4223	217.1217	200.5005	201.3198	
PZ	kmol/hr		5.4592	8.08E-08		5.7048	0.7675	5.4162	5.4418	0.4445	
H_2O	kmol/hr		175.1306	16.4312		174.8173	172.4397	192.3085	175.7403	173.9793	
CO_2	kmol/hr		4.08E-05	7.23E-05		0.0052	0.1047	8.04E-06	7.23E-06	0.0023	
H2S	kmol/hr		0	0		0	0	0	0	0	
H_3O^+	kmol/hr		6.81E-10	4.68E-09		1.90E-08	1.09E-07	2.21E-10	1.99E-10	6.97E-09	
OH^-	kmol/hr		0.0019	6.44E-07		0.0011	0.0003	0.0024	0.0022	0.0003	
HCO_3^-	kmol/hr		0.1217	0.0027		0.4716	4.2855	0.0718	0.0622	2.2635	
CO_3^{2-}	kmol/hr		0.0395	1.48E-05		0.0038	0.0479	0.0789	0.0647	0.5303	
PZH^+	kmol/hr		0.6385	2.70E-06		0.8421	1.0042	0.5789	0.5535	0.5960	
$PZCOO^-$	kmol/hr		1.9516	8.83E-09		1.3535	1.5858	2.1122	2.1200	1.9460	
$PZ(COO-)_2$	kmol/hr		0.1506	2.03E-10		0.2862	3.3045	0.0961	0.0895	4.1284	
HPZCOO	kmol/hr		0.0105	2.00E-07		0.0239	1.5452	0.0073	0.0057	1.0923	
AMP	kmol/hr		14.5560	7.61E-05		14.8205	4.5831	14.4519	14.4410	3.0502	
$AMPH^+$	kmol/hr		1.8379	0.0027		1.5791	11.6617	1.9775	1.9594	13.1076	
$AMPCOO^-$	kmol/hr		0.0208	2.58E-08		0.0151	0.0895	0.0199	0.0201	0.1765	
N_2	kmol/hr		0	0.0001		7.03E-15	0.0023	0	2.97E-09	0.0023	
O_2	kmol/hr		5.10E-05	2.70E-05		5.10E-05	0.0005	0	5.10E-05	0.0005	
Mole Fractions											
PZ			0.0273	4.92E-09		0.0285	0.0038	0.0249	0.0271	0.0022	
H_2O			0.8760	0.9997		0.8744	0.8561	0.8857	0.8765	0.8642	
CO_2			2.04E-07	4.40E-06		2.60E-05	0.0005	3.70E-08	3.60E-08	1.13E-05	
H2S			0	0		0	0	0	0	0	
H_3O^+			3.40E-12	2.84E-10		9.50E-11	5.42E-10	1.02E-12	9.94E-13	3.46E-11	
OH^-			9.59E-06	3.92E-08		5.31E-06	1.38E-06	1.10E-05	1.09E-05	1.63E-06	
HCO_3^-			0.0006	0.0002		0.0024	0.0213	0.0003	0.0003	0.0112	
CO_3^{2-}			0.0002	9.01E-07		1.90E-05	0.0002	0.0004	0.0003	0.0026	
PZH^+			0.0032	1.64E-07		0.0042	0.0050	0.0027	0.0028	0.0030	
$PZCOO^-$			0.0098	5.37E-10		0.0068	0.0079	0.0097	0.0106	0.0097	
$PZ(COO-)_2$			0.0008	1.24E-11		0.0014	0.0164	0.0004	0.0004	0.0205	
HPZCOO			5.25E-05	1.22E-08		0.0001	0.0077	3.36E-05	2.84E-05	0.0054	
AMP			0.0728	4.63E-06		0.0741	0.0228	0.0666	0.0720	0.0152	
$AMPH^+$			0.0092	0.0002		0.0079	0.0579	0.0091	0.0098	0.0651	
$AMPCOO^-$			0.0001	1.57E-09		7.55E-05	0.0004	9.15E-05	0.0001	0.0009	
N_2			0	8.13E-06		3.51E-17	1.14E-05	0	1.48E-11	1.14E-05	
O_2			2.55E-07	1.64E-06		2.55E-07	2.24E-06	0	2.54E-07	2.24E-06	
Mass Flows	kg/hr		5435.9835	296.4331		5435.9835	5975.5531	5750.0001	5447.1486	5975.5532	
PZ	kg/hr		470.2415	6.96E-06		491.3913	66.1075	466.5325	468.7406	38.2845	
H_2O	kg/hr		3155.0263	296.0119		3149.3822	3106.5492	3464.4915	3166.0100	3134.2858	

Liquid Phase	UNIT	ABGASOUT	CLD-LEAN	CLDGAS	CO2OUT	HOT-LEAN	HOT-RICH	LEAN-LM	LEANLM-C	RICH-LM	SAT-FG
CO_2	kg/hr		0.0018	0.0032		0.2288	4.6089	0.0004	0.0003	0.1002	
H2S	kg/hr		0	0		0	0	0	0	0	
H_3O^+	kg/hr		1.29E-08	8.89E-08		3.61E-07	2.08E-06	4.21E-09	3.79E-09	1.33E-07	
OH^-	kg/hr		0.0326	1.09E-05		0.0180	0.0047	0.0407	0.0371	0.0056	
HCO_3^-	kg/hr		7.4276	0.1627		28.7773	261.4887	4.3800	3.7948	138.1108	
CO_3^{2-}	kg/hr		2.3729	0.0009		0.2278	2.8758	4.7376	3.8852	31.8209	
PZH^+	kg/hr		55.6434	0.0002		73.3847	87.5106	50.4514	48.2385	51.9419	
$PZCOO^-$	kg/hr		252.0255	1.14E-06		174.7905	204.7953	272.7711	273.7693	251.3069	
$PZ(COO-)_2$	kg/hr		25.9329	3.49E-08		49.2644	568.8391	16.5500	15.4120	710.6780	
HPZCOO	kg/hr		1.3669	2.60E-05		3.1131	201.1070	0.9482	0.7413	142.1550	
AMP	kg/hr		1297.4840	0.0068		1321.0595	408.5252	1288.2086	1287.2382	271.8827	
$AMPH^+$	kg/hr		165.6766	0.2428		142.3507	1051.2393	178.2622	176.6289	1181.5850	
$AMPCOO^-$	kg/hr		2.7498	3.41E-06		1.9935	11.8230	2.6258	2.6509	23.3170	
N_2	kg/hr		0	0.0037		1.97E-13	0.0643	0	8.31E-08	0.0643	
O_2	kg/hr		0.0016	0.0009		0.0016	0.0144	0	0.0016	0.0144	
Mass Fractions											
PZ			0.0865	2.35E-08		0.0904	0.0111	0.0811	0.0861	0.0064	
H_2O			0.5804	0.9986		0.5794	0.5199	0.6025	0.5812	0.5245	
CO_2			3.30E-07	1.07E-05		4.21E-05	0.0008	6.16E-08	5.84E-08	1.68E-05	
H2S			0	0		0	0	0	0	0	
H_3O^+			2.38E-12	3.00E-10		6.64E-11	3.47E-10	7.32E-13	6.96E-13	2.22E-11	
OH^-			6.00E-06	3.69E-08		3.32E-06	7.92E-07	7.08E-06	6.81E-06	9.36E-07	
HCO_3^-			0.0014	0.0005		0.0053	0.0438	0.0008	0.0007	0.0231	
CO_3^{2-}			0.0004	3.00E-06		4.19E-05	0.0005	0.0008	0.0007	0.0053	
PZH^+			0.0102	7.93E-07		0.0135	0.0146	0.0088	0.0089	0.0087	
$PZCOO^-$			0.0464	3.85E-09		0.0322	0.0343	0.0474	0.0503	0.0421	
$PZ(COO-)_2$			0.0048	1.18E-10		0.0091	0.0952	0.0029	0.0028	0.1189	
HPZCOO			0.0003	8.77E-08		0.0006	0.0337	0.0002	0.0001	0.0238	
AMP			0.2387	2.29E-05		0.2430	0.0684	0.2240	0.2363	0.0455	
$AMPH^+$			0.0305	0.0008		0.0262	0.1759	0.0310	0.0324	0.1977	
$AMPCOO^-$			0.0005	1.15E-08		0.0004	0.0020	0.0005	0.0005	0.0039	
N_2			0	1.26E-05		3.62E-17	1.08E-05	0	1.53E-11	1.08E-05	
O_2			3.00E-07	2.91E-06		3.00E-07	2.42E-06	0	3.00E-07	2.42E-06	
Volume Flow	cum/hr		5.4065	0.2987		5.7186	5.7317	5.6570	5.3544	5.4570	

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